

ADCMT 7352A OPERATION MANUAL



Quick Links

[Scpi Command Reference](#)

Table of Contents

Contents of this Manual

Product Overview

If a Fault Occurs

Removing the Case

Power Fuse

Electromagnetic Interference

Note When Turning on the Power

Inspection on Delivery

Standard Accessories

Installation Environment

Protecting Against Electrostatic Discharge

Electrostatic Countermeasures for the Work Floor and Human Body

Electrostatic Countermeasures for the Workbench

ESD Countermeasures

Accessories

Input Cable

Power Requirements

Checking the Power Supply Voltage

Power Supply Voltage

Set Power Supply Voltage Indication

Changing the Power Supply Voltage

Connecting the Power Cable

Panel Description

Front Panel Description

Display Section

Power Switch

Measurement Function Keys

Display Section Description

Measurement Range Selection Keys

Sampling Selection Keys

Calculation Selection Keys

MENU/ENTER Key

SHIFT/EXIT/LOCAL Key

Right or Left Display Section Selection Key

Ach Measurement Input Terminals

Bch Measurement Input Terminals

CAT II (Marking)

4.1.2 Rear Panel Description

Rear Panel Description (7352E)

Basic Operation

Measurement Functions

Ach DC Voltage Measurement (DCV-Ach)

Maximum Allowable Applied Voltage (DCV-Ach)

Ach AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)

DC Voltage Measurement (DCV-Ach)

Measurement Range and Input Impedance (DCV-Ach)

Ach Resistance Measurement (2W -Ach and LP-2W -Ach)

AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)

Maximum Allowable Applied Voltage (ACV-Ach, ACV(AC+DC)-Ach)

Resistance Measurement (2W -Ach and LP-2W -Ach)

Maximum Allowable Applied Voltage (2W -Ach, LP-2W -Ach)

Ach DC Current Measurement (DCI-Ach)

Maximum Allowable Applied Current and Protection Function (DCI-Ach)

Ach AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)

AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)

Maximum Allowable Applied Current and Protection Function (ACI-Ach and ACI(AC+DC)-Ach)

Ach Continuity Measurement

Continuity Measurement-Ach

Maximum Allowable Applied Voltage (Continuity-Ach)

Ach Diode Measurement

Diode Measurement-Ach

Maximum Allowable Applied Voltage (Diode-Ach)

Ach Frequency Measurement (FREQ-Ach)

Maximum Allowable Applied Voltage (FREQ-Ach)

Frequency Measurement (FREQ-Ach)

Bch DC Voltage Measurement (DCV-Bch)

Maximum Allowable Applied Voltage (DCV-Bch)

Measurement Range and Input Impedance (DCV-Bch)

Bch DC Current Measurement (DCI-Bch)

Maximum Allowable Applied Current and Protection Function (DCI-Bch)

Bch AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)

AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)

Maximum Allowable Applied Current and Protection Function (ACI-Bch and ACI(AC+DC)-Bch)

Temperature Measurement

Maximum Allowable Applied Voltage (TEMP)

Setting the Measurement Range

Changing the Measurement Speed and Display Digit

Measurement Speed during the HOLD OFF (Trigger source: IMMED)

Measurement Speed during the HOLD ON (Trigger source: EXT) (7352A)

Measurement Speed during the HOLD ON (Trigger source: BUS) GPIB
(7352A)/USB

Measurement Speed during the HOLD ON (Trigger source: BUS) RS-232
(7352A)

How to Use Dual Display

Sampling Rate and Display Digits

How to Use Dual Display by Remote Operation

Menu

How to Navigate the Menus

Menu Operation

Menu Operation Key and Function

Menu List

Measurement Functions

Continuity Measurement

Temperature Measurement

Measurement Range and Auto-Range

Measurement Function and Range Structure

Auto-range

Auto-Zero Operation

Auto-Zero

Display Mode Setting

Display Digit Setting

Sampling Operation

Sampling Operations Combining with Function Settings

Single Measurement Sequence

Dual Sequential Measurement Sequence

Dual Synchronous Measurement Sequence

5.8 Trigger Function

Trigger Model

5.8.1 Trigger System Operation

Trigger Source Selection

Trigger Delay (Td)

Trigger Delay Time, Setting Resolution, and Accuracy

Trigger Delay Operation

Calculation Functions

Calculation System Diagram

NULL Calculation

Smoothing Calculation

Scaling Calculation

dB and dBm Calculations

Comparator Calculation

PASS Condition Setting and PASS/FAIL Judgment

MAX/MIN Calculation

Statistical Calculation

5.9.8 Calculation between 2 Measurements

5.10 Saving and Loading the Measurement Condition Settings

Auto-Loading when Turning on the Power

Saving the Setting Parameters

Loading the Setting Parameters

5.11 Storing and Recalling Measurement Data

Memory Store

Memory Recall

Memory Clear

5.12 System Settings

Disabling Functions

Relationship according to Disabling Function

Front Panel Disabling Function

Error Queue

Interface Selection

GPIB (7352A)

Overview

Interface Functions

Precautions in Use of GPIB

Standard Bus Cable

6.2.3 GPIB Setting

EIA-232] (7352A)

Overview

Setting

Output Data Format

USB

Overview

USB Specifications

Setting Up the USB Port

Connecting to a Personal Computer

USBid Setting

External Trigger Terminal (TRIGGER IN) (7352A)

Simple Equivalent Circuit of Trigger Input Terminal

Command Reference

Selecting the Command Language

Compatibility

6.6.2 Output Data Format

ADC Command Reference

SCPI Command Reference

Status Register Structure

Status Register Structure

Status Register Structure

Status Byte Register Structure

Status Byte Register (STB)

Standard Event Status Register (ESR)

Measurement Event Register (MER)

Questionable Event Register (QER)

Operation Event Register (OER)

Sample Programs (GPIB commands used in ADC CORPORATION)

Sample Programs (SCPI commands for GPIB)

Sample Programs (RS-232)

Interface

Sample Programs (USB)

DC Voltage Measurement (DCV)

Ach DC Voltage Measurement (DCV-Ach)

Bch DC Voltage Measurement (DCV-Bch)

AC Voltage Measurement

Ach AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)

Ach Resistance Measurement (2W -Ach and LP-2W -Ach)

DC Current Measurement (DCI)

Ach DC Current Measurement (DCI-Ach)

Bch DC Current Measurement (DCI-Bch)

AC Current Measurement (ACI, ACI(AC+DC))

Ach AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)

Bch AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)

Ach Frequency Measurement (FREQ-Ach)

Temperature Measurement (TEMP)

Ach Diode Measurement-Ach

Temperature Measurement

Ach Continuity Measurement-Ach

7.1.10 Calculation Functions

Interface Specifications

General Specifications

Accessories

Replacing Fuses

Replacing a Power Fuse

Replacing a Protection Fuse

How to Check Protection Fuses

How to Replace Protection Fuses

Input Terminal Protection Fuse

Cleaning

Cleaning the Outside

Cleaning Others

Calibration

Preparing for Calibration

8.3.2 Calibration Standards

Calibration Point

Calibration Point

8.3.4 Calibration Procedure

Replacing Parts with Limited Life

Parts with Limited Life

Product Disposal and Recycle

Storage

Transportation

Notes Regarding Repair, Replacement, and Periodic Calibration

Before Asking for Repair

Contacting ADC CORPORATION for Repair or Calibration

Address and Phone Number

System Recovery Procedure

Initial Values of Setting Parameters

8.10 Self-Test

Self-Test Items

Display and Key Tests

8.11 Error Message List

Other ManualsLib Projects

7352A/E

Digital Multimeter

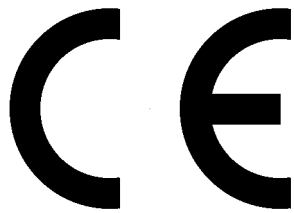
Operation Manual

MANUAL NUMBER FOE-8440254B00

Applicable Models
7352A
7352E



Certificate of Conformity



This is to certify, that

Digital Multimeter

7352A /7352E

instrument, type, designation

complies with the provisions of the EMC Directive, 2004/108/EC in accordance with EN61326

and Low Voltage Directive, 2006/95/EC in accordance with EN61010.

ADC Corp.

Japan

ROHDE&SCHWARZ

Europe GmbH

Munich, Germany

TABLE OF CONTENTS

1.	INTRODUCTION	1-1
1.1	Contents of this Manual	1-1
1.2	Product Overview	1-2
2.	PRECAUTIONS	2-1
2.1	If a Fault Occurs	2-1
2.2	Removing the Case	2-1
2.3	Power Fuse	2-1
2.4	Electromagnetic Interference	2-2
2.5	Note When Turning on the Power	2-2
3.	SETUP	3-1
3.1	Inspection on Delivery	3-1
3.2	Installation Environment	3-3
3.2.1	Operating Environment	3-3
3.2.2	Protecting Against Electrostatic Discharge	3-4
3.3	Accessories	3-5
3.3.1	Input Cable	3-5
3.3.2	Caution when Connecting Peripherals	3-7
3.4	Power Requirements	3-8
3.4.1	Checking the Power Supply Voltage	3-8
3.4.2	Changing the Power Supply Voltage	3-9
3.4.3	Connecting the Power Cable	3-11
4.	QUICK START	4-1
4.1	Panel Description	4-1
4.1.1	Front Panel Description	4-1
4.1.1.1	Display Section	4-2
4.1.1.2	Power Switch	4-2
4.1.1.3	Measurement Function Keys	4-2
4.1.1.4	Measurement Range Selection Keys	4-3
4.1.1.5	Sampling Selection Keys	4-3
4.1.1.6	Calculation Selection Keys	4-3
4.1.1.7	MENU/ENTER Key	4-3
4.1.1.8	SHIFT/EXIT/LOCAL Key	4-4
4.1.1.9	Right or Left Display Section Selection Key	4-4
4.1.1.10	Ach Measurement Input Terminals	4-4
4.1.1.11	Bch Measurement Input Terminals	4-4
4.1.1.12	CAT II (Marking)	4-5
4.1.2	Rear Panel Description	4-6
4.2	Basic Operation	4-8
4.2.1	Measurement Functions	4-8
4.2.1.1	Ach DC Voltage Measurement (DCV-Ach)	4-8
4.2.1.2	Ach AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)	4-9

Table of Contents

4.2.1.3	Ach Resistance Measurement (2W Ω -Ach and LP-2W Ω -Ach)	4-10
4.2.1.4	Ach DC Current Measurement (DCI-Ach)	4-12
4.2.1.5	Ach AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)	4-13
4.2.1.6	Ach Continuity Measurement	4-14
4.2.1.7	Ach Diode Measurement	4-15
4.2.1.8	Ach Frequency Measurement (FREQ-Ach)	4-16
4.2.1.9	Bch DC Voltage Measurement (DCV-Bch)	4-18
4.2.1.10	Bch DC Current Measurement (DCI-Bch)	4-19
4.2.1.11	Bch AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)	4-20
4.2.1.12	Temperature Measurement	4-21
4.2.2	Setting the Measurement Range	4-22
4.2.3	Changing the Measurement Speed and Display Digit	4-22
4.2.4	How to Use Dual Display	4-26
4.2.4.1	How to Use Dual Display by Remote Operation	4-27
5.	MENU OPERATION AND FUNCTION DESCRIPTION	5-1
5.1	Menu	5-1
5.1.1	How to Navigate the Menus	5-1
5.1.2	Menu List	5-4
5.2	Measurement Functions	5-8
5.2.1	Continuity Measurement	5-8
5.2.2	Temperature Measurement	5-8
5.3	Measurement Range and Auto-Range	5-9
5.3.1	Measurement Function and Range Structure	5-9
5.3.2	Auto-range	5-10
5.4	Auto-Zero Operation	5-11
5.5	Display Mode Setting	5-12
5.6	Display Digit Setting	5-12
5.7	Sampling Operation	5-13
5.7.1	Single Measurement	5-14
5.7.2	Dual Sequential Measurement	5-15
5.7.3	Dual Synchronous Measurement	5-16
5.7.4	Sampling Count Setting	5-17
5.8	Trigger Function	5-18
5.8.1	Trigger System Operation	5-19
5.8.2	Trigger Source Selection	5-20
5.8.3	Trigger Delay (Td)	5-21
5.8.4	Trigger Count Setting	5-22
5.9	Calculation Functions	5-23
5.9.1	NULL Calculation	5-25
5.9.2	Smoothing Calculation	5-26
5.9.3	Scaling Calculation	5-28
5.9.4	dB and dBm Calculations	5-30
5.9.5	Comparator Calculation	5-32
5.9.6	MAX/MIN Calculation	5-35
5.9.7	Statistical Calculation	5-37
5.9.8	Calculation between 2 Measurements	5-38
5.10	Saving and Loading the Measurement Condition Settings	5-39

5.10.1	Auto-Loading when Turning on the Power	5-39
5.10.2	Saving the Setting Parameters	5-39
5.10.3	Loading the Setting Parameters	5-39
5.11	Storing and Recalling Measurement Data	5-40
5.11.1	Memory Store	5-40
5.11.2	Memory Recall	5-40
5.11.3	Memory Clear	5-40
5.12	System Settings	5-41
5.12.1	Buzzer Setting	5-41
5.12.2	Disabling Functions	5-41
5.12.3	Front Panel Disabling Function	5-43
5.12.4	Error Queue	5-45
6.	HOW TO USE THE INTERFACE	6-1
6.1	Interface Selection	6-1
6.2	GPIB (7352A)	6-2
6.2.1	Overview	6-2
6.2.2	Precautions in Use of GPIB	6-3
6.2.3	GPIB Setting	6-4
6.3	RS-232[EIA-232] (7352A)	6-5
6.3.1	Overview	6-5
6.3.2	RS-232 Setting	6-6
6.3.3	RS-232 Output Data Format	6-8
6.4	USB	6-9
6.4.1	Overview	6-9
6.4.2	USB Specifications	6-9
6.4.3	Setting Up the USB Port	6-9
6.4.3.1	Connecting to a Personal Computer	6-9
6.4.3.2	USBid Setting	6-9
6.5	External Trigger Terminal (TRIGGER IN) (7352A)	6-10
6.6	Command Reference	6-11
6.6.1	Selecting the Command Language	6-11
6.6.1.1	Compatibility	6-12
6.6.2	Output Data Format	6-13
6.6.3	ADC Command Reference	6-18
6.6.4	SCPI Command Reference	6-31
6.6.5	Status Register Structure	6-43
6.7	Sample Programs	6-52
6.7.1	Sample Programs (GPIB commands used in ADC CORPORATION)	6-52
6.7.2	Sample Programs (SCPI commands for GPIB)	6-56
6.7.3	Sample Programs (RS-232)	6-60
6.7.4	Sample Programs (USB)	6-62
7.	SPECIFICATIONS	7-1
7.1	Specifications	7-1
7.1.1	DC Voltage Measurement (DCV)	7-1
7.1.1.1	Ach DC Voltage Measurement (DCV-Ach)	7-1
7.1.1.2	Bch DC Voltage Measurement (DCV-Bch)	7-2

Table of Contents

7.1.2	AC Voltage Measurement	7-3
7.1.2.1	Ach AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)	7-3
7.1.3	Ach Resistance Measurement (2W Ω -Ach and LP-2W Ω -Ach)	7-5
7.1.4	DC Current Measurement (DCI)	7-6
7.1.4.1	Ach DC Current Measurement (DCI-Ach)	7-6
7.1.4.2	Bch DC Current Measurement (DCI-Bch)	7-7
7.1.5	AC Current Measurement (ACI, ACI(AC+DC))	7-8
7.1.5.1	Ach AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)	7-8
7.1.5.2	Bch AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)	7-10
7.1.6	Ach Frequency Measurement (FREQ-Ach)	7-12
7.1.7	Temperature Measurement (TEMP)	7-13
7.1.8	Ach Diode Measurement-Ach	7-13
7.1.9	Ach Continuity Measurement-Ach	7-14
7.1.10	Calculation Functions	7-15
7.2	Interface Specifications	7-16
7.3	General Specifications	7-17
7.4	Accessories	7-18
8.	MAINTENANCE	8-1
8.1	Replacing Fuses	8-1
8.1.1	Replacing a Power Fuse	8-1
8.1.2	Replacing a Protection Fuse	8-2
8.1.2.1	How to Check Protection Fuses	8-2
8.1.2.2	How to Replace Protection Fuses	8-3
8.2	Cleaning	8-4
8.2.1	Cleaning the Outside	8-4
8.2.2	Cleaning Others	8-4
8.3	Calibration	8-5
8.3.1	Preparing for Calibration	8-5
8.3.2	Calibration Standards	8-6
8.3.3	Calibration Point	8-7
8.3.4	Calibration Procedure	8-9
8.4	Replacing Parts with Limited Life	8-12
8.5	Product Disposal and Recycle	8-13
8.6	Storage	8-15
8.7	Transportation	8-15
8.8	Notes Regarding Repair, Replacement, and Periodic Calibration	8-16
8.8.1	Before Asking for Repair	8-16
8.8.2	Contacting ADC CORPORATION for Repair or Calibration	8-17
8.8.3	Address and Phone Number	8-17
8.9	System Recovery Procedure	8-18
8.10	Self-Test	8-22
8.10.1	Self-Test	8-22
8.10.2	Display and Key Tests	8-23
8.11	Error Message List	8-24

7352A DIMENSIONAL OUTLINE DRAWING	EXT-1
---	-------

7352E DIMENSIONAL OUTLINE DRAWING	EXT-2
ALPHABETICAL INDEX	I-1

LIST OF ILLUSTRATIONS

No.	Title	Page
3-1	Operating Environment	3-3
3-2	Electrostatic Countermeasures for the Work Floor and Human Body	3-4
3-3	Electrostatic Countermeasures for the Workbench	3-4
3-4	Set Power Supply Voltage Indication	3-8
3-5	Connecting the Power Cable	3-11
4-1	Front Panel Description	4-1
4-2	Display Section Description	4-2
4-3	Rear Panel (7352A)	4-6
4-4	Rear Panel Description (7352E)	4-7
4-5	DC Voltage Measurement (DCV-Ach)	4-9
4-6	AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)	4-10
4-7	Resistance Measurement (2W Ω -Ach and LP-2W Ω -Ach)	4-11
4-8	DC Current Measurement (DCI-Ach)	4-12
4-9	AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)	4-13
4-10	Continuity Measurement-Ach	4-14
4-11	Diode Measurement-Ach	4-15
4-12	Frequency Measurement (FREQ-Ach)	4-17
4-13	DC Voltage Measurement (DCV-Bch)	4-18
4-14	DC Current Measurement (DCI-Bch)	4-19
4-15	AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)	4-20
4-16	Temperature Measurement	4-21
5-1	Menu Operation	5-1
5-2	Single Measurement Sequence	5-15
5-3	Dual Sequential Measurement Sequence	5-16
5-4	Dual Synchronous Measurement Sequence	5-17
5-5	Trigger Model	5-18
5-6	Trigger Delay Operation	5-22
5-7	Calculation System Diagram	5-24
6-1	Simple Equivalent Circuit of Trigger Input Terminal	6-10
6-2	Status Register Structure (1/2)	6-44
6-3	Status Register Structure (2/2)	6-45
6-4	Status Byte Register Structure	6-46
8-1	How to Check Protection Fuses	8-2
8-2	Input Terminal Protection Fuse	8-3

LIST OF TABLES

No.	Title	Page
3-1	Standard Accessories	3-2
3-2	ESD Countermeasures	3-4
3-3	Accessories	3-5
3-4	Power Supply Voltage	3-8
4-1	Maximum Allowable Applied Voltage (DCV-Ach)	4-8
4-2	Measurement Range and Input Impedance (DCV-Ach)	4-9
4-3	Maximum Allowable Applied Voltage (ACV-Ach, ACV(AC+DC)-Ach)	4-10
4-4	Maximum Allowable Applied Voltage (2W Ω -Ach, LP-2W Ω -Ach)	4-11
4-5	Maximum Allowable Applied Current and Protection Function (DCI-Ach)	4-12
4-6	Maximum Allowable Applied Current and Protection Function (ACI-Ach and ACI(AC+DC)-Ach)	4-13
4-7	Maximum Allowable Applied Voltage (Continuity-Ach)	4-14
4-8	Maximum Allowable Applied Voltage (Diode-Ach)	4-15
4-9	Maximum Allowable Applied Voltage (FREQ-Ach)	4-16
4-10	Maximum Allowable Applied Voltage (DCV-Bch)	4-18
4-11	Measurement Range and Input Impedance (DCV-Bch)	4-18
4-12	Maximum Allowable Applied Current and Protection Function (DCI-Bch)	4-19
4-13	Maximum Allowable Applied Current and Protection Function (ACI-Bch and ACI(AC+DC)-Bch)	4-20
4-14	Maximum Allowable Applied Voltage (TEMP)	4-21
4-15	Measurement Speed during the HOLD OFF (Trigger source: IMMED)	4-22
4-16	Measurement Speed during the HOLD ON (Trigger source: EXT) (7352A)	4-23
4-17	Measurement Speed during the HOLD ON (Trigger source: BUS) GPIB (7352A)/USB	4-24
4-18	Measurement Speed during the HOLD ON (Trigger source: BUS) RS-232 (7352A)	4-25
4-19	Sampling Rate and Display Digits	4-26
5-1	Menu Operation Key and Function	5-3
5-2	Measurement Function and Range Structure	5-9
5-3	Auto-Zero	5-11
5-4	Sampling Operations Combining with Function Settings	5-14
5-5	Trigger Delay Time, Setting Resolution, and Accuracy	5-21
5-6	PASS Condition Setting and PASS/FAIL Judgment	5-34
5-7	Relationship according to Disabling Function	5-42
6-1	Interface	6-1
6-2	Interface Functions	6-2
6-3	Standard Bus Cable	6-3
6-4	Status Byte Register (STB)	6-47
6-5	Standard Event Status Register (ESR)	6-48
6-6	Measurement Event Register (MER)	6-49
6-7	Questionable Event Register (QER)	6-50
6-8	Operation Event Register (OER)	6-51
7-1	Accessories	7-18

List of Tables

No.	Title	Page
8-1	Calibration Standards	8-6
8-2	Calibration Point	8-7
8-3	Parts with Limited Life	8-12
8-4	Before Asking for Repair	8-16
8-5	Initial Values of Setting Parameters	8-19
8-6	Self-Test Items	8-22

1. INTRODUCTION

This chapter describes the contents of this manual and the product overview of the 7352 Series Digital Multimeter.

1.1 Contents of this Manual

This manual can be used by novices or experienced users of this instrument. This manual can be read from Chapter 1 to learn more about this instrument or it can be used as a reference. Refer to the table of contents, which is found at the beginning of each chapter, and directly jump to the required section.

The contents of each chapter are as follows.

Chapter 1. INTRODUCTION	This chapter describes the contents of this manual and the product overview.
Chapter 2. PRECAUTIONS	This chapter describes precautions when using this instrument. Read this chapter before using this instrument.
Chapter 3. SETUP	This chapter describes how to setup this instrument. After setting up this instrument in an appropriate location, turn on the power and check that this instrument starts correctly.
Chapter 4. QUICK START	This chapter describes the function of each section on the panels of this instrument and how to operate this instrument.
Chapter 5. MENU OPERATION AND FUNCTION DESCRIPTION	This chapter describes the menu structure and functions of this instrument.
Chapter 6. HOW TO USE THE INTERFACE	This chapter describes how to use the interface of this instrument.
Chapter 7. SPECIFICATIONS	This chapter describes the specifications of this instrument.
Chapter 8. MAINTENANCE	This chapter describes how to care for this instrument such as cleaning, calibration, and storage to maintain the high performance and smooth functioning of this instrument. This chapter also describes how to identify problems and the relevant procedures to follow.

1.2 Product Overview

This digital multimeter includes an integration type AD converter.

This digital multimeter is ideal for use as a stand-alone product or in a system.

The main features of this instrument are as follows:

- The 2ch synchronous measurement can be performed by the use of the twin AD converter.
- Insulated structure between Ach and Bch input terminals allows for floating measurement.
- Two types of measured values or calculation results are simultaneously displayed on the dual display.
- Simple and easy panel operations, with flexible and various system functions by using menus.
- Large and visible fluorescent display.
- Wide dynamic range: Maximum display of 199999 on both displays.
- High sensitivity: 1 μ V for measuring DC voltage.
- Wide dynamic range: 10 pA to 10 A for measuring DC current.
- Low power resistance measurement function that reduces the self-heating effect of the device.
- This instrument is equipped with a USB, GPIB, RS-232, and trigger input and can easily be connected to a PC. (USB is available only for the 7352E).
- A large variety of calculation functions that can be combined.
- Four files of measurement parameters can be saved.
- Conventional and SCPI command systems are included as standard with this instrument.

2. PRECAUTIONS

This chapter describes precautions when using this instrument. Read this chapter before using this instrument.

CAUTION: *Using this instrument while it is stood on end or stacked with other products is unstable and dangerous.
Do not push or place heavy loads on this instrument.*

2.1 If a Fault Occurs

If any smoke, smell, or noise emanates from this instrument, disconnect this instrument from the power supply by removing the power cable. Contact an ADC CORPORATION sales representative immediately.

2.2 Removing the Case

The case of this instrument should only be opened by ADC CORPORATION service engineers.

WARNING: *This instrument contains high-voltage and high temperature parts which may cause electrical shocks or burns.*

2.3 Power Fuse

This instrument is protected from overcurrent by a power fuse.

The power fuse, which is located on the rear panel, cuts the power supply if an overcurrent flows. If the power fuse was blown, remove the power cable and separate between the power supply and this instrument. There may be some problems in this instrument and request an ADC CORPORATION sales representative to repair it.

2.4 Electromagnetic Interference

2.4 Electromagnetic Interference

This instrument may cause electromagnetic interference and affect television and radio. If this instrument's power is turned off and any electromagnetic interference that may be present is reduced, then this instrument has caused the interference.

Electromagnetic interference from this instrument may be prevented by the following precautions.

- Changing the direction of the antenna of the television or radio.
- Placing this instrument on the opposite side of the television or radio.
- Placing this instrument away from the television or radio.
- Using a different power source for the television or radio, and this instrument.

In an area where the electric field strength is high, high-frequency noise will occur on a DUT or input cable, and that noise may affect the measured value.

In this case, shield the DUT and input cable from high-frequency noise.

2.5 Note When Turning on the Power

When turning on the power, do not connect any DUTs.

3. SETUP

This chapter describes how to set up this instrument on delivery. The following topics are covered in this chapter.

- 3.1 Inspection on Delivery
- 3.2 Installation Environment
- 3.3 Accessories
- 3.4 Power Requirements

3.1 Inspection on Delivery

After receiving the product, inspect the outside and the accessories according to the following procedure.

1. Check that the shipping container and the cushioning material are not damaged.

IMPORTANT: *If the shipping container or the cushioning material is damaged, keep them until the following inspections are complete.*

2. Check that the outside of the product is not damaged.

WARNING: *If any outside components of the product such as the cover, panel (front or rear), power switch, or connector are damaged, do not turn on the power. Doing so may cause an electrical shock.*

3. Check that the standard accessories listed in Table 3-1 are complete and not damaged.

If any of the following occur, contact ADC CORPORATION sales representative.

- The shipping container or the cushioning material is damaged, or signs of stress are found.
- The outside of the product is damaged.
- The standard accessories are incomplete or are damaged.
- Defects are found in the operation check.

3.1 Inspection on Delivery

Table 3-1 Standard Accessories

Name	Model	Quantity	Remarks
Power cable	A01402	1	*1
Input cable (Red, Black)	CC010001	Each 1	
Power Fuse (For 100V/120V)	DFT-AAR315A-1	1	*2
Power Fuse (For 220V/240V)	DFT-AAR25A-1		
Overcurrent protection fuse	DFS-AN2A-1	1	
Operation manual	E7352A/E	1	This manual

*1: The power cable included in this instrument depends on the option that was specified when this instrument was purchased (see “Safety Summary”).

Quote the model name or option number when ordering the power cable.

*2: Either one is included according to the specified option.

NOTE: *Quote the model name of the accessory when ordering.*

3.2 Installation Environment

This section describes the environment into which this instrument should be installed.

3.2.1 Operating Environment

Install this instrument in an environment in which the following conditions are satisfied.

- Ambient temperature: 0°C to +50°C (operating temperature)
-25°C to +70°C (storage temperature)
- Relative humidity: 85% or less with no condensation
- An area free from corrosive gas
- An area away from direct sunlight
- A dust-free area
- An area free from vibrations
- A low noise area

Although this instrument has been designed to withstand a certain amount of noise from the AC power line, it should be used in a low noise area. Use a noise cut filter if ambient noise is unavoidable.

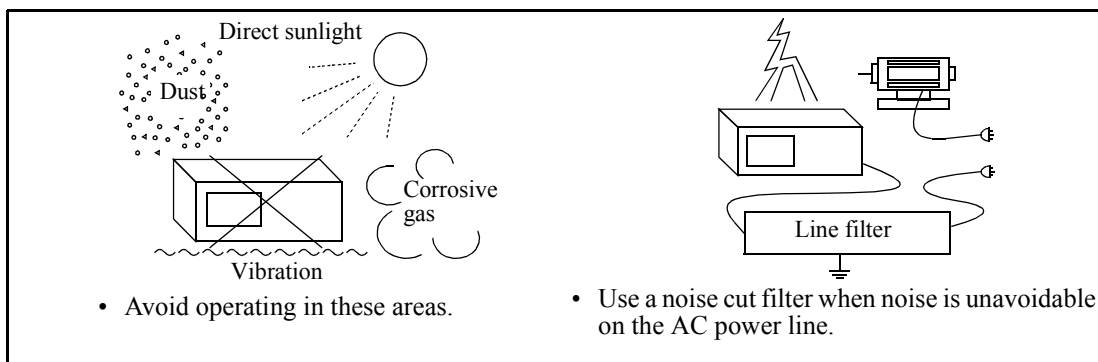


Figure 3-1 Operating Environment

NOTE: *Warm-up*

After this instrument reaches room temperature, turn on the power and warm up for 60 minutes or more to perform the high accuracy measurement.

3.2.2 Protecting Against Electrostatic Discharge

3.2.2 Protecting Against Electrostatic Discharge

To prevent semiconductors from being damaged by electrostatic discharge (ESD), the precautions shown below should be taken. We recommend combining two or more countermeasures to prevent damage from ESD.

(Static electricity can be generated easily by the movement of a person or the friction against insulation.)

Table 3-2 ESD Countermeasures

Human Body	Use a wrist strap (See Figure 3-2).
Work floor	Install a conductive mat, use conductive shoes, and connect to earth. (See Figure 3-2)
Workbench	Install a conductive mat and connect to earth. (See Figure 3-3)

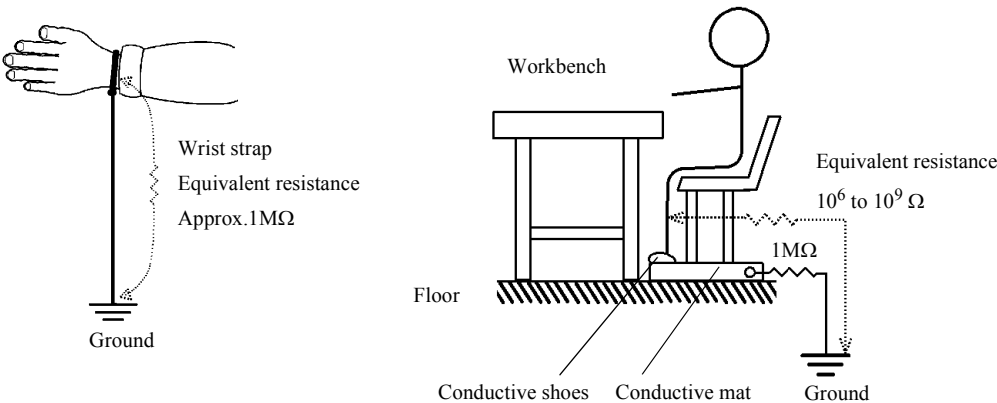


Figure 3-2 Electrostatic Countermeasures for the Work Floor and Human Body

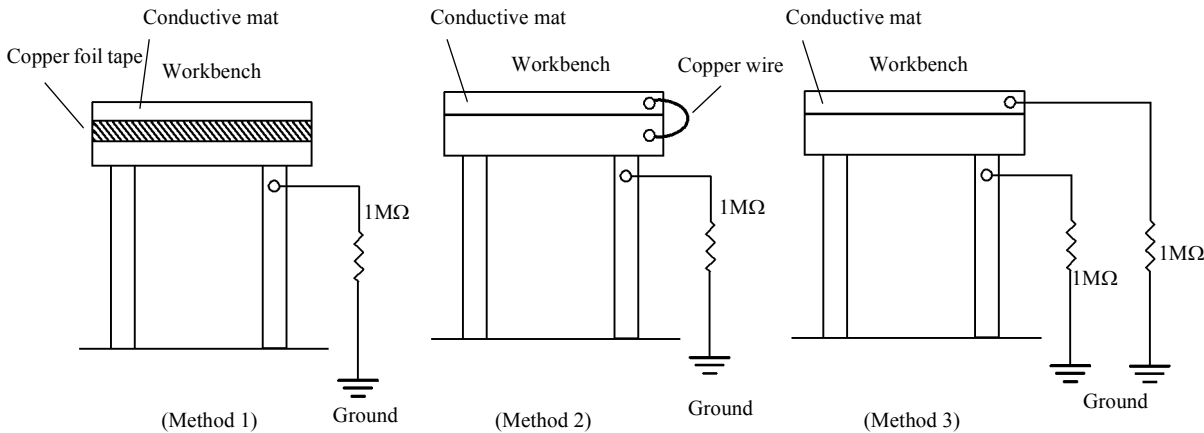


Figure 3-3 Electrostatic Countermeasures for the Workbench

3.3 Accessories

Accessories for this instrument are shown in Table 3-3.

This section describes accessories necessary to use this instrument.

Table 3-3 Accessories

Name	Model	Remarks
Input cable	CC010001	Standard accessory
	A01001	Shielded cable 33 VAC and 70 VDC or less
Alligator clip adapter	CC015001	33 VAC and 70 VDC or less
RS-232 cable	A01265	
Terminal adapter	TR1111	
JIS rack-mount set	A02263	
	A02264	Twins
EIA rack-mount set	A02463	
	A02464	Twins
Panel-mount set	A02039	
	A02040	Twins
Sheath type thermocouple	TR1101-100	T(CC) type
	TR1101-130	K(CA) type

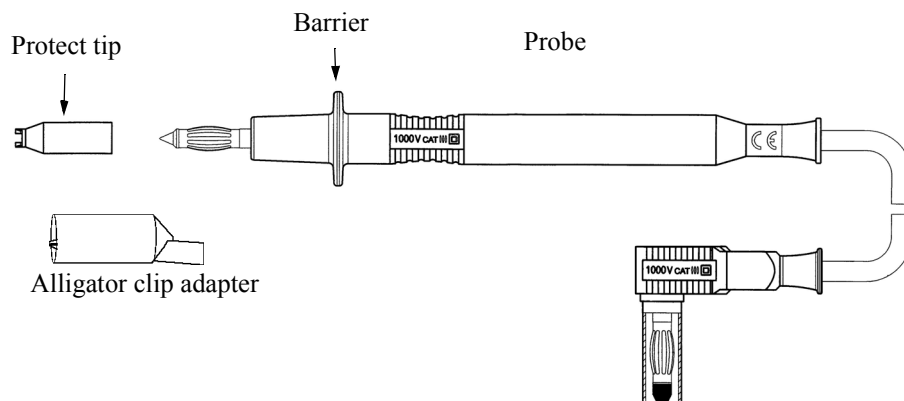
3.3.1 Input Cable

- The standard input cable (CC010001) consists of red and black and it can be used easily.

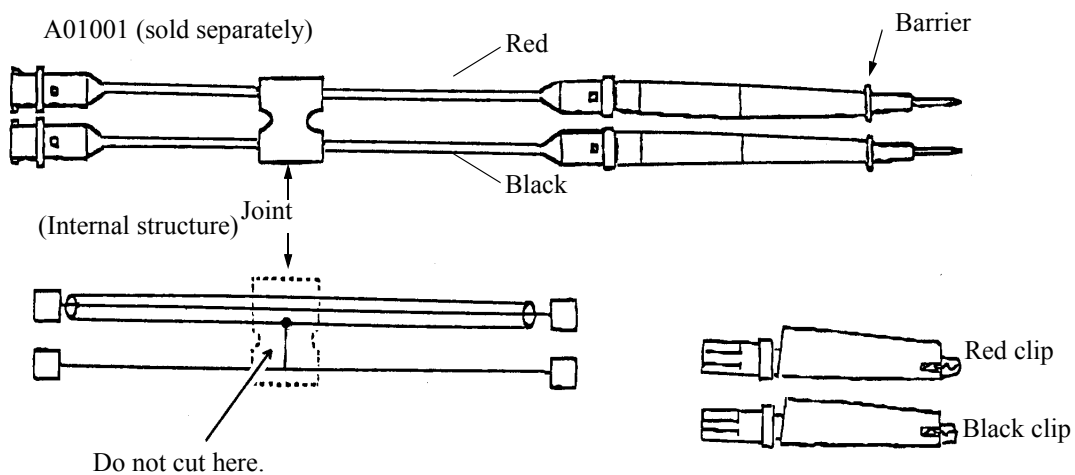
CAUTION: *Handle the input cable with caution because it has a sharp tip. The rating of the alligator clip adapter is 33 VAC and 70 VDC or less.*

WARNING: *Must not touch the metal part of the input cable to prevent any damage of electrical shock.*

3.3.1 Input Cable



2. To reduce the effects of short-term stability in high resistance (in the order of $M\Omega$) or high sensitivity (in the order of μV) measurements, use the optional input cable A01001 in which the HI side is shielded.



The red wire of A01001 is shielded. Do not cut the joint.

The rating of the probe decreases to 33 VAC and 70 VDC when the alligator clip adapter is attached.

3.3.2 Caution when Connecting Peripherals

When making up a system, which consists of two or more devices, be careful of the following:

1. Before connecting this instrument to a controller and peripherals, check that each device operates correctly according to each manual.
2. Do not use too long a cable to connect the instrument or controller.
3. Use shielded cables which are connected between peripherals.

Setting up the USB port

To use the USB interface of this instrument, the USB driver of the ADC instrument must be installed in a personal computer.

The USB driver of the ADC instrument can be downloaded from the following ADC home page.

URL <http://www.adc-tech.co.jp>

For more information on how to install the driver, refer to the operation manual that is included in the driver file.

Compatible OS: English-language version of Windows XP Professional

Compatible Languages: MS Excel2000 VBA
MS Visual Basic 6.0
MS Visual C++6.0

3.4 Power Requirements

3.4 Power Requirements

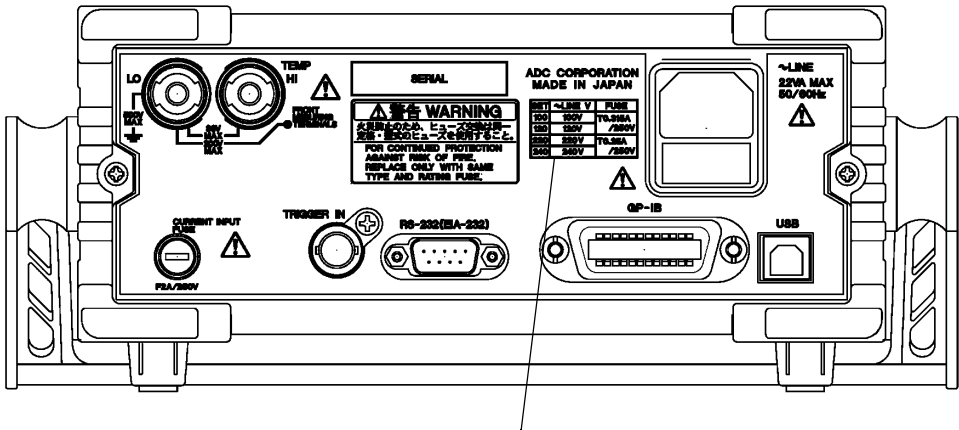
This section describes power requirements and how to connect the power cable.

3.4.1 Checking the Power Supply Voltage

Check that the power supply voltage setting on the rear panel of this instrument is the same as the AC power supply voltage to be used.

Table 3-4 Power Supply Voltage

Power supply		Voltage setting	Power supply fuse
Voltage	frequency		
90 V - 110 V	48-66 Hz	100 V	T315 mA/250 V
108 V - 132 V		120 V	
198 V - 242 V		220 V	T250 mA/250 V
216 V - 250 V		240 V	



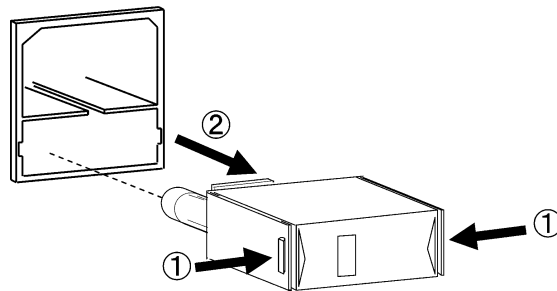
The set power supply voltage indication of this instrument (on the rear panel).

Figure 3-4 Set Power Supply Voltage Indication

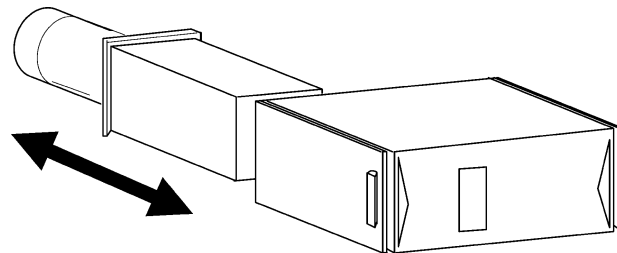
3.4.2 Changing the Power Supply Voltage

1. Remove the fuse holder assembly from the rear panel.

Push both side of the fuse holder assembly by using flathead screwdrivers (①) and then pull it out (②).



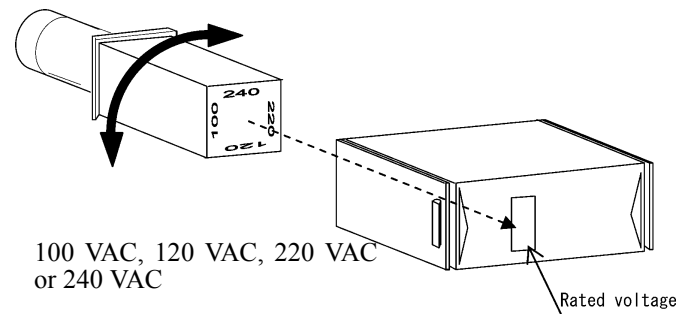
2. Remove the power supply voltage selector from the fuse holder assembly.



100 VAC or 120 VAC: T315 mA fuse

220 VAC or 240 VAC: T250 mA fuse

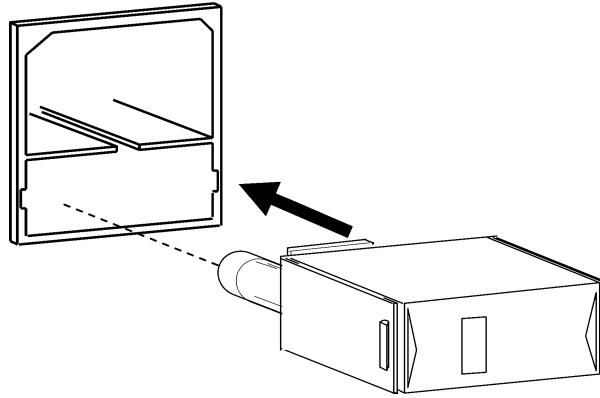
3. Rotate the power supply selector to be looked the correct voltage from the window.



4. Replace a fuse that is compliant with the specification (see Table 3-1).

3.4.2 Changing the Power Supply Voltage

5. Replace the fuse holder assembly into the rear panel.



6. Check that the correct power supply voltage is indicated in the window.

3.4.3 Connecting the Power Cable

This instrument includes a three-core power cable with a grounding conductor. To prevent accidents caused by electric shocks, only use the included power cable and securely connect to the ground through a three-pin power outlet.

1. Check that the included power cable is not damaged.

WARNING: *Never use a damaged power cable. Doing so may cause an electrical shock.*

2. Connect the AC power connector on the rear panel of this instrument to a three-pin power outlet that has a ground terminal by using the included power cable (See Figure 3-5).

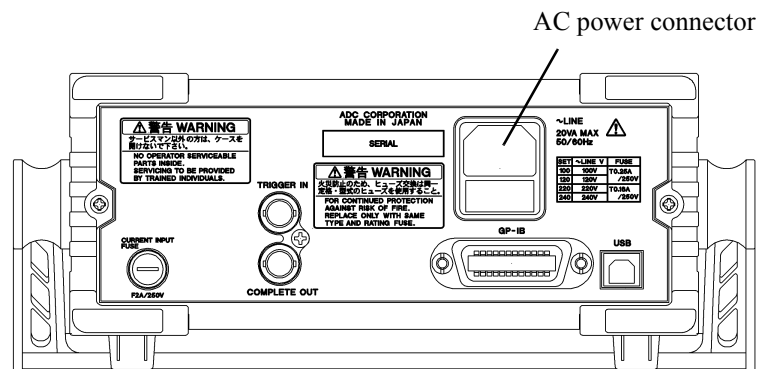


Figure 3-5 Connecting the Power Cable

WARNING:

1. Use a suitable power cable for the power supply voltage.
 2. To prevent any danger of electrical shock, connect the power cable to a three-pin power outlet that is connected to a ground terminal. The instrument will not be grounded if an extension cord, which does not include a ground terminal, is used.
-

4. QUICK START

The high speed measurement and high accuracy measurement can be performed in this digital multimeter. To make the performing operations easier, this instrument includes the following functions on the front panel.

Setting the measurement function

Setting the measurement range

Sampling conditions

Setting the NULL and smoothing calculations to ON or OFF.

Chapter 4 describes the front and rear panels and panel operations which do not use MENU. Other settings are set in the layered menus. For more information on function descriptions and how to navigate the menus, refer to Chapter 5.

4.1 Panel Description

4.1.1 Front Panel Description

The front panel consists of the display section, operation keys, and measurement input terminal section.

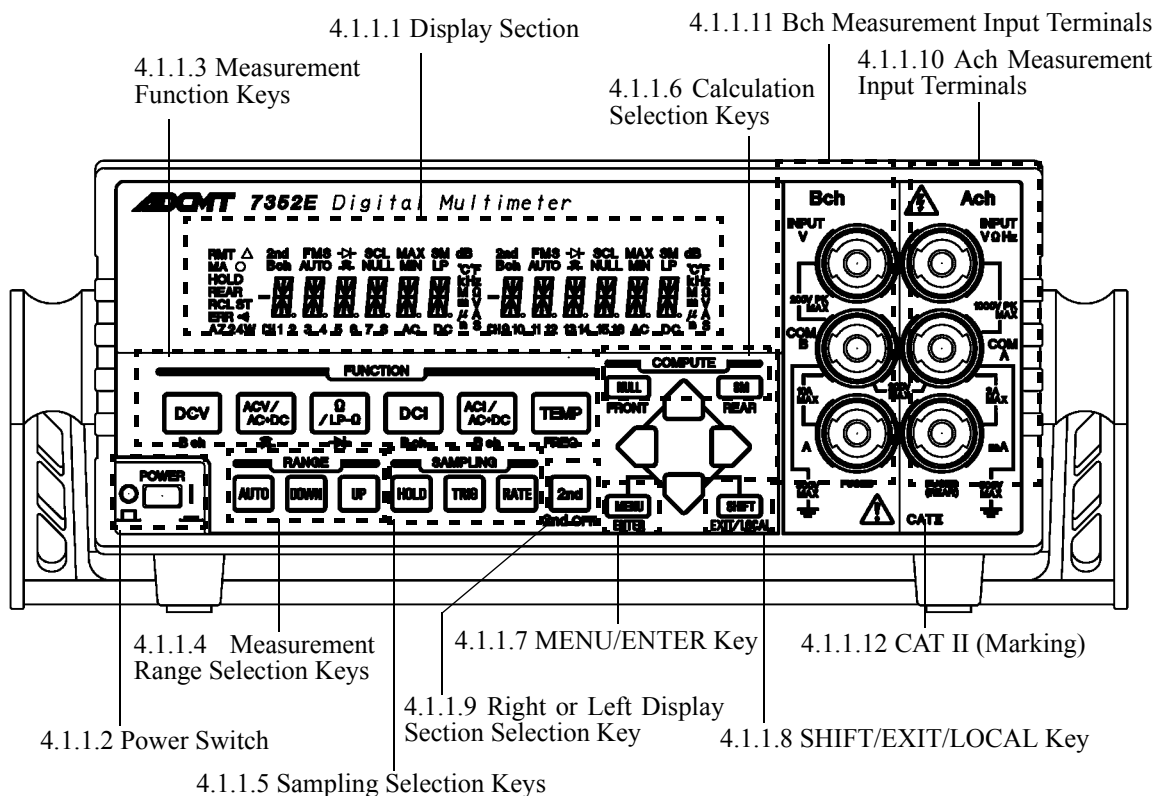


Figure 4-1 Front Panel Description

4.1.1 Front Panel Description

4.1.1.1 Display Section

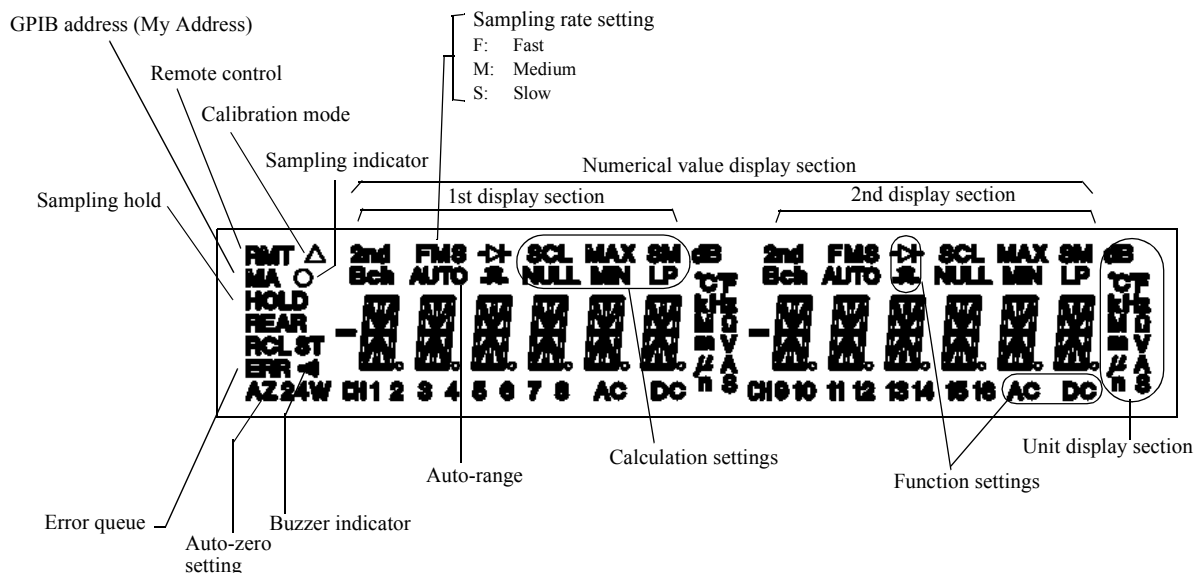


Figure 4-2 Display Section Description

4.1.1.2 Power Switch

Pressing the power switch turns on the power and the power is supplied to this instrument.

Re-pressing the power switch turns off the power and the power is shut down.

4.1.1.3 Measurement Function Keys

1. DCV-Ach Selects the Ach DC voltage measurement (DCV).
2. ACV/AC+DC-Ach Selects the Ach AC voltage measurement (ACV). AC coupling and AC+DC coupling are alternately switched by pressing this key while ACV is selected.
3. Ω /LP- Ω -Ach Selects the Ach resistance measurement ($2W\Omega$) in which the two-wire method is used. $2W\Omega$ and LP- Ω (low power resistance measurement) are alternately switched by pressing this key while $2W\Omega$ is selected.
4. DCI-Ach Selects the Ach DC current measurement (DCI).
5. ACI/AC+DC-Ach Selects the Ach AC current measurement (ACI). AC coupling and AC+DC coupling are alternately switched by pressing this key while ACI is selected.
6. TEMP Selects the temperature measurement (TEMP).
7. DCV-Bch(SHIFT + DCV) Selects the Bch DC voltage measurement (DCV).

- | | |
|--------------------------------|---|
| 8. Continuity(SHIFT + ACV) | Selects the Ach continuity measurement. |
| 9. Diode(SHIFT + Ω) | Selects the Ach diode measurement. |
| 10. DCI-Bch(SHIFT + DCI) | Selects the Bch DC current measurement (DCI). |
| 11. ACI/AC+DC-Bch(SHIFT + ACI) | Selects the Bch AC current measurement (ACI).
AC coupling and AC+DC coupling are alternately switched by pressing SHIFT and ACI while ACI-Bch is selected. |
| 12. FREQ(SHIFT + TEMP) | Selects the Ach AC voltage frequency measurement (FREQ). |

4.1.1.4 Measurement Range Selection Keys

- | | |
|---------|--|
| 1. AUTO | Switches the measurement range between auto-range and fixed range. |
| 2. DOWN | Decreases the measurement range by one level and fixes the range. |
| 3. UP | Increases the measurement range by one level and fixes the range. |

4.1.1.5 Sampling Selection Keys

- | | |
|---------|---|
| 1. HOLD | Holds or releases the sampling mode. |
| 2. TRIG | Starts the measurement when the sampling mode is set to HOLD. |
| 3. RATE | Sets the sampling rate to high (FAST), medium (MED), or low speed (SLOW1 or SLOW2). |

4.1.1.6 Calculation Selection Keys

- | | |
|---------|--|
| 1. NULL | Sets or cancels the NULL calculation. |
| 2. SM | Sets or cancels the smoothing calculation. |

4.1.1.7 MENU/ENTER Key

- | | |
|--|--|
| 1. MENU | Settings are configured by the MENU operation. |
| 2. ENTER(For MENU operation) | Saves the settings. |
| 3.     | Selects a menu item and its layer. |

4.1.1 Front Panel Description

4.1.1.8 SHIFT/EXIT/LOCAL Key

- | | | |
|----|-----------------------------|---|
| 1. | SHIFT | Sets or cancels the SHIFT mode. |
| 2. | EXIT(For MENU operation) | Exits from MENU and returns to the measurement state display. (Settings in the process of being changed are not saved.) |
| 3. | LOCAL(For remote operation) | Changes to the LOCAL operation when the instrument is in the remote state. |

NOTE: *If the LLO (Local LockOut) command is set from the GPIB interface, this instrument does not change to the LOCAL operation.*

4.1.1.9 Right or Left Display Section Selection Key

- | | | |
|----|-------------|---|
| 1. | 2nd | The right display section (the second display) is enabled and the '2nd' indicator is turned on. Since then, operations while the '2nd' indicator is on affect the right display. Pressing this key again activates the left display section (the first display) and the '2nd' indicator is turned off. Since then, operations while the '2nd' indicator is off affect the left display. |
| 2. | SHIFT + 2nd | The right display section (the second display) is disabled and turned off. |

4.1.1.10 Ach Measurement Input Terminals

- | | | |
|----|---------------------|--|
| 1. | INPUT VΩHz terminal | HI terminal used for the Ach DC voltage, AC voltage, resistance, diode, continuity, and frequency measurements |
| 2. | COM A terminal | LO terminal used for the Ach DC voltage, AC voltage, resistance, DC current, AC current, diode, continuity, and frequency measurements |
| 3. | mA terminal | HI terminal used for the Ach DC current and AC current measurements |

4.1.1.11 Bch Measurement Input Terminals

- | | | |
|----|------------------|--|
| 1. | INPUT V terminal | HI terminal used for the Bch DC voltage measurement |
| 2. | COM B terminal | LO terminal used for the Bch DC voltage, DC current, and AC current measurements |
| 3. | A terminal | HI terminal used for the Bch DC current and AC current measurements |

4.1.1.12 CAT II (Marking)

This display shows that the safety of this instrument satisfies the measurement category II which is compliant with IEC 61010-1 Ed.2. Measurement category II is for measurements performed on circuits directly connected to the low voltage equipment.

4.1.2 Rear Panel Description

4.1.2 Rear Panel Description

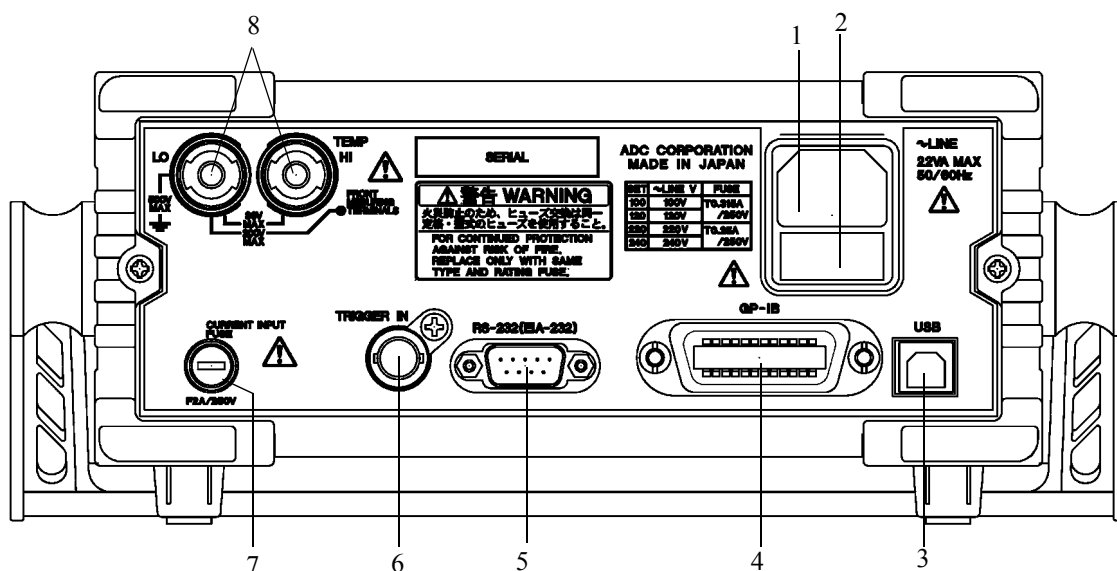
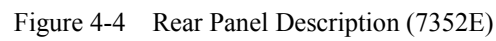


Figure 4-3 Rear Panel (7352A)

- | | |
|---|--|
| 1. Power connector | Connector for the AC power supply
The included power cable is connected here. |
| 2. Fuse holder assembly | The power supply voltage can be selected from 100 V, 120 V, 220 V, and 240 V. A slow-blow fuse is stored here. |
| 3. USB connector | Connector for USB
This connector is used to output data and for remote control. |
| 4. GPIB connector (7352A) | Connector for GPIB
This connector is used to output data and for remote control. |
| 5. RS-232 connector (7352A) | Connector for RS-232
This connector is used to output data and for remote control. |
| 6. TRIGGER IN (7352A) | This connector is used to input the external trigger. |
| 7. Current input terminal protection fuse | Protects the current input terminal (A _{ch}). |
| 8. Measurement input terminals | TEMP HI and LO terminals
A thermocouple for measuring temperature is connected here. |



4.2 Basic Operation

4.2 Basic Operation

When power is supplied (the power switch is turned on), this instrument automatically executes a self-test and then measurement can be performed.

For more information on the details of the self-test, refer to Chapter 8, “MAINTENANCE.”

When the self-test is complete without any abnormality, this instrument enters into the measurement state. If an abnormality occurs, the instrument enters into the error display state. Check the error number by operating menus and exit from here.

For more information on navigating menus, refer to Section 5.1, “Menu.”

This instrument saves the set conditions when the power is turned off. The factory defaults are used for descriptions in Chapter 4. For more information on how to initialize this instrument, refer to Chapter 8.

Depending on the measurement function, the Bch indicator is turned on or off as described below:

The Bch indicator is on for the Bch measurement function.

The Bch indicator is off for the Ach measurement function or the temperature measurement.

4.2.1 Measurement Functions

4.2.1.1 Ach DC Voltage Measurement (DCV-Ach)

1. Connect a DUT to the Ach COM and V terminals on the front panel.
2. Press **DCV**.

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electrical shock due to the failure of this instrument may occur.

Table 4-1 Maximum Allowable Applied Voltage (DCV-Ach)

Input terminal		Maximum allowable applied voltage
Ach	Between VΩHz and COM A terminals	1000 V _{peak}
	Between COM A terminal and the chassis	500 V
	Between COM A and COM B terminals	200 V

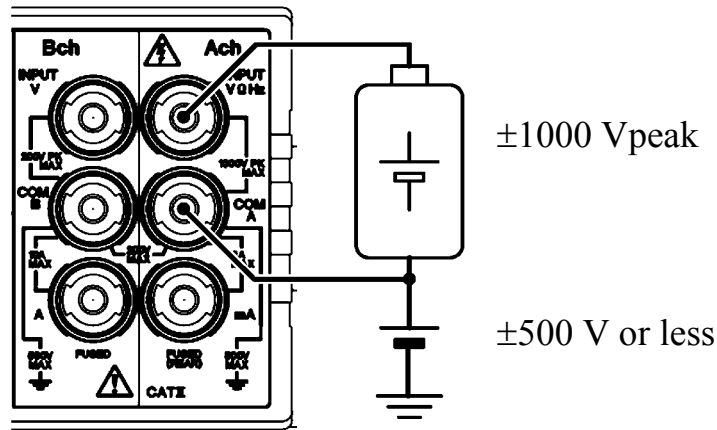


Figure 4-5 DC Voltage Measurement (DCV-Ach)

Table 4-2 Measurement Range and Input Impedance (DCV-Ach)

Measurement range	Input impedance
200 mV, 2000 mV	1 GΩ or more
20 V, 200 V, 1000 V	10 MΩ ± 1%

4.2.1.2 Ach AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)

For an AC voltage superimposed on a DC voltage, select the ACV-Ach function that measures an effective value of the AC voltage only, or the ACV(AC+DC)-Ach function that measures a root-mean-square value of the AC and DC voltages.

1. Connect a DUT to the Ach COM and V terminals on the front panel.
2. <ACV-Ach measurement>

Press **ACV/AC+DC**.

The AC indicator is activated to show the AC coupling.

An effective value of the AC voltage with AC coupling is measured.

<ACV(AC+DC)-Ach measurement>

Re-press **ACV/AC+DC**.

ACV and DCV are measured and the root-mean-square value ($\sqrt{\text{ACV}^2 + \text{DCV}^2}$) is displayed.

The AC and DC indicators are activated to show the AC+DC coupling.

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electrical shock due to the failure of this instrument may occur.

4.2.1 Measurement Functions

Table 4-3 Maximum Allowable Applied Voltage (ACV-Ach, ACV(AC+DC)-Ach)

Input terminal		Maximum allowable applied voltage
Ach	Between VΩHz and COM A terminals	1000 V _{peak} , 700 V _{rms} , 2.2×10^7 VHz
	Between COM A terminal and the chassis	500 V
	Between COM A and COM B terminals	200 V

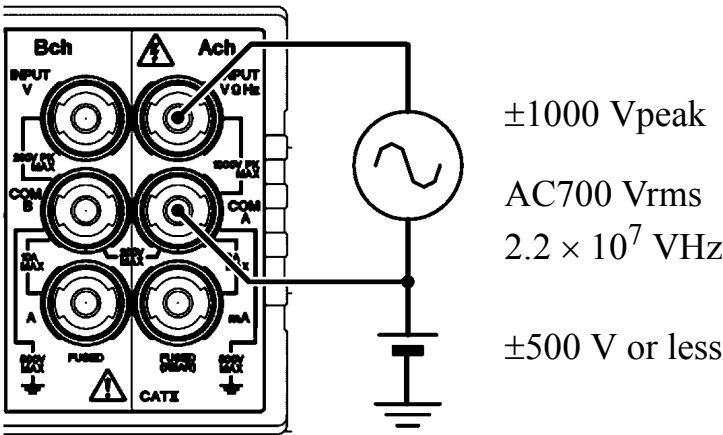


Figure 4-6 AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)

Measurement method	TrueRMS
Input range	5% or more of a full scale
Crest factor	3 (This value is limited by the maximum allowable applied voltage.)
Filter response time	Approx. 1 sec (Time until the measurement value reaches to within 0.1% of the final value)

4.2.1.3 Ach Resistance Measurement (2WΩ-Ach and LP-2WΩ-Ach)

1. Connect a DUT to the Ω and COM terminals on the front panel.
2. Press **Ω/LP-Ω**.
Resistance is measured by using the two-wire method.
3. Re-press **Ω/LP-Ω**.
The LP indicator is activated to show the low power resistance measurement.
The resistance is measured in low power where the measurement current (IS) is reduced.

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electrical shock due to the failure of this instrument may occur.

Table 4-4 Maximum Allowable Applied Voltage (2WΩ-Ach, LP-2WΩ-Ach)

Input terminal		Maximum allowable applied voltage
Ach	Between VΩHz and COM A terminals	1000 V _{peak}
	Between COM A terminal and the chassis	500 V
	Between COM A and COM B terminals	200 V

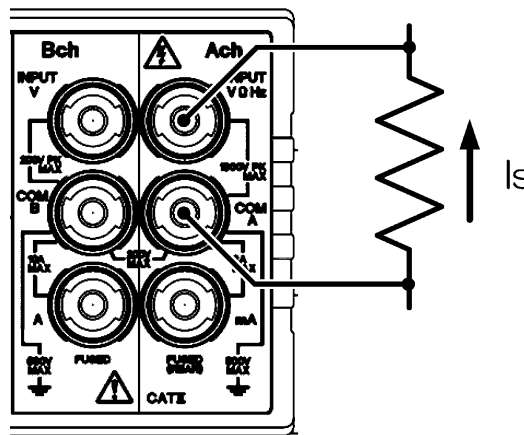


Figure 4-7 Resistance Measurement (2WΩ-Ach and LP-2WΩ-Ach)

The measurement result includes the input cable resistance and contact resistance. To reject these resistance effects, short the tip of the input cable and set the NULL calculation to ON before the measurement.

Ensure the following when in the high-resistance measurement to reduce the effect of the induction noise.

- Shielding the DUT.
- Using the A01001 input cable (sold separately) whose HI side is shielded.
- Securing the input cable to prevent movement.

4.2.1 Measurement Functions

4.2.1.4 Ach DC Current Measurement (DCI-Ach)

1. Connect a DUT to the Ach mA and COM A terminals on the front panel.
2. Press **DCI**.

Table 4-5 Maximum Allowable Applied Current and Protection Function (DCI-Ach)

Input terminal		Maximum allowable applied current	Protection function
Ach	Between mA and COM A terminals	2 A	F2A/250V fuse (Rear panel)

WARNING: Do not measure a current source whose potential is more than 250 V.
This instrument may be damaged and accidents causing injury may occur.

The mA terminal is protected from any applied over-current by a fast-blow fuse.
For more information on how to replace the current protection fuse, refer to Chapter 8, “MAINTENANCE.”

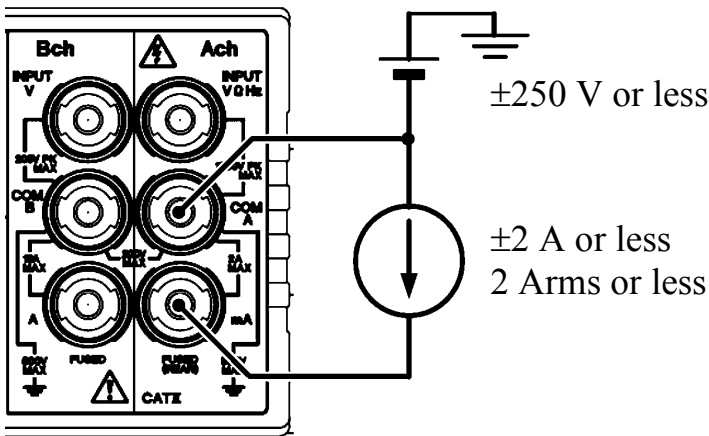


Figure 4-8 DC Current Measurement (DCI-Ach)

NOTE: Applying current causes the temperature to rise in the input terminals and internal circuits, so thermo-electromotive force is generated by the temperature rise.
While in other function measurements, especially high-sensitivity measurements, wait until the temperature stabilizes after applying the current.

4.2.1.5 Ach AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)

For an AC current superimposed on a DC current, select the ACI-Ach function that measures an effective value of the AC current only, or the ACI(AC+DC)-Ach function that measures a root-mean-square value of the AC and DC currents.

1. Connect a DUT to the Ach mA and COM terminals on the front panel.

2. <ACI-Ach measurement>

Press **[ACI]**.

The AC indicator is activated to show the AC coupling.

An effective value of the AC current with AC coupling is measured.

<ACI(AC+DC)-Ach measurement>

Re-press **[ACI]**.

ACI and DCI are measured and the root-mean-square value ($\sqrt{ACI^2 + DCI^2}$) is displayed.

The AC and DC indicators are activated to show the AC+DC coupling.

Table 4-6 Maximum Allowable Applied Current and Protection Function (ACI-Ach and ACI(AC+DC)-Ach)

Input terminal		Maximum allowable applied current	Protection function
Ach	Between mA and COM A terminals	2 A	F2A/250V fuse (Rear panel)

WARNING: Do not apply current that exceeds the maximum allowable applied current.
Do not measure a current source whose potential is more than 250 V.
Fire or electrical shock due to the failure of this instrument may occur.

The mA terminal is protected from any applied over-current by a fast-blow fuse.

For more information on how to replace the current protection fuse, refer to Chapter 8, "MAINTENANCE."

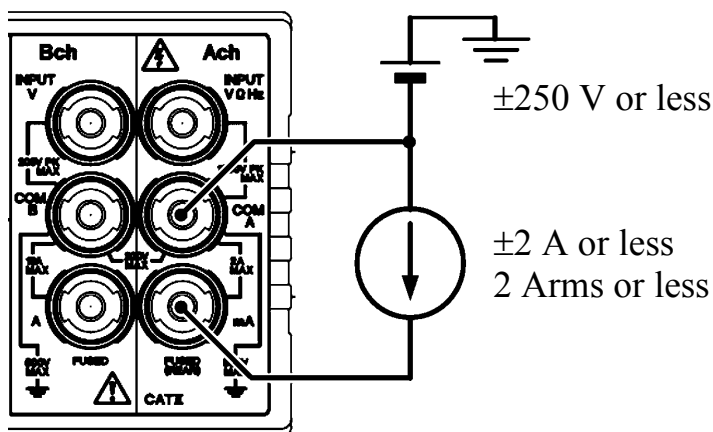


Figure 4-9 AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)

4.2.1 Measurement Functions

Measurement method	TrueRMS
Input range	5% or more of a full scale
Crest factor	3 (However, this value is limited by the current protection fuse rating.)
Filter response time	Approx. 1 sec (Time until the measurement value reaches to within 0.1% of the final value)

4.2.1.6 Ach Continuity Measurement

1. Connect a DUT to the Ach Ω and COM terminals on the front panel.
2. Press **SHIFT** and **ACV/AC+DC**.

The continuity indicator () is activated.

The resistance is measured using the two wire method. If the measured value is equal to the continuity threshold constant or less, the buzzer sounds.

Measurement range	2000 Ω
Measurement current	1 mA
Continuity threshold constant	1 Ω to 1000 Ω

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electrical shock due to the failure of this instrument may occur.

Table 4-7 Maximum Allowable Applied Voltage (Continuity-Ach)

Input terminal		Maximum allowable applied voltage
Ach	Between V Ω Hz and COM A terminals	1000 V _{peak}
	Between COM A terminal and the chassis	500 V
	Between COM A and COM B terminals	200 V

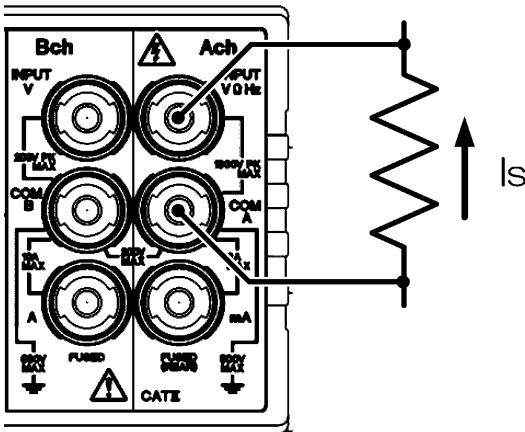


Figure 4-10 Continuity Measurement-Ach

4.2.1.7 Ach Diode Measurement

1. Connect a DUT to the Ach V and COM terminals on the front panel.
2. Press **SHIFT** and **Ω /LP- Ω** .

The diode indicator ($\rightarrow|$) is activated.

A current of approximately 1 mA is applied from the COM terminal to the V terminal, and the voltage between both terminals is measured.

Measurement range 2000 mV

Measurement current 1 mA

WARNING: *Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electrical shock due to the failure of this instrument may occur.*

Table 4-8 Maximum Allowable Applied Voltage (Diode-Ach)

Input terminal		Maximum allowable applied voltage
Ach	Between V Ω Hz and COM A terminals	1000 V _{peak}
	Between COM A terminal and the chassis	500 V
	Between COM A and COM B terminals	200 V

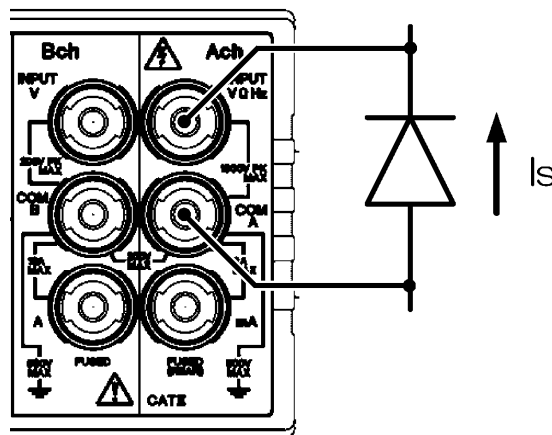


Figure 4-11 Diode Measurement-Ach

4.2.1 Measurement Functions

4.2.1.8 Ach Frequency Measurement (FREQ-Ach)

1. Connect a DUT to the Ach V and COM terminals on the front panel.
2. Press **SHIFT** and **TEMP**.

The frequency in the ACV-Ach measurement function is displayed.

The DC component of the input signal is cut by the AC coupling and the frequency is measured at the zero-cross point.

Measurement method Reciprocal

Input voltage range 100 mVrms to 700 Vrms and 10% or more of each range (Sine-wave)
The range does not describe the frequency range, but rather the input voltage range.

Measure the input voltage by using the ACV function and then measure the frequency by using an appropriate voltage range depending on the input voltage. Even if the input voltage exceeds the measurement range, the measurement is performed but the accuracy of the measured value may be out of the specifications.

Gate time

Sampling rate	Gate time	Measurement frequency range	Maximum measurement period	Maximum display
SLOW	1000 ms	1 Hz to 300 kHz	2.2 s	999999
MED	100 ms	10 Hz to 300 kHz	220 ms	99999
FAST	10 ms	100 Hz to 300 kHz	22 ms	9999

If the input signal cannot be detected due to the period of the input signal is longer than the gate time, 0 Hz is displayed.

NOTE: When the sampling rate is set to SLOW, the sampling rate indicator alternates between the dark "S" and bright "S".

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electric shocks due to the failure of this instrument may occur.

Table 4-9 Maximum Allowable Applied Voltage (FREQ-Ach)

Input terminal		Maximum allowable applied voltage
Ach	Between VΩHz and COM A terminals	1000 V _{peak} , 700 V _{rms} , 2.2×10^7 VHz
	Between COM A terminal and the chassis	500 V
	Between COM A and COM B terminals	200 V

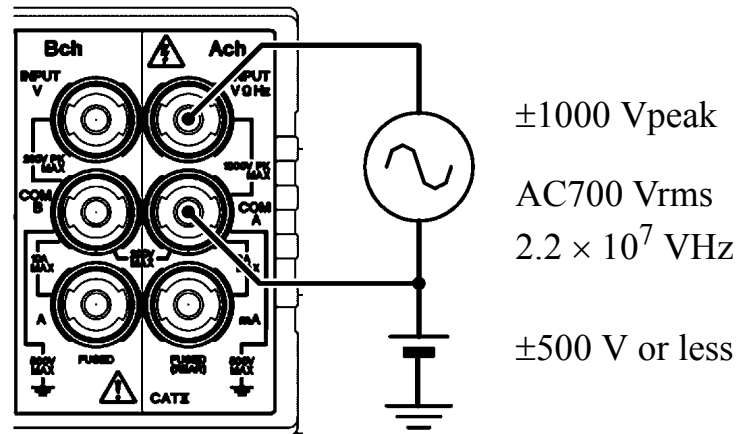


Figure 4-12 Frequency Measurement (FREQ-Ach)

4.2.1 Measurement Functions

4.2.1.9 Bch DC Voltage Measurement (DCV-Bch)

1. Connect a DUT to the Bch COM and V terminals on the front panel.
2. Press **SHIFT** and **DCV**.
The Bch and DC indicators are activated.

WARNING: Do not apply voltage that exceeds the maximum allowable applied voltage. Fire or electrical shock due to the failure of this instrument may occur.

Table 4-10 Maximum Allowable Applied Voltage (DCV-Bch)

Input terminal		Maximum allowable applied voltage
Bch	Between V and COM B terminals	200 V _{peak}
	Between COM B terminal and the chassis	500 V
	Between COM A and COM B terminals	200 V

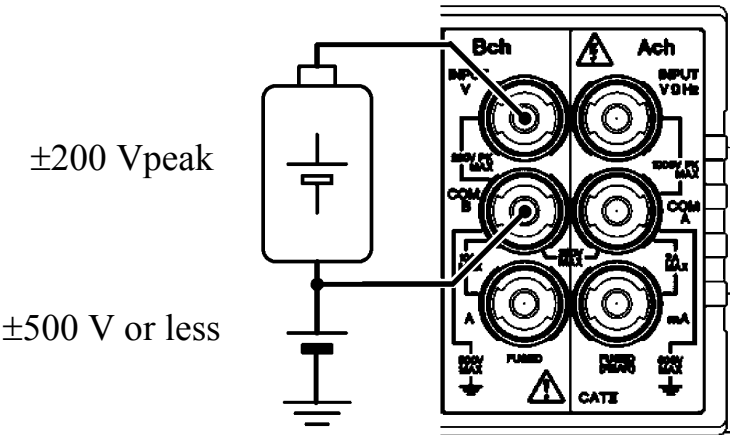


Figure 4-13 DC Voltage Measurement (DCV-Bch)

Table 4-11 Measurement Range and Input Impedance (DCV-Bch)

Measurement range	Input impedance
200 mV, 2000 mV	1 GΩ or more
20 V, 200 V	10 MΩ ± 1%

4.2.1.10 Bch DC Current Measurement (DCI-Bch)

1. Connect a DUT to the Bch A and COM terminals on the front panel.
2. Press **SHIFT** and **DCI**.
The Bch and DC indicators are activated.

Table 4-12 Maximum Allowable Applied Current and Protection Function (DCI-Bch)

Input terminal		Maximum allowable applied current	Protection function
Bch	Between A and COM B terminals	10 A	F15A/250V fuse (Inside of this instrument)

WARNING: Do not measure a current source whose potential is more than 250 V.
This instrument may be damaged and accidents causing injury may occur.

The A terminal is protected from any applied over-current by a fast-blow fuse.

For more information on how to replace the current protection fuse, refer to Chapter 8, “MAINTENANCE.”

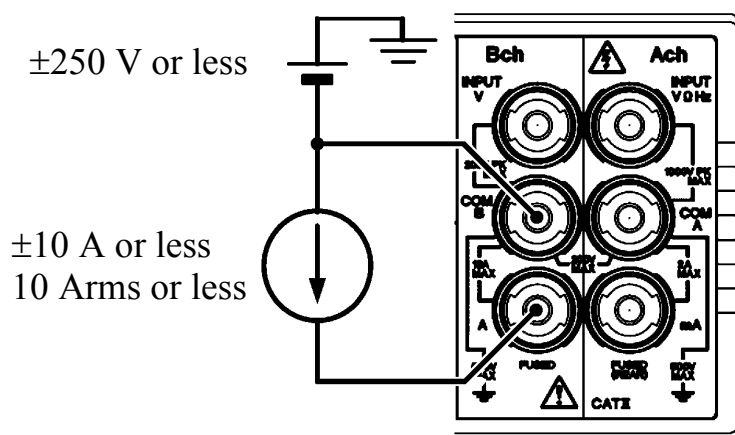


Figure 4-14 DC Current Measurement (DCI-Bch)

Restrictions on the auto-range

The 10 A range is a fixed range.

NOTE: Applying current causes the temperature to rise in the input terminals and internal circuits, so thermo-electromotive force is generated by the temperature rise.
While in other function measurements, especially high-sensitivity measurements, wait until the temperature stabilizes after applying the current.

4.2.1 Measurement Functions

4.2.1.11 Bch AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)

For an AC current superimposed on a DC current, select the ACI-Bch function that measures an effective value of the AC current only, or the ACI(AC+DC)-Bch function that measures a root-mean-square value of the AC and DC currents.

1. Connect a DUT to the Bch A and COM terminals on the front panel.
2. <ACI-Bch measurement>

Press **SHIFT** and **ACI**.

The Bch and AC indicators are activated to show the AC coupling.

An effective value of the AC current with AC coupling is measured.

<ACI(AC+DC)-Bch measurement>

Re-press **SHIFT** and **ACI**.

ACI and DCI are measured and the root-mean-square value ($\sqrt{ACI^2+DCI^2}$) is displayed.

The Bch, AC, and DC indicators are activated to show the AC+DC coupling.

Table 4-13 Maximum Allowable Applied Current and Protection Function (ACI-Bch and ACI(AC+DC)-Bch)

Input terminal		Maximum allowable applied current	Protection function
Bch	Between A and COM B terminals	10 A	F15A/250V fuse (Inside of this instrument)

WARNING: Do not apply current that exceeds the maximum allowable applied current.
Do not measure a current source whose potential is more than 250 V.
Fire or electrical shock due to the failure of this instrument may occur.

The A terminal is protected from any applied over-current by a fast-blow fuse.

For more information on how to replace the current protection fuse, refer to Chapter 8, “MAINTENANCE.”

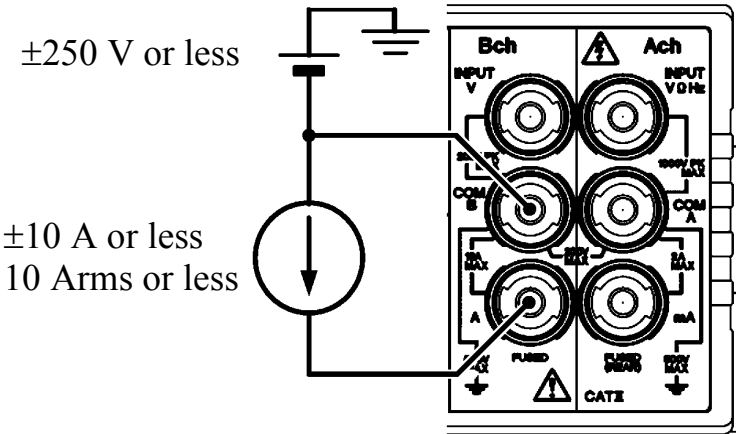


Figure 4-15 AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)

Measurement method	TrueRMS
Input range	5% or more of a full scale
Crest factor	3 (However, this value is limited by the current protection fuse rating.)
Filter response time	Approx. 1 sec (Time until the measurement value reaches to within 0.1% of the final value)
Restrictions on the auto-range	
The 10 A range is a fixed range.	

4.2.1.12 Temperature Measurement

Connect a thermocouple to the TEMP HI and LO terminals on the rear panel.

Press **TEMP**.

The '°C' indicator is activated.

Temperatures at the terminals are internally compensated.

For the thermocouple, select K or T type from 'TEMP' in the **MEAS** category of **MENU**.

Table 4-14 Maximum Allowable Applied Voltage (TEMP)

Input terminal		Maximum allowable applied voltage
TEMP	Between TEMP HI and TEMP LO terminals	36 V _{peak}
	Between TEMP HI/LO terminals and the chassis	500 V
	Between TEMP HI/LO terminals and COM A terminal	200 V
	Between TEMP HI/LO terminals and COM B terminal	200 V

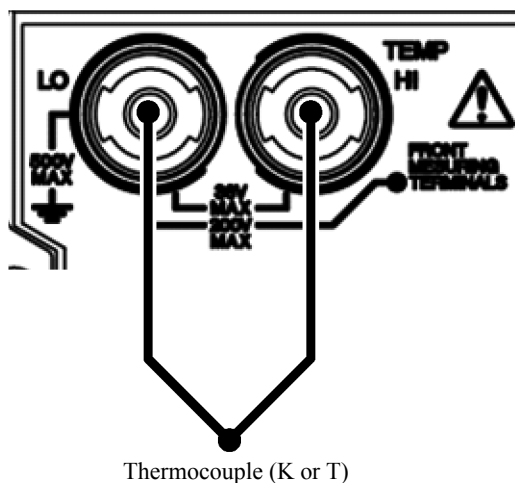


Figure 4-16 Temperature Measurement

4.2.2 Setting the Measurement Range

4.2.2 Setting the Measurement Range

- UP** Increases the measurement range by one level and fixes the range.
- DOWN** Decreases the measurement range by one level and fixes the range.
- AUTO** Switches the measurement range between auto-range and fixed range.

4.2.3 Changing the Measurement Speed and Display Digit

Press **RATE**.

The measurement speed and display digit are switched as shown in Table 4-15 to Table 4-19.

For frequency measurements, the sampling rate is used to set the gate time. For more information on frequency measurements (FREQ), refer to Section 4.2.1, "Measurement Functions." In addition, refer to Section 5.7, "Sampling Operation."

Table 4-15 Measurement Speed during the HOLD OFF (Trigger source: IMMED)

• Single measurement

Measurement only for the left display

Sampling rate	Measurement speed (Measurement time)				Integration time
	DCV-Ach DCV-Bch DCI-Ach DCI-Bch 2WΩ-Ach LP-2WΩ-Ach	ACV-Ach ACI-Ach ACI-Bch Continuity -Ach Diode-Ach	ACV(AC+DC) -Ach ACI(AC+DC) -Ach ACI(AC+DC) -Bch TEMP	FREQ -Ach	
FAST	140 times/s (7.1 ms)		30 times/s (33 ms)	45 times/s (22 ms)	2 ms
MED (50 Hz)	40 times/s (25 ms)		19 times/s (52 ms)	4.5 times/s (220 ms)	1 PLC
MED (60 Hz)	46 times/s (21.7 ms)		22 times/s (45.4 ms)	4.5 times/s (220 ms)	1 PLC
SLOW1	9.5 times/s (105 ms)		4.7 times/s (212 ms)	0.4 times/s (2200 ms)	100 ms
SLOW2	4.9 times/s (205 ms)		2.4 times/s (412 ms)	-	200 ms

• Dual synchronous measurement

Measurement when the Ach function is set for the left display and the Bch function is set for the right display

Sampling rate	Measurement speed (Measurement time) *1			Integration time
	Ach		Bch	
	DCV-Ach DCI-Ach 2WΩ-Ach LP-2WΩ-Ach	ACV-Ach ACI-Ach Continuity -Ach Diode-Ach	DCV-Bch DCI-Bch ACI-Bch	
FAST	123 times/s (8.1 ms)			2 ms
MED (50 Hz)	38 times/s (26.3 ms)			1 PLC
MED (60 Hz)	43.5 times/s (23.0 ms)			1 PLC
SLOW1	9.5 times/s (105 ms)			100 ms
SLOW2	4.9 times/s (205 ms)			200 ms

Conditions: Auto-range: OFF, Auto-zero: OFF, Calculation: OFF, and Display: OFF

*1: For dual synchronous measurements, if ACV(AC+DC)-Ach, ACI(AC+DC)-Ach, ACI(AC+DC)-Bch, TEMP, or FREQ-Ach is set for the Ach or Bch function, the measurement time is the same as the single measurement time for the set function.

4.2.3 Changing the Measurement Speed and Display Digit

Table 4-16 Measurement Speed during the HOLD ON (Trigger source: EXT) (7352A)

- Single measurement

Measurement only for the left display

Sampling rate	Measurement speed (Measurement time)				Integration time
	DCV-Ach DCV-Bch DCI-Ach DCI-Bch 2WΩ-Ach LP-2WΩ-Ach	ACV-Ach ACI-Ach ACI-Bch Continuity -Ach Diode-Ach	ACV(AC+DC) -Ach ACI(AC+DC) -Ach ACI(AC+DC) -Bch TEMP	FREQ -Ach	
FAST	80 times/s (12.5 ms)		45 times/s (22 ms)	38 times/s (26 ms)	2 ms
MED (50 Hz)	32 times/s (31 ms)		17 times/s (58 ms)	8.6 times/s (116 ms)	1 PLC
MED (60 Hz)	37 times/s (27 ms)		19 times/s (52 ms)	8.6 times/s (116 ms)	1 PLC
SLOW1	9 times/s (110 ms)		4.5 times/s (218 ms)	0.9 times/s (1016 ms)	100 ms
SLOW2	4.7 times/s (210 ms)		2.3 times/s (418 ms)	-	200 ms

Conditions: Auto-range: OFF, Auto-zero: OFF, Calculation: OFF, Display: OFF, and Trigger delay: 0 sec

- Dual synchronous measurement

Measurement when the Ach function is set for the left display and the Bch function is set for the right display

Sampling rate	Measurement speed (Measurement time) *1			Integration time
	Ach		Bch	
	DCV-Ach DCI-Ach 2WΩ-Ach LP-2WΩ-Ach	ACV-Ach ACI-Ach Continuity -Ach Diode-Ach	DCV-Bch DCI-Bch ACI-Bch	
FAST	64.5 times/s (15.5 ms)			2 ms
MED (50 Hz)	29.4 times/s (34 ms)			1 PLC
MED (60 Hz)	33.3 times/s (30 ms)			1 PLC
SLOW1	8.8 times/s (113 ms)			100 ms
SLOW2	4.7 times/s (210 ms)			200 ms

Conditions: Auto-range: OFF, Auto-zero: OFF, Calculation: OFF, and Display: OFF

*1: For the dual synchronous measurement, if ACV(AC+DC)-Ach, ACI(AC+DC)-Ach, ACI(AC+DC)-Bch, TEMP, or FREQ-Ach is set for the Ach or Bch function, the measurement time is the same as the single measurement time for the set function.

4.2.3 Changing the Measurement Speed and Display Digit

Table 4-17 Measurement Speed during the HOLD ON (Trigger source: BUS) GPIB (7352A)/USB

- Single measurement

Measurement only for the left display

Sampling rate	Measurement speed (Measurement time)				Integration time
	DCV-Ach DCV-Bch DCI-Ach DCI-Bch 2W Ω -Ach LP-2W Ω -Ach	ACV-Ach ACI-Ach ACI-Bch Continuity -Ach Diode-Ach	ACV(AC+DC)-Ach ACI(AC+DC)-Ach ACI(AC+DC)-Bch TEMP	FREQ -Ach	
FAST	74 times/s (13.5 ms)		42 times/s (23.5 ms)	38 times/s (26 ms)	2 ms
MED (50 Hz)	31 times/s (32 ms)		16 times/s (60 ms)	8.6 times/s (116 ms)	1 PLC
MED (60 Hz)	35 times/s (28 ms)		18 times/s (53 ms)	8.6 times/s (116 ms)	1 PLC
SLOW1	9 times/s (111 ms)		4.5 times/s (219 ms)	0.9 times/s (1016 ms)	100 ms
SLOW2	4.7 times/s (211 ms)		2.3 times/s (419 ms)	-	200 ms

Conditions: Auto-range: OFF, Auto-zero: OFF, Calculation: OFF, Display: OFF, and Trigger delay: 0 sec

- Dual synchronous measurement

Measurement when the Ach function is set for the left display and the Bch function is set for the right display

Sampling rate	Measurement speed (Measurement time) *1			Integration time
	Ach		Bch	
	DCV-Ach DCI-Ach 2W Ω -Ach LP-2W Ω -Ach	ACV-Ach ACI-Ach Continuity -Ach Diode-Ach	DCV-Bch DCI-Bch ACI-Bch	
FAST	60 times/s (16.5 ms)			2 ms
MED (50 Hz)	28 times/s (35 ms)			1 PLC
MED (60 Hz)	32 times/s (31 ms)			1 PLC
SLOW1	8 times/s (114 ms)			100 ms
SLOW2	4.7 times/s (214 ms)			200 ms

Conditions: Auto-range: OFF, Auto-zero: OFF, Calculation: OFF, and Display: OFF

*1: For the dual synchronous measurement, if ACV(AC+DC)-Ach, ACI(AC+DC)-Ach, ACI(AC+DC)-Bch, TEMP, or FREQ-Ach is set for the Ach or Bch function, the measurement time is the same as the single measurement time for the set function.

4.2.3 Changing the Measurement Speed and Display Digit

Table 4-18 Measurement Speed during the HOLD ON (Trigger source: BUS) RS-232 (7352A)

- Single measurement

Measurement only for the left display

Sampling rate	Measurement speed (Measurement time)				Integration time
	DCV-Ach DCV-Bch DCI-Ach DCI-Bch 2WΩ-Ach LP-2WΩ-Ach	ACV-Ach ACI-Ach ACI-Bch Continuity -Ach Diode-Ach	ACV(AC+DC) -Ach ACI(AC+DC) -Ach ACI(AC+DC) -Bch TEMP	FREQ -Ach	
FAST	19 times/s (51 ms)		18 times/s (53 ms)	38 times/s (26 ms)	2 ms
MED (50 Hz)	14 times/s (69 ms)		11 times/s (90 ms)	8.6 times/s (116 ms)	1 PLC
MED (60 Hz)	15 times/s (65 ms)		11 times/s (85 ms)	8.6 times/s (116 ms)	1 PLC
SLOW1	6 times/s (150 ms)		3 times/s (251 ms)	0.9 times/s (1016 ms)	100 ms
SLOW2	4 times/s (248 ms)		2 times/s (449 ms)	-	200 ms

Conditions: Auto-range: OFF, Auto-zero: OFF, Calculation: OFF, Display: OFF, and Trigger delay: 0 sec

- Dual synchronous measurement

Measurement when the Ach function is set for the left display and the Bch function is set for the right display

Sampling rate	Measurement speed (Measurement time) *1			Integration time
	Ach		Bch	
	DCV-Ach DCI-Ach 2WΩ-Ach LP-2WΩ-Ach	ACV-Ach ACI-Ach Continuity -Ach Diode-Ach	DCV-Bch DCI-Bch ACI-Bch	
FAST	18.5 times/s (54 ms)			2 ms
MED (50 Hz)	13.9 times/s (68 ms)			1 PLC
MED (60 Hz)	14.7 times/s (68 ms)			1 PLC
SLOW1	6 times/s (153 ms)			100 ms
SLOW2	4 times/s (251 ms)			200 ms

Conditions: Auto-range: OFF, Auto-zero: OFF, Calculation: OFF, and Display: OFF

*1: For the dual synchronous measurement, if ACV(AC+DC)-Ach, ACI(AC+DC)-Ach, ACI(AC+DC)-Bch, TEMP, or FREQ-Ach is set for the Ach or Bch function, the measurement time is the same as the single measurement time for the set function.

- Dual sequential measurement

Measurement when the same channel function of Ach or Bch is set for the right and left displays

The measurement time is expressed as the following formula:

Measurement time for the left display + Time for function change (left to right) + Measurement time for the right display + Time for function change (right to left) + Data processing time

4.2.4 How to Use Dual Display

Table 4-19 Sampling Rate and Display Digits

Sampling rate	DCV-Ach DCI-Ach 2W Ω -Ach LP-2W Ω -Ach Continuity-Ach Diode-Ach	ACV-Ach ACI-Ach	ACV (AC+DC)-Ach ACI (AC+DC)-Ach	
	DCV-Bch DCI-Bch	ACI-Bch	ACI(AC+DC)-Bch	TEMP
FAST	4 1/2-digit display	4 1/2-digit display	3 1/2-digit display	4 1/2-digit display
MED	5 1/2-digit display	4 1/2-digit display	3 1/2-digit display	4 1/2-digit display
SLOW1	5 1/2-digit display	5 1/2-digit display	4 1/2-digit display	4 1/2-digit display
SLOW2	5 1/2-digit display	5 1/2-digit display	4 1/2-digit display	4 1/2-digit display

In addition, refer to Section 5.7, “Sampling Operation.”

HOLD setting

HOLD Sets the sampling mode to HOLD or releases it from HOLD.

TRIG Starts the measurement when the sampling mode is set to HOLD.

4.2.4 How to Use Dual Display

The 7352A/E is capable to display two measurement or calculation results at the same time.

Two measurement points can be simultaneously measured and displayed by combining the Ach, Bch, and temperature measurement functions.

How to use the second display (right display)

- Pressing **2nd** on the front panel activates the ‘2nd’ indicator.
While the ‘2nd’ indicator is on, setting operations are for the second display (right display).
- Select a function or calculation for the second display (right display).

How to turn the second display (right display) off

- Pressing **SHIFT** on the front panel activates the ‘SHIFT’ indicator.
- Pressing **2nd** on the front panel turns the second display (right display) off.

Example: In cases where the Bch DC voltage (DCV-Bch) is displayed on the left display and the Ach AC voltage (AC coupling) (ACV-Ach) is displayed on the right display

Key operation:

1. Press **SHIFT** to enter the SHIFT mode.
2. Press **DCV** to set the DCV-Bch function for the first display (left display).
3. Press **2nd** to activate the second display (right display). The '2nd' indicator is activated.
4. Press **ACV** to set the ACV-Ach function for the second display (right display).

4.2.4.1 How to Use Dual Display by Remote Operation

Example: In cases where the Bch DC voltage (DCV-Bch) is displayed on the left display and the Ach AC voltage (AC+DC coupling) (ACV(AC+DC)-Ach) is displayed on the right display

In cases where commands used in ADC CORPORATION are used

1. Set the DCV-Bch function for the first display (left display).
 "DSP1, F12"
 DSP1: The first display selection
 F12: Bch DC voltage measurement (DCV-Bch)
2. Set the ACV(AC+DC)-Ach function for the second display (right display).
 "DSP2, F7"
 DSP2: The second display selection
 F7: AC voltage (AC+DC coupling) measurement (ACV(AC+DC)-Ach)

In cases where SCPI commands are used

1. Set the DCV-Bch function for the first display (left display).
 "SENSe:FUNCTION "VOLTage:BDC",(@1)"
2. Set the ACV(AC+DC)-Ach function for the second display (right display).
 "SENSe:FUNCTION "VOLTage:ACDC",(@2)"

5. MENU OPERATION AND FUNCTION DESCRIPTION

5.1 Menu

The functions and conditions in this instrument are set by using layered menus.

5.1.1 How to Navigate the Menus

Menus have a three-layer structure. (Some menus have a four-layer structure.)

First layer	Category layer	Selects the menu category.
Second layer	Selection layer	Selects the setting parameters in the category.
Third layer	Entry/Execution layer	Used to enter the values or select the setting.

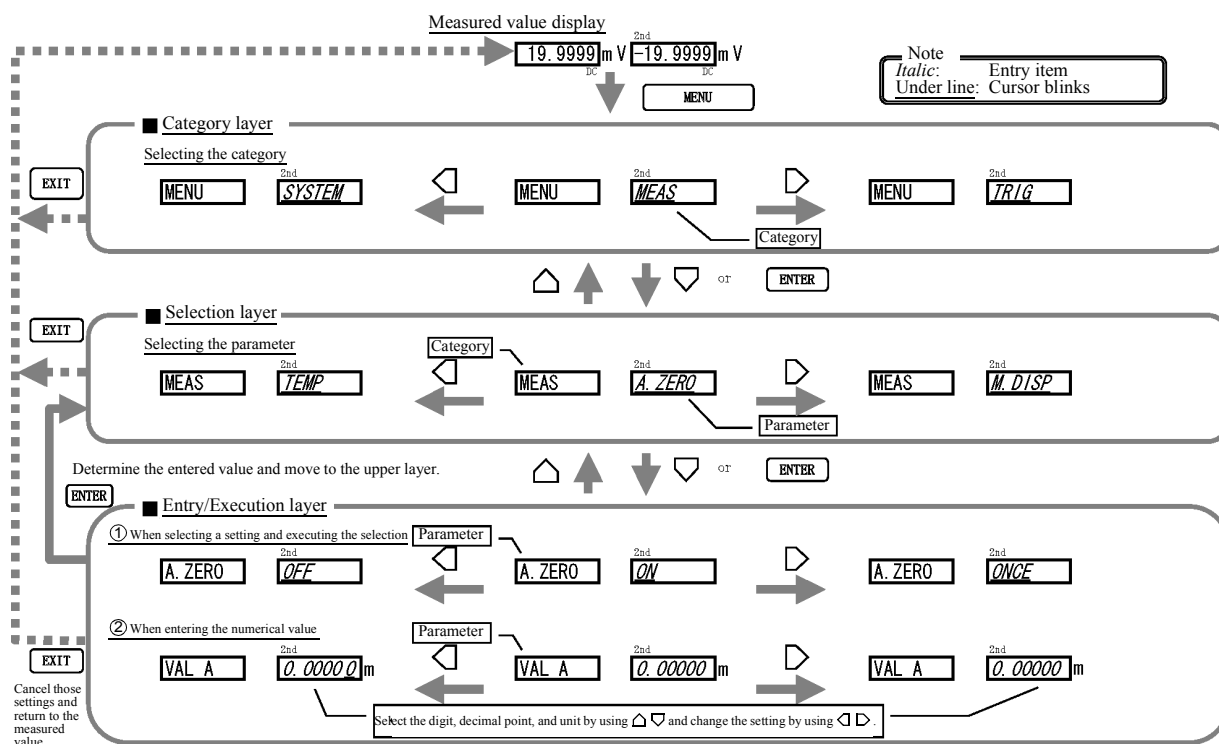


Figure 5-1 Menu Operation

- Press **MENU**.
The Category layer in which **MEAS** to **SYSTEM** be selected is displayed.
- Select the category by using $\triangleleft$ and $\triangleright$.

5.1.1 How to Navigate the Menus

3. Press .
The Selection layer is displayed.
4. Select the parameters by using  and .
5. Press .
The Entry/Execution layer is displayed.
6. Set the parameters by using , , , , and the **TRIG** key.
7. Pressing **ENTER** determines the entered parameter settings and moves to the Selection layer.
If pressing **EXIT/LOCAL** before ENTER, the entered settings are canceled and MENU is quitted.
“DONE” after “BUSY” is displayed during the process.

Setting the selectable parameters

For the selectable parameter that is set by selecting the item, select the parameter by using  and , and determine it by pressing **ENTER**.

Setting the numerical parameters

For a numerical parameter set by entering a numerical value, select either numerical digit, decimal point position, or unit prefix by using  and .

The operation of  and  varies depending on the target items.

If there is not a blinking target item, the layer is moved to the Selection layer by using .

If a numerical digit blinks, the numerical value can be increased or decreased.

If a decimal point blinks, the decimal point can be moved.

If the unit display section blinks, the unit prefix can be changed in the following order: n, μ , m, none, k, and M.

Pressing **TRIG** performs the measurement and can set the result as the value of the parameter.

Although any numerical value can be entered into the numerical parameter, it is rounded off to the resolution digit of each parameter.

Table 5-1 shows the key function in each layer.

Table 5-1 Menu Operation Key and Function

	Category layer	Selection layer	Entry/Execution layer		
			Numerical value parameter		Selectable parameter/ execution item
			Parameter: Dark display, No blinking	Target items: Numerical value/ decimal point/unit	
	-	Moves to the Category layer.	Moves to the Selection layer.	Increases the value.	Moves to the Selection layer.
	Moves to the Selection layer.	Moves to the Entry/Execution layer.	-	Decreases the value.	-
	Moves the target item to the left category.	Moves the target item to the left parameter.	Moves the target item to the numerical value.	Moves the target item to the left.	Displays the previous selection.
	Moves the target item to the right category.	Moves the target item to the right parameter.	Moves the target item to the unit.	Moves the target item to the right.	Displays the next selection.
TRIG	-	-	Performs the measurement and reflects the result to the parameter. *1		-
ENTER	Moves to the Selection layer.	Moves to the Entry/Execution layer.	Determines the entered value and moves to the Selection layer.		Determines the selection and moves to the Selection layer.
EXIT	Exiting the MENU system	Exiting the MENU system	Exiting the MENU system		

*1: Only the constant that is used to calculate the HIGH, LOW, and NULL values

Parameters that can be set on the right and left displays are shown in Section 5.1.2, “Menu List.”

These parameters can be set for the left or right display section that is selected during the menu operation.

2nd is disabled during the menu operation.

5.1.2 Menu List

5.1.2 Menu List

	Category layer	Selection layer	Entry/Execution layer	Each setting for right and left	
(a)	MENU	MEAS	A.ZERO	Auto-zero OFF/ON/ONCE	
			M.DISP	Display mode OFF/ON	
			DIGIT	Number of display digits. 5 1/2 to 3 1/2	●
			Ω . BEEP	The threshold in the continuity measurement function 1 Ω to 1000 Ω	
			TEMP	Temperature sensor selection K TYPE/T TYPE	
	TRIG		T.DELAY	Trigger delay 0 s to 3600 s	● *1
			N.SMPL	Sampling count 1 to 16000	
			SOURCE *2	Trigger source selection IMMED/MANUAL/EXT(7352A)/BUS	
			T.COUNT	Trigger count 1 to 50000	
			T.D.EACH	Each trigger delay for the right and left displays OFF/ON	
	CALC		COMP	Comparative calculation OFF/ON	
			MATH	Other calculations OFF/dB/dBm/SCL	●
			MAX/MIN	MAX/MIN calculation and selecting the display OFF/ON/MAX/MIN	●
			RESULT	Display of the MAX/MIN calculation result SAMPLE/MAX/MIN/AVG	●
			PASS	PASS condition settings (*2) LOGOHI/___HI/___GO___/LO___/LO___HI/LOGO___/ ___GOHI/_____	
			BEEP	Calculation Result and buzzer (*2) OFF/PASS/FAIL	
				Calculation Result and buzzer (*2) OFF/HI___LO/___GO___/HI___/___LO	
			SM.CNT	Number of times smoothing is performed 2 to 100 times	●
			2MEAS *2	Calculation for two measurement mode OFF/(M1+M2)/(M1-M2)/(M1*M2)/(M1/M2)	
	*1 Trigger delay setting for each right and left display is available in T.D.EACH setting to ON.				
	*2 It is available for only 7532A and 7352E.				

	Category layer	Selection layer	Entry/Execution layer	Each setting for right and left
(a)	CONST	HIGH	HIGH constant (COMP) -999999.E+6 to +999999.E+6	
		LOW	LOW constant (COMP) -999999.E+6 to +999999.E+6	
		ΔNULL	NULL constant (For only NULL ON) -999999.E+6 to +999999.E+6	●
		VAL A	Constant A (SCL) -999999.E+6 to +999999.E+6	●
		VAL B	Constant B (SCL) -999999.E+6 to +999999.E+6	●
		VAL C	Constant C (SCL) -999999.E+6 to +999999.E+6	●
		VAL dB	Constant D (dB/dBm) 0.00001E-9 to 999999.E+6	●
	PARAM	P.ON LD	Selection of parameters which are loaded when the power is turned on P.OFF/USER0	
		SAVE	Parameter save-file selection USER0/USER1/USER2/USER3	
		LOAD	Parameter load-file selection USER0/USER1/USER2/USER3/DFLT	
	INIT	PARAM	Initialization DFLT	
	MEMORY	STORE	Measurement memory: OFF/ON	
		RECALL	Measurement memory recall Recall setting for the data number	●
		STAT	Statistical calculations and measurement result readings	●
(b)				

5.1.2 Menu List

	<u>Category layer</u>	<u>Selection layer</u>	<u>Entry/Execution layer</u>	<u>Each setting for right and left</u>
(b)	I/F	BUS	Interface selection (This selection differs depending on the model.) GPIB/USB/RS232	
		GP.ADR	GPIB address (Only when GPIB is selected for the interface) 0 to 30	
		USB ID	USB ID (Only when USB is selected for the interface) 1 to 127	
		BAUD	Baud rate selection (Only when RS-232 is selected for the inter- face) 9600/300/600/1200/2400/4800	
		DATA	Data length selection (Only when RS-232 is selected for the inter- face) 8 BIT/7 BIT	
		PARITY	Parity selection (Only when RS-232 is selected for the interface) NONE/ODD/EVEN	
		STOP	Stop bit selection (Only when RS-232 is selected for the interface) 1 BIT/2 BIT	
		ECHO	Echo selection (Only when RS-232 is selected for the interface) OFF/ON	
		HEADER	Header OFF/ON	
		T.ONLY	Addressable/Talk only (Only when GPIB or RS-232 is selected for the interface) OFF/ON	
		LANG *3	Command language selection (This selection differs depending on the interface selection.) ADC/SCPI/R6452	
		CONT	Initial value selection of the Continue setting OFF/ON (Only when SCPI is selected as the command language.)	
(c)		*3: Selectable command languages are different depending on the interface. For more information, refer to Section 6.6.1, "Selecting the Command Language."		

<u>Category layer</u>		<u>Selection layer</u>	<u>Entry/Execution layer</u>	<u>Each setting for right and left</u>
(c)	SYSTEM	BEEP	Buzzer setting OFF/ON	
		TEST	Self-Test	
		P/KEY	Display and key tests	
		F.INHIB	Disable function Each function can be set to ENA (Enabling) or DIS (Disabling).	
		P.LOCK	Panel lock OFF/ON	
		PWD	Password (For panel lock) 4 digits password	
		CAL	CAL mode OFF/ON	
		QUEUE	Error record display No.01 to No.20	

5.2 Measurement Functions

For more information on the measurement function description, refer to Chapter 4.

This chapter describes the parameters set in MENU that is used in the measurement function.

5.2.1 Continuity Measurement

Setting the continuity threshold constant

The continuity threshold constant can be selected in the range between 1 Ω and 1000 Ω from 'Ω.BEEP' in **MEAS** category in **MENU**.

Press **ENTER** to apply the setting.

5.2.2 Temperature Measurement

Setting the temperature sensor (thermocouple)

Select a thermocouple type from 'TEMP' in the **MEAS** category of **MENU**.

K TYPE

T TYPE

Press **ENTER** to apply the setting.

5.3 Measurement Range and Auto-Range

5.3.1 Measurement Function and Range Structure

Table 5-2 shows the range structure corresponding to the measurement function.

Table 5-2 Measurement Function and Range Structure

Ach								
DCV-Ach	ACV-Ach (AC+DC)-Ach	2WΩ-Ach	LP-2WΩ-Ach	DCI-Ach	ACI-Ach ACI(AC+DC)-Ach	FREQ-Ach	Continuity-Ach	Diode-Ach
				2000 nA				
				20 μA				
200 mV	200 mV	200 Ω	200 Ω	200 μA	200 μA	200 mV		
2000 mV	2000 mV	2000 Ω	2000 Ω	2000 μA	2000 μA	2000 mV	2000 Ω	2000 mV
20 V	20 V	20 kΩ	20 kΩ	20 mA	20 mA	20 V		
200 V	200 V	200 kΩ	200 kΩ	200 mA	200 mA	200 V		
1000 V	700 V	2000 kΩ	2000 kΩ	2000 mA	2000 mA	700 V		
		20 MΩ	20 MΩ					
		200 MΩ						

Bch			
DCV-Bch	DCI-Bch	ACI-Bch ACI(AC+DC)-Bch	TEMP
200 mV			
2000 mV			Fixed
20 V			
200 V			
	10 A	10 A	

NOTE: The range settings for each function are saved.

For the frequency function, the range is not switched in the frequency range but switched in the input voltage range.

5.3.2 Auto-range

5.3.2 Auto-range

When Auto-range is set to ON, the measurement range of this instrument is automatically set to the optimum range depending on the input voltage.

The judgment value used when the range is changed

Range DOWN When the displayed value is less than 10% of the displayed range (less than 200 V if the DCV 1000 V range and 700 V range are both selected)

Range UP When the displayed value exceeds the maximum display

Auto-range is set to OFF under the following conditions.

- Frequency measurement
- Diode measurement
- Continuity measurement
- The 10 A range for the DC and AC current measurements
- Temperature measurement

5.4 Auto-Zero Operation

Auto-zero function eliminates the offset error in the measurement system of this instrument.

If Auto-zero is set to ON, the internal offset is measured and the offset error is eliminated because the offset is subtracted from the input measurement value. However, the measurement time is doubled.

Set Auto-zero in 'A.ZERO' in **MEAS** in **MENU**.

Auto-zero is performed at the current measurement range in the measurement function and range shown in the following table.

Table 5-3 Auto-Zero

Function	Auto-zero
DCV-Ach	Can be performed
ACV-Ach AC+DCV-Ach	-
2W Ω -Ach LP-2W Ω -Ach	Can be performed
DCI-Ach	Can be performed
ACI-Ach ACI(AC+DC)-Ach	-
TEMP	Can be performed
FREQ-Ach	-
Diode-Ach	Can be performed
Continuity-Ach	Can be performed
DCV-Bch	Can be performed
DCI-Bch	Can be performed
ACI-Bch ACI(AC+DC)-Bch	-

Auto zero can be performed only once according to the measurement condition or measurement timing.

If ONCE is set and Auto-zero is performed, the offset error is eliminated and then Auto-zero is turned off.

5.5 Display Mode Setting

5.5 Display Mode Setting

Setting the display mode

Select either ON or OFF in 'M.DISPLAY' in **MEAS** category in **MENU**.

Press **ENTER** to apply the setting.

A time of approximately 1 ms is required for the output to be displayed. (The time may vary depending on conditions.)

Select OFF if sampling at high speed.

An error is displayed even if the display is set to OFF.

5.6 Display Digit Setting

Setting the number of display digits

Set the number of display digits in 'DIGIT' in the **MEAS** category in **MENU**.

Press **ENTER** to apply the setting.

This setting is disabled in remote output but display.

5.7 Sampling Operation

This instrument performs the following three types of sampling operation depending on the setting.

- **Single measurement**
Measurement is performed when only the left measurement display is used.
- **Dual sequential measurement**
Measurement is performed when functions of the same channel, Ach or Bch, are set for the right and left measurement displays.
When one trigger event is received, measurements are sequentially performed from the left side to the right side.
Relays may operate depending on combinations of functions and ranges on the right and left displays. In this case, because relays operate during each measurement trigger, note the life span of relays. (For more information, refer to Section 8.4, “Replacing Parts with Limited Life.”)
Because relays do not operate for the dual synchronous measurement, using combinations of the dual synchronous measurement is recommended.
- **Dual synchronous measurement**
Measurement is performed when functions of the different channel, Ach or Bch, are set for the right and left measurement displays.
When one trigger event is received, measurements of the right and left sides are simultaneously performed.

For each sampling operation, common features are as follows:

- Measurements are performed when the trigger event is received.
- When each measurement is complete, the data output starts.
- The measured data is output to the display or remote interface.
- The measured data can be stored in the data memory.

Table 5-4 shows sampling operations in which function settings are combined.

5.7.1 Single Measurement

Table 5-4 Sampling Operations Combining with Function Settings

			Right side display											
			OFF	Ach							Bch			
				DCV	ACV ACV(AC+DC)	Ω LP- Ω	DCI	ACI ACI(AC+DC)	Continuity	Diode	FREQ	DCV	DCI	ACI ACI(AC+DC)
Left side display	Ach	DCV	Single measure- ment	Sequential measurement (Two functions are alternately measured by switching them.)							Synchronous measurement			
		ACV												
		ACV(AC+DC)												
		W												
		LP- Ω												
		DCI												
		ACI												
		ACI(AC+DC)												
		Continuity												
	Bch	Diode	Single measure- ment	Synchronous measurement (Two functions are simultaneously measured.)							Sequential measurement			
		FREQ												
		DCV												
		DCI												
		ACI												
		ACI(AC+DC)												
		TEMP												

5.7.1 Single Measurement

Measurement is performed by using only the left measurement display.
The sampling operation is the same as a conventional digital multimeter.
See Table 5-4.
Figure 5-2 shows the single measurement sequence.

The following shows the operation procedure to set the single measurement during dual measurement.

Key operation		Display section	
1	Dual display is shown.	Measured value	Measured value
2	Press SHIFT .	Measured value	SHIFT
3	Press 2nd . The measurement for the right side display is stopped and the single measurement is set.	DCV-Ach measured value	

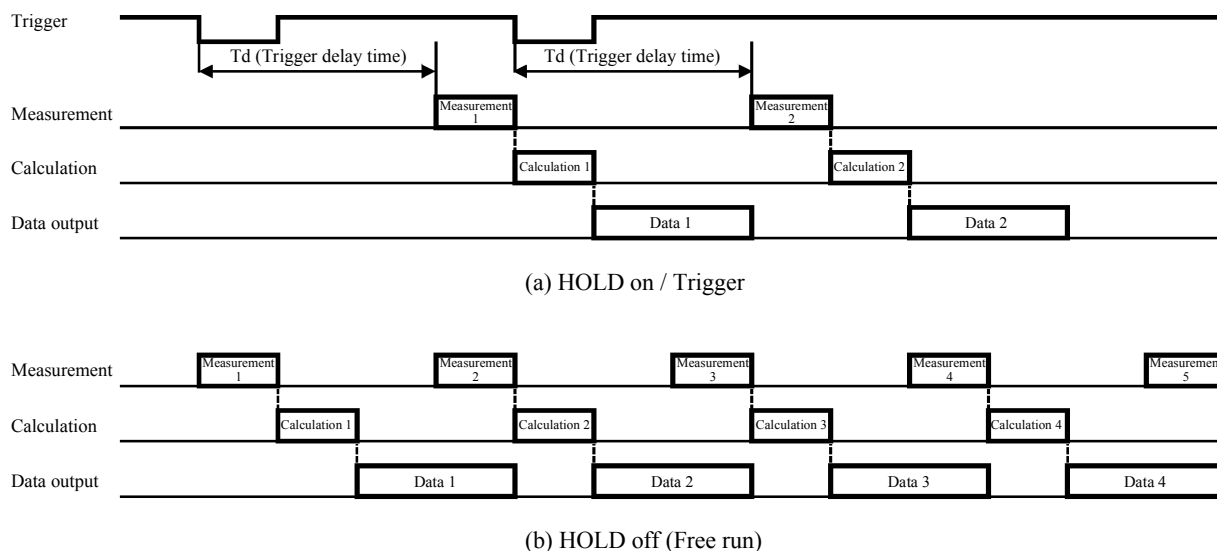


Figure 5-2 Single Measurement Sequence

5.7.2 Dual Sequential Measurement

Measurement is performed when functions of the same channel, Ach or Bch, are set for the right and left measurement displays.

For more information on combinations of related functions, see Table 5-4.

When one trigger event is received, measurements are performed according to the following order:

Trigger → Measurement for the left side display → Changing the function and range → Measurement for the right side display → Data output

Figure 5-3 shows the dual sequential measurement sequence.

The following shows an operation example to set the dual sequential measurement during single measurement.

- | | <u>Key operation</u> |
|---|---|
| 1 | The left display shows the state of the DCV function. |
| 2 | Press 2nd . |
| 3 | Press DCI . The set function is displayed on the right side display section.
In this case, the DCI-Ach function is displayed. |

<u>Display section</u>	
DCV-Ach measured value	
DCV-Ach measured value	2nd
DCV-Ach measured value	DCI-Ach measured value

5.7.3 Dual Synchronous Measurement

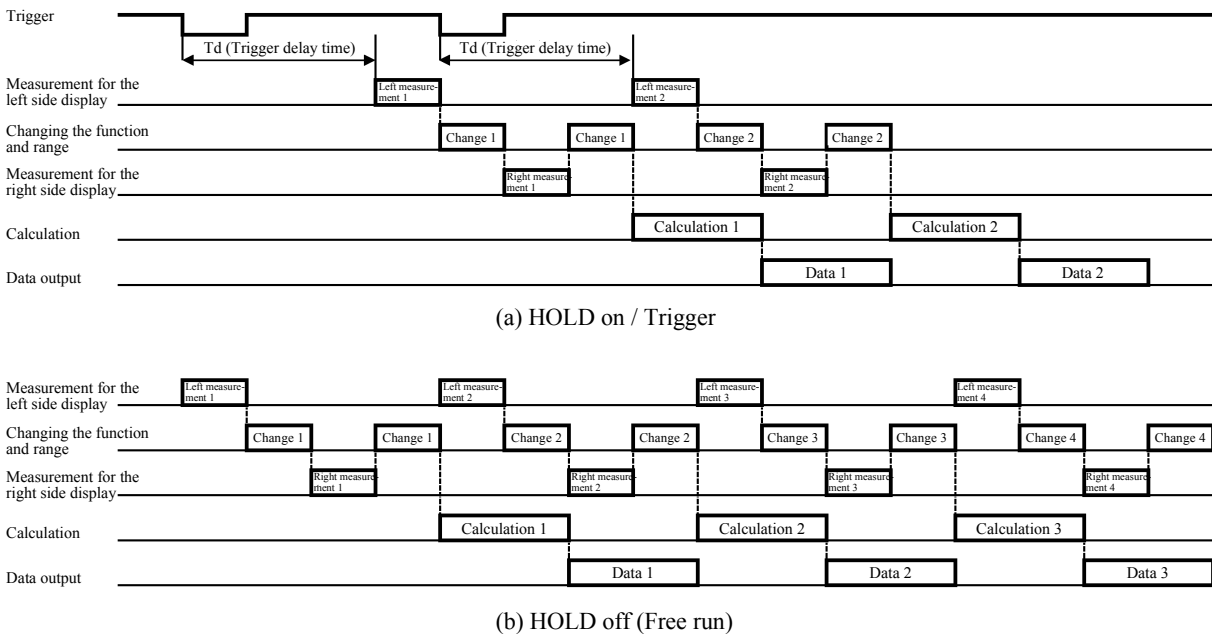


Figure 5-3 Dual Sequential Measurement Sequence

5.7.3 Dual Synchronous Measurement

Measurement is performed when functions of the different channel, Ach or Bch, are set for the right and left measurement displays.

For more information on combinations of related functions, see Table 5-4.

Because Ach and Bch each have a separate AD converter, when one trigger event is received, measurements of the right and left sides are simultaneously performed as described below:

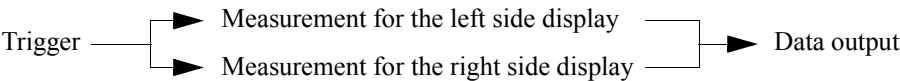
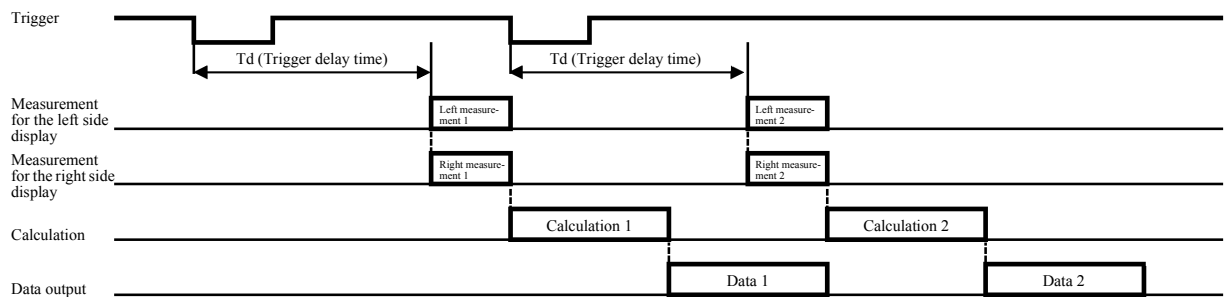


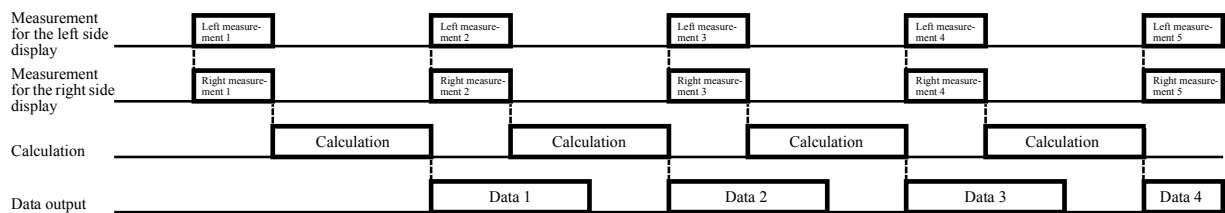
Figure 5-4 shows the dual synchronous measurement sequence.

The following shows an operation example to set the dual synchronous measurement during single measurement.

Operation		Display section	
1	The left display shows the state of the DCV-Ach function.	DCV-Ach measured value	
2	Press 2nd .	DCV-Ach measured value	2nd
3	Press SHIFT .	SHIFT	2nd
4	Press DCI . The set function is displayed on the right side display section. In this case, the DCI-Bch function is displayed.	DCV-Ach measured value	DCI-Bch measured value



(a) HOLD on / Trigger



(b) HOLD off (Free run)

Figure 5-4 Dual Synchronous Measurement Sequence

5.7.4 Sampling Count Setting

Although sampling is usually performed once when a trigger is detected, sampling can be performed more than once.

The sampling count is set from 'N.SMPL' in the **TRIG** category of **MENU**.

For more information on the operation, refer to Section 5.1, "Menu."

Sampling count : 1 to 16000 samples/trigger

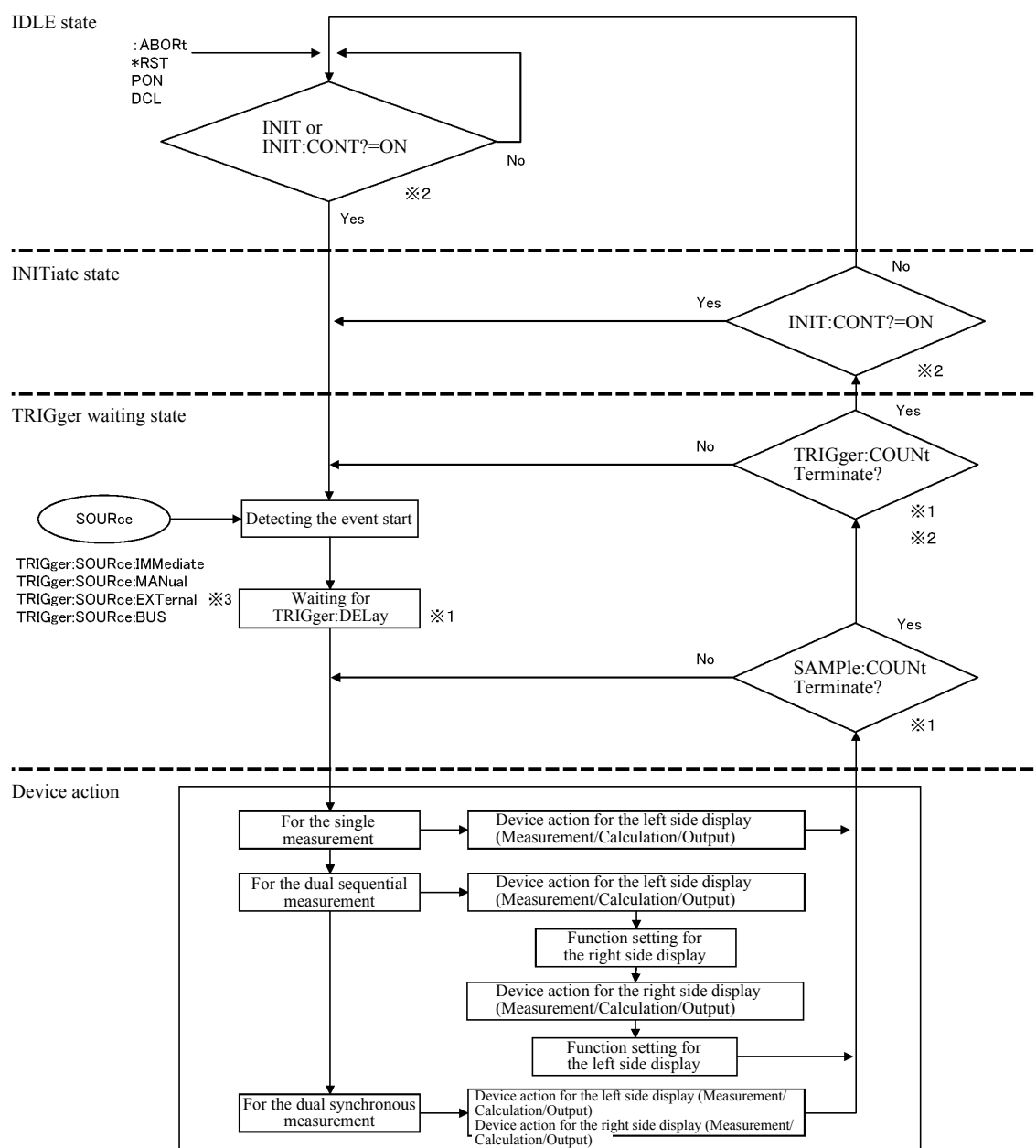
The set sampling count is common to the right and left displays.

5.8 Trigger Function

5.8 Trigger Function

The trigger system of this instrument allows trigger detection, the sampling start to be delayed, and multiple samplings per trigger.

Figure 5-5 shows the trigger model.



*1: This process is PASS (Always “Yes”) for IMMEDIATE.

*2: This process is PASS (Always “Yes”) for LOCAL.

*3: TRIGger:SOURce:EXtErnal can be used for the 7352A only.

Figure 5-5 Trigger Model

5.8.1 Trigger System Operation

1. IDLE state

The trigger system is in the IDLE state before the trigger sequence starts, after the trigger sequence is terminated, and while the processes of this system are not performed such as a reset state.

2. INITiate state

The INITiate state means that the trigger system operation starts.

The IDLE state is changed to the INITiate state by using either of the following trigger system commands: Starting the trigger system (INITiate), Setting CONTinuous to ON (INITiate:CONTinuous {ON})

If CONTinuous is set to ON when the measurement is complete, the trigger system enters the INITiate state without returning to the IDLE state and waits for the next trigger.

3. TRIGger waiting state

When the trigger event is detected from the selected trigger source, the device action starts after the trigger delay time.

4. Device action

The measurement and calculations are performed.

5.8.2 Trigger Source Selection

The trigger source is selected in 'SOURCE' in the **TRIG** category in **MENU**.

Trigger source

IMMED	Performs continuous measurement with a time period that is set by the sampling rate.
MANUAL	Detects an event when the TRIG key is pressed or the trigger signal is input from the external trigger connector.
EXT	Detects an event when the trigger signal is input from the external trigger connector.(7352A)
BUS	Detects an event when the *TRG command or the GET command for GPIB is received from the enabled remote interface bus.

The following shows the parameters whose settings automatically change depending on the trigger source setting.

IMMED	HOLD: OFF
MANUAL	HOLD: ON
EXT	HOLD: ON
BUS	HOLD: ON

Trigger key

If the **TRIG** key on the front panel is pressed when this instrument is in the local state and HOLD ON, the measurement starts. The trigger key can be used regardless of the trigger source selection.

Trigger event buffer

If triggers are input while this instrument is in the measurement state, one trigger event is buffered. In this case, the operation is the same as when the measurement is complete at the same time a trigger is generated.

5.8.3 Trigger Delay (Td)

Time from a trigger input to sampling start can be delayed by setting the trigger delay.

Because timing with other devices can be adjusted by using this function, the measurement can be started after the input becomes stable.

Table 5-5 shows the trigger delay time and setting resolution.

Table 5-5 Trigger Delay Time, Setting Resolution, and Accuracy

Trigger delay setting	Resolution *1	Delay time accuracy	
		Ext trigger	Bus trigger
0 to 600 ms	10 μ s	0.8 ms or less + resolution	3.7 ms or less + resolution
600.1 m to 6 s	100 μ s	0.8 ms or less + resolution	3.7 ms or less + resolution
6001 m to 60 s	1 ms	0.8 ms or less + resolution	3.7 ms or less + resolution
61 s to 3600 s	1 s	1 s	1 s

*1: If a value that is less than the resolution is set by using the panel or a remote command, the setting value is rounded to the resolution.

The trigger delay can be selected from a common setting and each setting for the right and left displays.

- Setting the common trigger delay for the right and left displays

The factory default trigger delay is set to the common setting for the right and left displays (disabled for each set on the right and left displays).

For the measurement operation started by the common trigger delay for the right and left displays, see Figure 5-6 (a).

The trigger delay time is set at 'T.DELAY' in **TRIG** of **MENU**.

In this case, the trigger delay time is set only for the first display, left side, regardless of the '2nd' indicator display.

- Setting each trigger delay for the right and left displays

For the measurement operation started by each trigger delay for the right and left displays, see Figure 5-6 (b).

To activate each trigger delay for the right and left displays, set 'T.D.EACH' to ON in **TRIG** of **MENU**. If set to OFF, it is disabled.

Set the trigger delay time for the second display, right side, at 'T.DELAY' in **TRIG** of **MENU** while the '2nd' indicator is activated by pressing **2nd**.

Set the trigger delay time for the first display, left side, while the '2nd' indicator is off by pressing **2nd** again.

Each trigger delay for the right and left displays is enabled only for the dual synchronous measurement where functions of Ach and Bch are set for the right and left displays.

In other cases, measurement is started by the trigger delay time set for the first display, left side.

For more information on the dual synchronous measurement, refer to Section 5.7, "Sampling Operation."

5.8.4 Trigger Count Setting

NOTE: *If a trigger delay time difference between the right and left measurements is set within 1.3 msec (for example, the same trigger delay time is set for the right and left measurements), up to 1.3 msec delay occurs between both measurement starts regardless of the set time.
If a delay occurrence is not desired, use the common trigger delay for the right and left displays.
For the HOLD off, the trigger delay is ignored.*

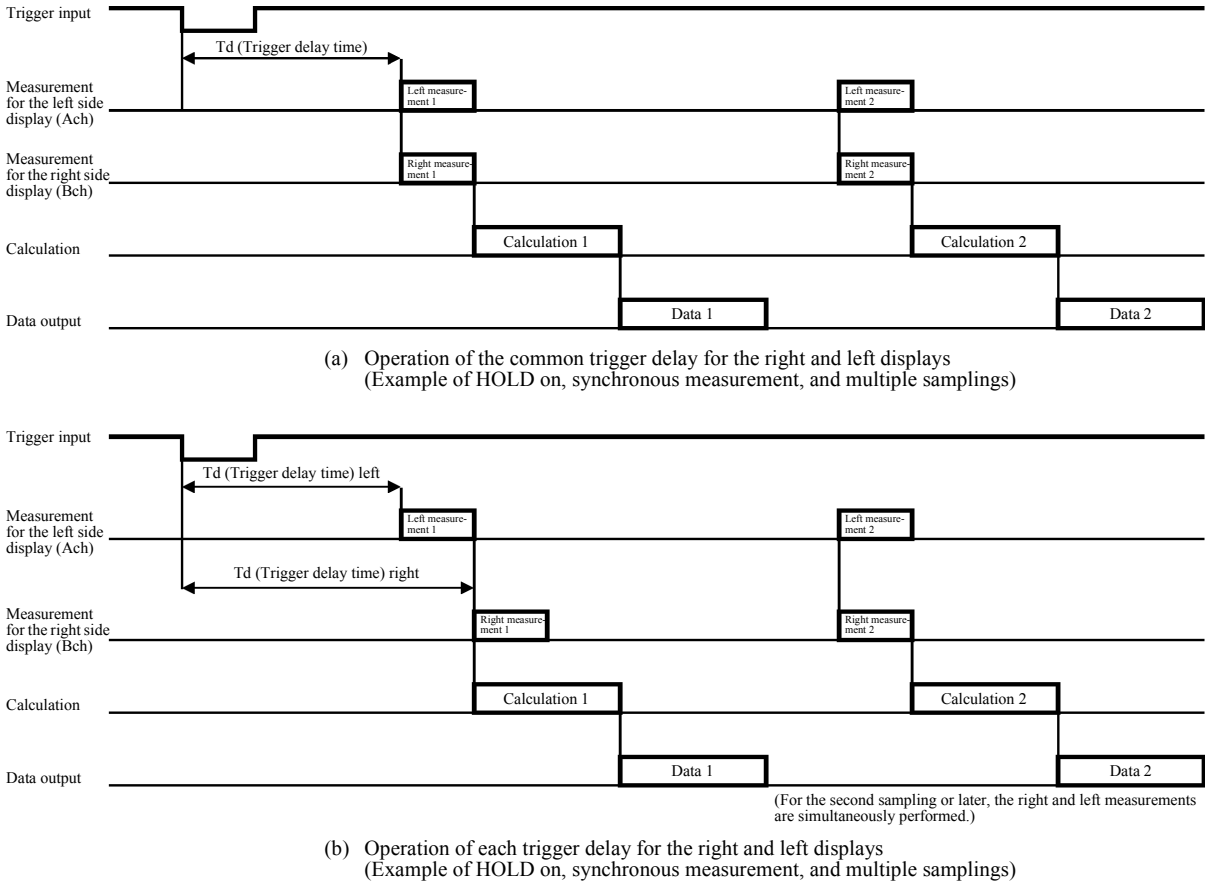


Figure 5-6 Trigger Delay Operation

5.8.4 Trigger Count Setting

Although the trigger system usually returns to the Idle state when a trigger is received, it can return to the trigger waiting state to receive triggers.

Trigger count 1 to 50000

The set trigger count is common to the right and left displays.

5.9 Calculation Functions

This instrument supports the following nine calculation functions.

Item	Function
NULL calculation	Subtracts a fixed value from a measurement value.
Smoothing calculation	Calculates the moving average.
Scaling calculation	Calculates $(\frac{M-B}{A}) \times C$.
dB calculation	Calculates $20 \log_{10} (\frac{M}{D})$.
dBm calculation	Calculates $10 \log_{10} (\frac{M^2}{D} \times \frac{1}{10^{-3}})$.
Comparator calculation	Performs a comparative calculation (HI/GO/LO).
MAX/MIN calculation	Calculates the maximum, minimum and average values.
Statistical calculation	Calculates the maximum value, minimum value, average value, standard deviation, and dispersion for the data memory.
Calculation between 2 measurements	Performs four arithmetic operations between measurements for the left side (first) and the right side (second) displays.

5.9 Calculation Functions

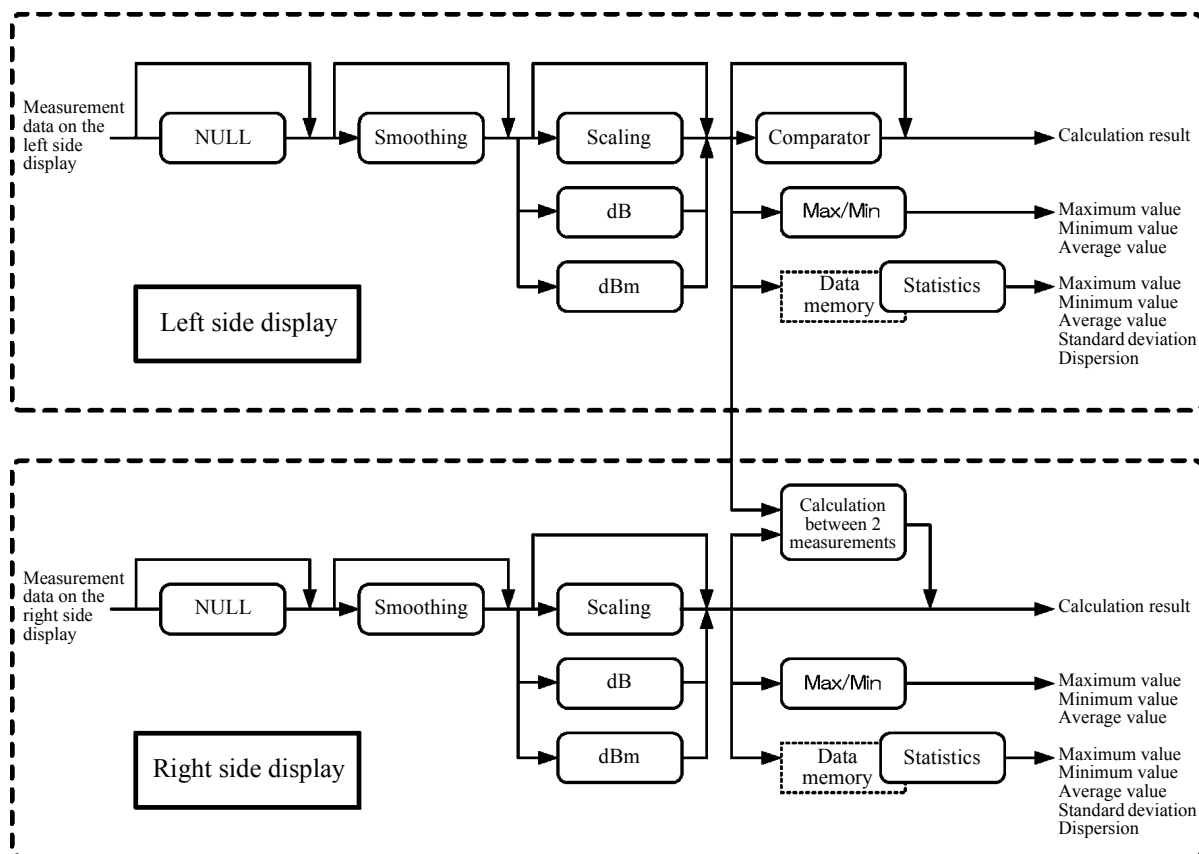


Figure 5-7 Calculation System Diagram

NOTE:

1. The following results may be obtained from calculating the data. This is because the internal resolutions of measurement values and setting values for the calculation are higher than the resolutions of the displayed data and remote output data.
 - If the buzzer is set to ON in the MAX/MIN calculation, the buzzer sounds when the MAX or MIN value is renewed. However the buzzer may sound even when the display and remote output data are not changed.
2. If the calculation is set to ON, the display of the calculation result is different from the ordinary measurement values and the decimal point and unit do not depend on the current measurement range.

<Example> DC voltage measurement, measurement range: 2000 mV, input voltage: 1 V
 Measurement value: 1000.00 mV
 Calculation Result: 1.00000 V
3. If a calculation is performed while the '2nd' indicator is on, it is applied to a measurement for the right display. For the comparator calculation and the calculation between 2 measurements, however, it is not applied.

5.9.1 NULL Calculation

1. Function

The NULL calculation subtracts the NULL constant from the measurement value and outputs the result.

$$\text{Measurement data output} = \text{Measured value} - \text{NULL constant}$$

Pressing **NULL** performs the calculation and the NULL indicator in the display section turns on. If **NULL** is pressed again while the NULL calculation is performed, the calculation is canceled and the NULL indicator turns off.

CAUTION: *The actual measured value is not displayed while the calculation is performed. Remember that any hazardous voltage present on the input connector or test lead may not be detected at this time.*

2. NULL constant

The measurement value when the **NULL** key is pressed is applied as the NULL constant. The NULL calculation cannot be performed when the overload (OL) occurs. The NULL constant can be checked in **MENU** during the calculation.



The setting ranges are shown below.

Setting range	Minimum setting value
-999999.E +6 to +999999.E + 6	0.00000E -9

The exponential part is set by using a unit prefix (n, μ, m, k, M).

3. Changing the NULL constant

1. Display the NULL constant according to item 2 (previous description).
2. After pressing , the blinking position of the following items, which can be changed, is moved by using  and .
 - From the most significant digit to the least significant digit in the numeric value
 - Unit prefix (exponential part setting)
 - Decimal point
3. Change the setting value by using  and  at the blinking position.
The unit prefix is switched according to the following order.
n ↔ μ ↔ m ↔ No unit prefix ↔ k ↔ M
4. Press **ENTER** to apply the setting.

5.9.2 Smoothing Calculation

4. Canceling the NULL calculation

The NULL calculation is canceled when any of the following conditions occur.

- The **NULL** key is pressed while the NULL calculation is performed.
- The *RST command is executed.

5.9.2 Smoothing Calculation

1. Function

The smoothing function is used when noise is superimposed on the measurement signal. Because measurement values of the specified number of times (smoothing count) are calculated by using the moving averaging method, the dispersion of those measurement values becomes small.

The smoothing calculation equation is shown below:

$$\text{Display} = (\text{Measured value 1} + , , , , + \text{Measured value N}) / N$$

Pressing **SM** performs the calculation and the SM indicator in the display section turns on continuously or blinks.

If **SM** is pressed again while the smoothing calculation is performed, the calculation is canceled and the SM indicator turns off.

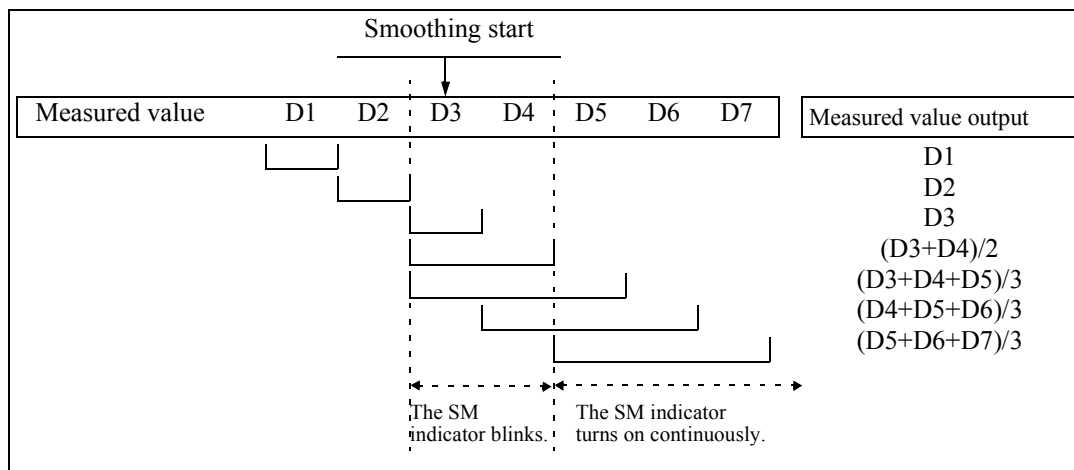
The SM indicator blinks from the smoothing start to the (N - 1)th measurement and turns on after the Nth measurement.

The (N + 1)th data is displayed as below:

$$\text{Display} = (\text{Measured value 2} + , , , , + \text{Measured value (N + 1)}) / N$$

The average of the measured values, which have been acquired, is displayed from the smoothing start to the (N - 1)th measurement.

When the smoothing count, N, is set to 3, the operation is as follows:



The smoothing calculation result, $D(sm)$, at the n th measurement, which exceeds the smoothing count, is shown below:

Smoothing calculation result $D(sm) = \frac{1}{T} \sum_{i=n-T+1}^n D_i$	$D(sm)$:	Smoothing calculation result at the n -th measurement
	D_i :	Measured value (before the smoothing calculation)
	T :	Smoothing count (2 to 100 can be set.)

2. Setting the smoothing count

The smoothing count is set in **MENU**.



The smoothing count can be set between 2 and 100.

3. Restarting the smoothing calculation

The smoothing calculation restarts from $N=1$ when any of the following conditions occur.

- The power is turned on.
- The smoothing count is changed.
- The NULL calculation is performed.

4. Canceling the smoothing calculation

The smoothing calculation is canceled when any of the following conditions occur.

- The **SM** key is pressed while the smoothing calculation is performed.
- The measurement function is changed.
- The *RST command is executed.

5. Smoothing calculation and overload (OL)

If the overload occurs during the smoothing process, this measured value is ignored.
(All measurement data except for OL data can be used in the smoothing calculation.)

5.9.3 Scaling Calculation

5.9.3 Scaling Calculation

1. Function

The scaling calculation equation is shown below.

$$\text{Display} = \frac{\text{Measured value M} - \text{Constant B}}{\text{Constant A}} \times \text{Constant C}$$

When SCL is selected from the MATH item in **MENU**, the SCL indicator in the display section turns on and the scaling calculation is performed.

If OFF is selected from the MATH item in **MENU** while the scaling calculation is performed, the calculation is canceled and the SCL indicator turns off.

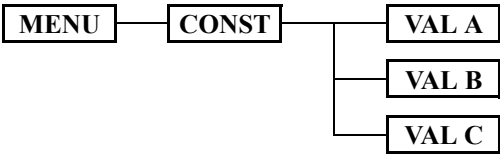


CAUTION: The actual measured value is not displayed while the scaling calculation is performed. Remember that any hazardous voltage present on the input connector or test lead may not be detected at this time.

2. Setting scaling constants

NOTE: Press the **EXIT/LOCAL** key if quitting the setting procedure.

1. Set Constants A, B, and C in **MENU**.



2. After selecting a constant to be set, the constant can be set by pressing **↵**.

3. Setting the numerical value of the constant

The setting ranges of Constants A, B, and C are shown below.

Constant	Setting range	Minimum setting value
A	-999999.E +6 to +999999.E + 6	0.00001E - 9
B		0.00000E - 9
C		

The exponential part is set by using a unit prefix (n, μ, m, k, M).

1. In the constant setting mode, the following items, which blink, can be selected by using $\square$ and $\square$.
 - From the most significant digit to the least significant digit in the numeric value
 - Unit prefix
 - Decimal point
2. Change the numeric value and unit prefix by using $\square$ and $\square$.
 The unit prefix is switched according to the following order.
 $n \leftrightarrow \mu \leftrightarrow m \leftrightarrow \text{No unit prefix} \leftrightarrow k \leftrightarrow M$
 Press the **TRIG** key in this mode if setting the measurement value as the scaling constant.
3. Press **ENTER** to apply the setting.
4. S OL (scaling over)

“ $\pm S OL$ ” is displayed if the scaling calculation result exceeds the values of $\pm 999.999E + 6$.
 In this time, the measurement range is not changed to the upper range even if the measurement range is set to Auto-range.
 (Auto-range is determined according to the measurement value acquired before the calculation is performed.)
5. Canceling the scaling calculation

The scaling calculation is canceled when any of the following conditions occur.

 - OFF is selected from the MATH item in **MENU** while the scaling calculation is performed.
 - The measurement function is changed.
 - The *RST command is executed.
 - The dB or dBm calculation is performed.

5.9.4 dB and dBm Calculations

5.9.4 dB and dBm Calculations

1. Function

The dB calculation, in which the decibel conversion is accomplished, can be performed only for the voltage and current measurements.

The dBm calculation, which is useful to calculate the power gain per 1 V, can be performed only for the voltage measurement.

If the zero is measured while the dB or dBm calculation is performed, the calculation error occurs and the error message "Err d" is displayed.

Only one calculation can be selected from the dB, dBm, and scaling calculations at a time.

The dB and dBm calculation equations are shown below:

dB: Display = 20 log₁₀ $\frac{| \text{Measured value} |}{\text{Constant D}}$

dBm: Display = 10 log₁₀ $\frac{(\text{Measured value})^2 / \text{Constant D}}{10^{-3}}$

When dB or dBm is selected from the MATH item in **MENU**, the dB indicator in the display section turns on and the calculation is performed.



CAUTION: *The actual measured value is not displayed while the dB or dBm calculation is performed. Remember that any hazardous voltage present on the input connector or test lead may not be detected at this time.*

2. Setting Constant D

The setting range of Constant D is shown below.

Setting range	Minimum setting value
0.00001E -9 to 999999.E + 6	0.00001E - 9

The exponential part is set by using a unit prefix (n, μ, m, k, M).

1. Set Constant D in **MENU**.



2. After the constant setting mode is enabled by pressing , the following items, which blink, can be selected by pressing and .

- From the most significant digit to the least significant digit in the numeric value

- Unit prefix
- Decimal point

3. Change the numeric value and unit prefix by using $\triangle$ and ∇ .

The unit prefix is switched according to the following order.

$n \leftrightarrow \mu \leftrightarrow m \leftrightarrow \text{No unit prefix} \leftrightarrow k \leftrightarrow M$

Press the **TRIG** key in this mode if setting the measured value as Constant D.

4. Press **ENTER** to apply the setting.

3. Canceling the dB and dBm calculations

The dB and dBm calculations are canceled when any of the following conditions occur.

- The scaling calculation is set while the dB or dBm calculation is performed.
- The measurement function is changed.
- The *RST command is executed.

5.9.5 Comparator Calculation

5.9.5 Comparator Calculation

1. Function

The comparator calculation equation is shown below.

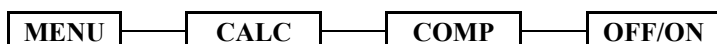
$HI = (\text{Measured value} > \text{HI setting value})$
 $LO = (\text{Measured value} < \text{LOW setting value})$
 $GO = (\text{LOW setting value} \leq \text{Measured value} \leq \text{HI setting value})$

(Processing the special data)

- The overload (OL) of plus (+) data is judged as HI.
- The overload (OL) of minus (-) data is judged as LO.

The result whether HI, GO, or LO is displayed by setting COMP in **MENU** to ON.

If the comparator calculation is set to ON while in the dual display, the single measurement is set. The comparator calculation is performed for the left measurement display.



The judgment result can be output to the display, buzzer, and status register.

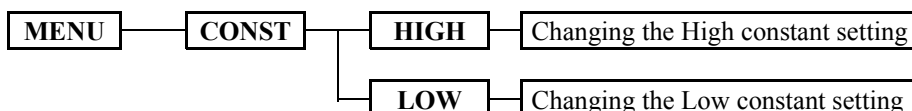
(The buzzer sounds only when BEEP is set to ON.)

The comparator calculation can be applied to all measurement functions.

However, when MAX/MIN calculation is set to ON, the comparator calculation is applied to the value before the MAX/MIN calculation.

2. Setting the judgment conditions

The setting examples of the High constant, Low constant, and buzzer are shown below.



3. Setting the numerical value of the constant

The setting ranges of the High and Low constants are shown below.

Constant	Setting range	Minimum setting value
High	-999999.E +6 to +999999.E + 6	0.00000E - 9
Low		

The exponential part is set by using a unit prefix (n, μ, m, k, M).

- After the constant setting mode is enabled by pressing $\square$, the following items, which blink, can be selected by pressing $\square$ and $\square$.
 - From the most significant digit to the least significant digit in the numeric value

- Unit prefix
 - Decimal point
2. Change the numeric value and unit prefix by using $\triangleleft$ and $\triangleright$ at the blinking position.
The unit prefix is switched according to the following order.
 $n \leftrightarrow \mu \leftrightarrow m \leftrightarrow \text{No unit prefix} \leftrightarrow k \leftrightarrow M$
Press the **TRIG** key in this mode if setting the measurement value as the High or Low constant.
 3. Press **ENTER** to apply the setting.

If the constants are set as “High constant < Low constant”, the calculation results may be judged as “HI and LO”. In this case, HI and LO are displayed at the same time.

4. Buzzer setting

The buzzer output according to the PASS/FAIL judgments can be set.

However, the buzzer output cannot be performed if $\bullet \cdot \cdot \cdot$ turns off.

The buzzer setting is performed in BEEP in **MENU**.



1. Press $\triangleright$ on the current setting display (BEEP) to enable the buzzer setting mode.
 2. Select OFF, FAIL, or PASS by using $\triangleleft$ and $\triangleright$.
- #### 5. PASS condition setting

In the PASS condition setting, each comparator calculation result (HI, GO, and LO) is specified as either PASS or not.

The PASS or FAIL judgment result can be known from the bit0 and bit1 values in the measurement event register (MER).

The judgment result is FAIL when bit0 in MER is 1, and is PASS when bit1 in MER is 1.

Table 5-6 shows the PASS condition setting and PASS or FAIL judgment.

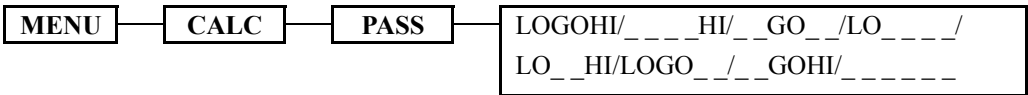
Example: In cases where the comparator calculation result HI is specified as the PASS condition but LO and GO are not specified as the PASS condition, if a result is HI when the comparator calculation is performed, the judgment is PASS. If a result is GO or LO, the judgment is FAIL.

5.9.5 Comparator Calculation

Table 5-6 PASS Condition Setting and PASS/FAIL Judgment

Judgment result	Hi	PASS	PASS	FAIL	FAIL	PASS	FAIL	PASS	FAIL
	Go	PASS	FAIL	PASS	FAIL	FAIL	PASS	PASS	FAIL
	Lo	PASS	FAIL	FAIL	PASS	PASS	PASS	FAIL	FAIL
PASS condition setting	Panel setting	LGH	--H	-G-	L--	L-H	LG-	-GH	---
	Remote command	LOP1	LOP0	LOP0	LOP1	LOP1	LOP1	LOP0	LOP0
		MIP1	MIP0	MIP1	MIP0	MIP0	MIP1	MIP1	MIP0
		HIP1	HIP1	HIP0	HIP0	HIP1	HIP0	HIP1	HIP0

The PASS conditions are set in **MENU**.



1. Press in the current set value display state (PASS) to enable the PASS setting mode.
2. Select the PASS condition by using and .
6. Canceling the comparator calculation
The comparator calculation is canceled when any of the following conditions occur.
 - The comparator calculation is set to OFF while the comparator calculation is performed.
 - The measurement function is changed.
 - The *RST command is executed.
7. Changing the measurement range and comparator operation
The comparator function operates continuously even if the measurement range is changed.
The reference values for the judgment are not changed because the values and the unit are set.

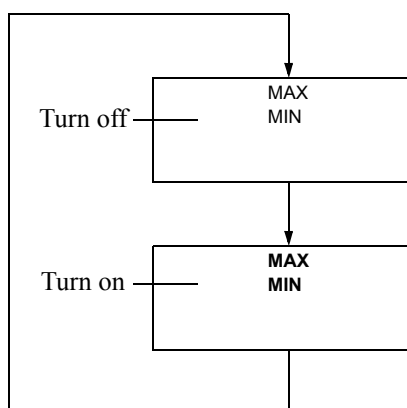
5.9.6 MAX/MIN Calculation

1. Function

1. This function calculates the maximum, minimum, and average values while the MAX/MIN calculation is performed.
2. The following calculations are performed at the same time in the MAX/MIN calculation.
 - Maximum value (MAX)
 - Minimum value (MIN)
 - Average value (AVE)
 - Measurement count
3. All measured data except for overload (OL) and calculation error data can be used for the calculation.
4. If BEEP is set to ON, the buzzer sounds when the MAX or MIN value is renewed. However, although the displayed value is not changed, the buzzer may sound. This is because the resolution of measured value is higher than the resolution of the displayed value.

2. Setting the calculation

The operation is changed as shown below by selecting MAX/MIN in **MENU**:



Canceling the calculation

The MAX and MIN indicators turn off.

Performing the calculation

Any one of the following indicator displays can be selected. (NOTE)

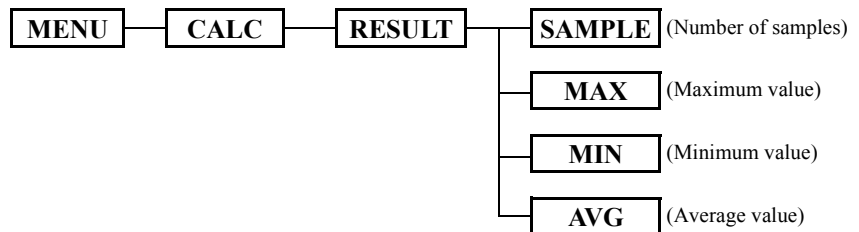
- The MAX and MIN indicators turn on.
- The MAX indicator blinks.
- The MIN indicator blinks.

NOTE: Refer to “4. MAX/MIN calculation value display function during the measurement” in this section.

5.9.6 MAX/MIN Calculation

3. How to call the MAX/MIN calculation result

1. Select RESULT in **MENU** to see the calculation results display.



2. Select an item to be displayed by pressing ◀ and ▶.

The measurement count, maximum value, minimum value, and average value are displayed in this order.

4. MAX/MIN calculation value display function during the measurement

1. The value while the MAX/MIN calculation is performed can be selected and displayed from the following three items.

- Measured value display (MAX and MIN indicator turn on.)
- Maximum value display (MAX indicator blinks.)
- Minimum value display (MIN indicator blinks.)

2. How to set

Select a display item according to “2. Setting the calculation” in this section.

The selected value is displayed by pressing **ENTER**.

5. Canceling the MAX/MIN calculation

The MAX/MIN calculation is canceled when any of the following conditions occur.

- The MAX/MIN calculation is set to OFF while the MAX/MIN calculation is performed.
- The measurement function is changed.
- The *RST command is executed.

6. Restarting the MAX/MIN calculation

The MAX/MIN calculation values are cleared and the MAX/MIN calculation restarts when any of the following conditions occur.

- The power is turned on.
- The MAX/MIN calculation is set to ON after it is set to OFF.
- Any calculation except for the comparator calculation is changed to ON or OFF.
- The NULL constant or the smoothing count is changed.

5.9.7 Statistical Calculation

1. Function

- This function calculates the maximum value, minimum value, average value, standard deviation, and dispersion of the measured data that is stored in the measurement memory.
- For a statistical calculation, the following calculations are performed at the same time:
 - Number of samples $S_{CNT} = (\text{Amount of data}) - (\text{Amount of overload})$, in the specified range
 - Maximum value $S_{MAX} = \text{Maximum value of data to be calculated}$
 - Minimum value $S_{MIN} = \text{Minimum value of data to be calculated}$
 - Average value $S_{AVE} = \frac{1}{S_{CNT}} \sum_{k=1}^{S_{CNT}} D_k$
 - Dispersion $S_{PTP} = |S_{MAX} - S_{MIN}|$
 - Standard deviation $S\sigma = \sqrt{\frac{1}{S_{CNT}-1} \sum_{k=1}^{S_{CNT}} (D_k - S_{AVE})^2}$
- All measured data except for overload (OL) and calculation error data can be used for the calculation.

2. Performing the calculation

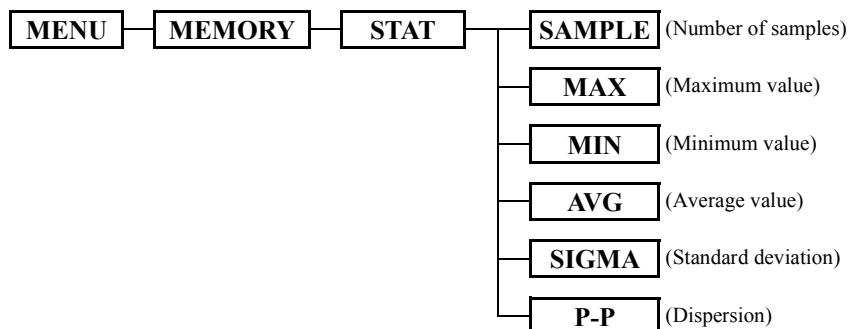
- Select STAT in **MENU**.



- Select STAT and press **ENTER** to perform the calculation.
If data does not exist in the measurement data memory, 'LAST' is displayed.

3. How to call the calculation result

- The calculation result is displayed by selecting STAT in **MENU**.



- After performing the statistical calculation, select an item to be displayed by pressing ◀ and ▶.

5.9.8 Calculation between 2 Measurements

1. Function

The calculation equation between 2 measurements is shown below:

For addition:	$[\text{Measurement for the left side display}] + [\text{Measurement for the right side display}]$
For subtraction:	$[\text{Measurement for the left side display}] - [\text{Measurement for the right side display}]$
For multiplication:	$[\text{Measurement for the left side display}] \times [\text{Measurement for the right side display}]$
For division:	$[\text{Measurement for the left side display}] \div [\text{Measurement for the right side display}]$

When one of the four arithmetic operations is selected by the 2MEAS item in **MENU**, the selected arithmetic display on the left side display section turns on and the calculation is performed.

If OFF is selected by the 2MEAS item in **MENU** while the calculation between 2 measurements is performed, the calculation is canceled and the arithmetic operation display on the left side display section turns off.



NOTE:

1. While the calculation between 2 measurements is performed, the actual measured values are not displayed. Remember that any hazardous voltage on the input connector and test lead may not be detected at this time.
2. For the single measurement, the calculation between 2 measurements cannot be used. Set this calculation on the dual display.

2. 2MEAS OL (Overload for the calculation between 2 measurement)

If a calculation result between 2 measurements exceeds a value of $\pm 999.999\text{E}+6$, “ $\pm\text{S OL}$ ” is displayed. At this time, the measurement range is not changed to the upper range even if it is set to the auto-range. (Auto-range is determined according to the measurement value acquired before the calculation is performed.)

3. Canceling the calculation between 2 measurements

The calculation between 2 measurements is canceled when any of the following conditions occur.

- When OFF is selected by the 2MEAS item in **MENU** while the calculation between 2 measurements is performed.
- When the measurement function is changed.
- When the *RST command is executed.
- When the comparator calculation is performed (ON).
- When the right side display is canceled (OFF) while the calculation between 2 measurements is performed.

4. Calculation between 2 measurements and OL (overload)

If the overload occurs during the calculation between 2 measurements, this measured value is ignored.

5. Zero division

If zero is measured on the right side display while division between 2 measurements is performed, a calculation error occurs and the error message “Err d” is displayed.

5.10 Saving and Loading the Measurement Condition Settings

In this instrument, four files of measurement condition setting parameters can be saved in nonvolatile memory USER0 to USER3.

For information on setting parameters, refer to Section 8.9, “System Recovery Procedure” and “Table 8-5 Initial Values of Setting Parameters.”

5.10.1 Auto-Loading when Turning on the Power

This instrument starts under the stored conditions when the power is turned off.

If the same startup conditions are always used, the measurement conditions stored in USER0 can be loaded when the power is turned on.

Select from the followings in ‘P.ON LD’ in the **PARAM** category in **MENU**.

P.OFF	Starts this instrument under the measurement conditions stored when the power is turned off.
-------	--

USER0	Loads the measurement conditions in USER0.
-------	--

Press **ENTER** to apply the setting.

5.10.2 Saving the Setting Parameters

Select a memory from USER0 to USER3 in ‘SAVE’ in the **PARAM** category in **MENU**.

Press **ENTER** to save the setting parameters into the selected memory.

5.10.3 Loading the Setting Parameters

Select a memory from USER0 to USER3 and DFLT in ‘LOAD’ in the **PARAM** category in **MENU**.

Press **ENTER** to load the setting parameters from the selected memory.

If DFLT is selected, the initial values are loaded.

For more information on initial values, refer to Section 8.9, “System Recovery Procedure”.

5.11 Storing and Recalling Measurement Data

5.11 Storing and Recalling Measurement Data

This instrument includes measurement data memory, in which 20,000 measurement results can be stored.
For the dual display, 10,000 measurement results can be stored for the right and left side displays respectively.

5.11.1 Memory Store

Select ON or OFF from 'STORE' in the **MEMORY** category of **MENU**.

Press **ENTER** to apply the setting and the 'ST' indicator to be activated.

The measured data is stored into the measurement data memory by closing MENU and starting the measurement.

When the number of measurement results reaches 20,000, the buzzer which is set to ON sounds and the 'ST' indicator blinks.
(For the dual display, use 10,000 in place of 20,000.)

5.11.2 Memory Recall

Enter a data number to be recalled at 'RECALL' in the **MEMORY** category of **MENU**.

Press **ENTER** to apply the setting and the 'RCL' indicator to be activated.

The data number can be increased and decreased by using ◀ and ▶ before recalling the data number.

5.11.3 Memory Clear

The measurement data memory is cleared when any of the following conditions occur.

- The power is turned on.
- The device clear command is executed from the GPIB interface.
- The *RST or ICL command is executed.
- The parameter initialization is performed in the **INIT** category of the **MENU** mode.
- The memory store setting of the measured data is switched from OFF to ON.
- The single and dual displays are switched.

5.12 System Settings

5.12.1 Buzzer Setting

Select ON or OFF from 'BEEP' in the **SYSTEM** category of **MENU**.

ON The buzzer sounds.

OFF The buzzer does not sound.

The buzzer does not sound regardless of the result of comparator calculation.

Press **ENTER** to apply the setting.

If an error occurs, the buzzer sounds even when BEEP is set to OFF.

5.12.2 Disabling Functions

To prevent incorrect functions from being selected because of malfunctions and to use this instrument safely, some functions can be disabled.

Select each function to be set to ENA or DIS from 'F.INHIB' in the **SYSTEM** category of **MENU**.

Press  to enter in the entry layer 2 and select ENA (Enabling) or DIS (Disabling) using  and .

Press **ENTER** to apply the setting.

- The relationship according to disabling the function setting
Although ENA or DIS can be set for each function, some functions are automatically disabled depending on the setting of other functions.
If a higher function setting in the dependency relation is disabled, the lower function setting cannot be changed.

5.12.2 Disabling Functions

Table 5-7 Relationship according to Disabling Function

Set function	Function that depends on the set function					
	Diode-Ach	ACV (AC+DC)-Ach	ACI (AC+DC)-Ach	Continuity- Ach	ACI-Ach	FREQ-Ach
DCV-Ach		○				
ACV-Ach		○	○		●	●
DCI-Ach			○			
ACI-Ach			○		/	●
2WΩ-Ach/ LP-2WΩ-Ach	◎			◎		
FREQ-Ach						/
Set function	Function that depends on the set function					
	TEMP	ACI (AC+DC)-Bch				
DCV-Bch	◎					
DCI-Bch		○				
ACI-Bch		○				

◎ : Automatically set to ENA or DIS according to the set function.

○ : Disabled if any of the set functions are set to DIS, and enabled if all of the related set functions are set to ENA.

● : Disabled if any of the set functions are set to DIS, and not enabled even if all of the related set functions are set to ENA.

- Canceling the setting

To cancel disabled settings and to enable settings of all functions, perform the initialization in the **INIT** category of **MENU** or perform the re-setting at 'F.INHIB' in the **SYSTEM** category of **MENU**.

5.12.3 Front Panel Disabling Function

This function disables all panel keys on the front panel in this instrument.

The following shows how to set the front panel disabling function.

Password setting

The password is required to enable or disable the front panel keys.

A four-digit number from 0000 to 9999 can be used for the password.

Set the password according to the following procedure.

- | <u>Operation</u> | <u>Display section</u> |
|--|------------------------|
| 1. Press MENU and select SYSTEM by using ◀ and ▶. | MENU SYSTEM |
| 2. Press ◀. | SYSTEM BEEP |
| 3. Select 'PWD' by using ◀ and ▶. | SYSTEM PWD |
| 4. Press ◀. | PWDnew |
| 5. Select the digit in the numerical value entry section by using ◀ and ▶.
Change the numerical value by using ▲ and ▼ to enter the password. | PWDnew |
| 6. Press ENTER . | PWDold |
| 7. Enter the current password according to the above procedure. | PWDold |
| 8. Press ENTER .
If the correct password is entered, PASS is displayed and the password is set.

If an incorrect password is entered, FAIL is displayed and the procedure returns to step 3.
Reset by using the correct password. | PWD PASS |
| 9. Press EXIT to exit from MENU. | PWD FAIL |

5.12.3 Front Panel Disabling Function

Panel lock setting

Set the panel lock according to the following procedure.

- Operation

1. Press **MENU** and select **SYSTEM** by using  and .

2. Press .

3. Select 'P.LOCK' by using  and .

4. Press .

5. Select ON by using  and  and press **ENTER**.

6. Select the digit in the numerical value entry section by using  and .

Change the numerical value by using  and  to enter the password.

7. Press **ENTER**.

If a correct password is entered, P.LOCK is displayed and exit from MENU.

If an incorrect password is entered, the procedure returns to step 3.

Re-set by using the correct password.
- | Display section | |
|-----------------|--------|
| MENU | SYSTEM |
| SYSTEM | BEEP |
| SYSTEM | P.LOCK |
| P.LOCK | OFF |
| P.LOCK | ON |
| PWD | |
| PWD | LOCK |
- 5-44

Canceling the panel lock

Cancel the panel lock according to the following procedure.

- | <u>Operation</u> | <u>Display section</u> | | | | |
|--|--|-------|------|-----|------|
| 1. If pressing any key while the panel lock is set, the entry layer of P.LOCK is set. | <table border="1" style="border-collapse: collapse; width: 100px; height: 40px;"><tr><td style="padding: 2px;">PWLck</td><td style="padding: 2px;"></td></tr></table> | PWLck | | | |
| PWLck | | | | | |
| 2. Select the digit in the numerical value entry section by using $\triangleleft$ and $\triangleright$.
Change the numerical value by using $\triangleup$ and $\triangledown$ to enter the password. | <table border="1" style="border-collapse: collapse; width: 100px; height: 60px;"><tr><td style="padding: 2px;">PWLck</td><td style="padding: 2px;"></td></tr></table> | PWLck | | | |
| PWLck | | | | | |
| 3. Press ENTER .

If a correct password is entered, PASS is displayed and the password is set.

If an incorrect password is entered, FAIL is displayed and the procedure returns to step 3. | <table border="1" style="border-collapse: collapse; width: 100px; height: 40px;"><tr><td style="padding: 2px;">PWD</td><td style="padding: 2px;">PASS</td></tr></table>
<table border="1" style="border-collapse: collapse; width: 100px; height: 40px;"><tr><td style="padding: 2px;">PWD</td><td style="padding: 2px;">FAIL</td></tr></table> | PWD | PASS | PWD | FAIL |
| PWD | PASS | | | | |
| PWD | FAIL | | | | |

NOTE: *The panel lock cannot be released if the password is forgotten. Do not forget the password. Copying the password onto paper is recommended.*

5.12.4 Error Queue

If any error occurs in this instrument, the error number is recorded and the 'ERR' indicator turns on.

Only the error number can be read from the front panel menu.

The error number and error message can be read from the GPIB or USB by using the query command.

Up to 20 errors can be recorded.

For more information on error numbers and solutions, refer to Section 8.11, "Error Message List."

Reading the error queue

Errors are displayed in order by pressing $\triangledown$ in 'QUEUE' in **SYSTEM** category in **MENU**.

Clearing the error queue

The error queue is cleared by reading the errors.

6. HOW TO USE THE INTERFACE

The 7352A and 7352E include the following standard interfaces according to the model.

7352A USB, GPIB, and RS-232 interfaces

7352E USB interface

However, both interfaces cannot be used at the same time. Select either one.

Table 6-1 Interface

Interface	7352A	7352E
GPIB	○	-
USB	○	○
RS-232	○	-

6.1 Interface Selection

The interface can be selected and set only from MENU on the front panel.

1. Because the setting of the interface selection is saved in the nonvolatile memory, the setting does not change even if the power is turned off or the interface is reset.
2. The address specific to the instrument is set in the interface. To identify each DMM on the USB interface, the address of the USB interface (USB.ID) is set.
The address is displayed by turning on the power or using the address setting in MENU.

The interface setting item and the initial setting when shipping are shown below.

Setting item	Initial setting when shipping
Header ON/OFF	ON
GPIB address/USB.ID	1
GPIB talker function	Addressable

Select the interface from GPIB and USB in 'BUS' in **I/F** in **MENU**.

Press **ENTER** to apply the setting.

Header ON/OFF

Select ON or OFF in 'HEADER' in **I/F** in **MENU**.

Press **ENTER** to apply the setting.

6.2 GPIB (7352A)

6.2 GPIB (7352A)**6.2.1 Overview**

The automatic measurement system can be set up easily by using GPIB (General Purpose Interface Bus) because measurement functions and parameters can be set and measurement data can be read by using the external control.

The measurement values are not affected by any external devices connected to this instrument, because the GPIB signals from this instrument are electrically isolated from the measurement signal system.

The remote commands are common to the USB commands.

- General Specifications

Standard: IEEE-488

Code: ASCII code

Logical level: Logic 0 "High" state +2.4 V or more

Logic 1 "Low" state +0.4 V or less

Table 6-2 Interface Functions

Code	Function
SH1	Source handshake function
AH1	Acceptor handshake function
T5	Basic talker function, Listener-specified talker cancel function, Talk-Only mode function, Serial polling function
L4	Basic listener function, Talker-specified listener cancel function
SR1	Service request function
RL1	Remote/Local switching function
PP0	No parallel polling function
DC1	Device clear function (The SDC and DCL commands can be used.)
DT1	Device trigger function (The GET command can be used.)
C0	No controller function
E2	Using the tri-state bus driver

6.2.2 Precautions in Use of GPIB

1. Do not use too long a bus cable to connect any instrument or controller. Ensure that the total cable length does not exceed 20 m. ADC CORPORATION uses the following cables as the standard bus cable.

Table 6-3 Standard Bus Cable

Length	Name
0.5 m	408JE-1P5
1 m	408JE-101
2 m	408JE-102
4 m	408JE-104

2. Bus cables have piggyback connectors. A piggyback connector has both a male connector and a female connector by itself. The male and female connectors can be stacked on top of each other.
When connecting bus cables, do not stack three or more connectors. Secure the connectors with the clamping screws.
3. Check the power requirements, grounding conditions, and setting conditions of each device before turning on the power of each device.
Ensure that all devices on the bus are turned on. If any device on the bus is not turned on, the system may not operate correctly.
4. Connecting or disconnecting cables
Turn off all devices on the bus before connecting or disconnecting GPIB cables. Check that the chassis of the devices on the bus are connected to each other and to the ground before connecting or disconnecting any GPIB cables.
5. ATN interruption while transmitting a message
If an ATN interruption request is received on the bus while the messages are transmitted among devices, the request is allowed and previous status is cleared.
6. Do not connect the controller when using the Talk-Only mode.
7. Up to 255 characters in one transmission can be recognized as a program command. If the program command exceeds 255 characters, an error occurs.
8. Retain LOW on the REN line for 5 ms or more after the program command has been sent.
9. The next command can be received before the execution of the INI(ADC), INITiate(SCPI), or *TRG command is complete.
To delay the execution of the next command until the current command is complete, use the *OPC, *OPC?, or *WAI command. Describe the *OPC, *OPC?, or *WAI command at the end of a line in the program.
Example “:INIT;*OPC”
 “:INIT;*OPC?”
 “:INIT;*WAI”

6.2.3 GPIB Setting

6.2.3 GPIB Setting

The following setting menu can be used when the GPIB is selected as the interface.

Address setting

1. Enter any address from 0 to 30 in 'GP.ADR' of **I/F** in **MENU**.
2. Press **ENTER** to apply the setting.

Talk-Only setting

1. Select the following from 'T.ONLY' of **I/F** in **MENU**.
ON: Talk-Only
OFF: Addressable
2. Press **ENTER** to apply the setting.

Command language selection

1. Select the following from 'LANG' of **I/F** in **MENU**.
ADC: Command used in ADC CORPORATION
SCPI: SCPI command
R6452: Command used in ADC CORPORATION's R6452
2. Press **ENTER** to apply the setting.

Continue setting (For SCPI)

1. Select the following from 'CONT' of **I/F** in **MENU**.
ON: CONTINUOUS ON
OFF: CONTINUOUS OFF
2. Press **ENTER** to apply the setting.

6.3 RS-232[EIA-232] (7352A)

6.3.1 Overview

By using the RS-232 interface, setting measurement functions and parameters of this instrument and reading the measured data can be remotely controlled, and the automatic measurement system can be easily built up.

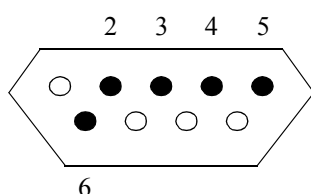
Because the interface is electrically isolated from the measurement signal system, measured values are not affected by external devices.

The remote commands are common to the GPIB commands.

- Specifications

Setting item		Factory default
Header of output data	: on, off	on
Talk-Only	: on, off	off
Baud rate	: 9600, 4800, 2400, 1200, 600, 300	9600
Parity	: even, odd, None	None
Number of data bits	: 8, 7	8
Number of stop bits	: 1, 2	1
Echo	: on, off	off

The 9-pin connector (DB-9 male connector) used for the RS-232 interface is located on the rear panel of this instrument.



Pin No.	Input/output	Description
2	Input	Receive data (RxD)
3	Output	Transmit data (TxD)
4	Output	Data terminal ready (DTR)
5	-	Signal ground (SG)
6	Input	Data set ready (DSR)

Up to 251 characters in one transmission can be recognized as a program code.

If a program code exceeds 251 characters, an error occurs.

If the DSR (data set ready) status is false when it is checked in this instrument, the transmit data (TxD) output is stopped.

Data transmission restarts when DSR is true.

CAUTION: This instrument cannot use the flow control performed by the X parameter (XON/XOFF).

6.3.2 RS-232 Setting

6.3.2 RS-232 Setting

Header of output data

1. Select the following from 'HEADR' of **I/F** in **MENU**.
ON/OFF
2. Press **ENTER** to apply the setting.

Talk-Only setting

1. Select the following from 'T.ONLY' of **I/F** in **MENU**.
ON: Talk-Only
OFF: Addressable
2. Press **ENTER** to apply the setting.

Command language selection

1. Select the following from 'LANG' of **I/F** in **MENU**.
ADC: Command used in ADC CORPORATION
R6452: Command used in ADC CORPORATION's R6452
2. Press **ENTER** to apply the setting.

Baud rate

1. Select the following from 'BAUD' of **I/F** in **MENU**.
9600 - 300 - 600 - 1200 - 2400 - 4800
2. Press **ENTER** to apply the setting.

Number of data bits

1. Select the following from 'DATA' of **I/F** in **MENU**.
8 BIT/7 BIT
2. Press **ENTER** to apply the setting.

Parity

1. Select the following from 'PARITY' of **I/F** in **MENU**.
NONE - ODD - EVEN

2. Press **ENTER** to apply the setting.

Number of stop bits

1. Select the following from 'STOP' of **I/F** in **MENU**.
1 BIT/2 BIT
2. Press **ENTER** to apply the setting.

Echo

1. Select the following from 'ECHO' of **I/F** in **MENU**.
ON/OFF
2. Press **ENTER** to apply the setting.

6.3.3 RS-232 Output Data Format

6.3.3 RS-232 Output Data Format

Part of the output data format differs between RS-232 and GPIB.

Data output from RS-232 is as follows:

1. Echo
2. Prompt
3. Measured data
4. Inquiry result (For inquiry command)

Data for each output and output format are as follows:

1. Echo output

The echo output is enabled when its interface setting is set to ON.

Identical received data is output. However, if <^C> (CONTROL C) and <LF> are entered, the following data is output:

(<LF>) + (Prompt) + (Delimiter)

NOTE: The delimiter has been set to <CR>...<LF> and it cannot be changed.

2. Prompt

An execution result for a command received by RS-232 is output by using a prompt.

First, <LF> is output, followed by a prompt and delimiter (<CR><LF>).

The following two prompts are used:

Prompt	Description
=>	A command was correctly received, analyzed, and executed.
?>	An error occurred when a command was received, analyzed, and executed.

3. Measured data output (For the Only mode)

For the Only mode, the measured data is output only if the RS-232 transmission is enabled and the transmission buffer is empty after the measurement was complete.

The delimiter (<CR><LF>) is output for each measured value.

The measurement data output format is the same between GPIB and RS-232.

However, for RS-232, the block delimiter has been set to <CR><LF> and it cannot be changed.

4. Output of an inquiry result executed by an inquiry command

For the output of an inquiry result executed by an inquiry command, <LF> is output first, followed by an inquiry result and delimiter, and finally a prompt at the end.

<LF> + Inquiry result + Delimiter (<CR><LF>) + <LF> + Prompt (=>) + Delimiter (<CR><LF>)

6.4 USB

6.4.1 Overview

A USB (Universal Serial Bus), which is compliant with the USB2.0 standard, is included in this instrument as standard.

By connecting multiple instruments to the USB ports on a personal computer, the functions of the instruments can be set, measurement data can be read from the personal computer, and the automatic measurement system can be set up easily.

NOTE: *This instrument may not operate with all personal computers, hubs, etc.*

6.4.2 USB Specifications

- Standard: Compliance with USB2.0
- Connector: USB B type (female)
- ID: Any number from 1 to 127 can be set as USBid.
- Remote/Local: Function is available.
- Input command: ASCII strings command which sets functions and the query command
- Output format: ASCII strings which describe measurement data and query reply output
- Driver: The USB driver for ADC instruments is used.

6.4.3 Setting Up the USB Port

6.4.3.1 Connecting to a Personal Computer

Connect the connection cable to the USB connector (B type) on the rear panel of this instrument and the USB connector on the personal computer.

Insert the connector completely when connecting.

Use the USB hub when connecting several instruments to a personal computer.

6.4.3.2 USBid Setting

The USBid setting menu can be used when USB is selected for the interface.

1. Enter any address from 1 to 127 in 'USBID' in **I/F** in **MENU**.
2. Press **ENTER** to apply the setting.

6.5 External Trigger Terminal (TRIGGER IN) (7352A)

6.5 External Trigger Terminal (TRIGGER IN) (7352A)

This instrument can be triggered by inputting a negative logic pulse from the external trigger terminal (TRIGGER IN) on the rear panel.

Select EXTERNAL as the trigger source when using this terminal.

Input the TTL level or contact signal as a trigger signal.

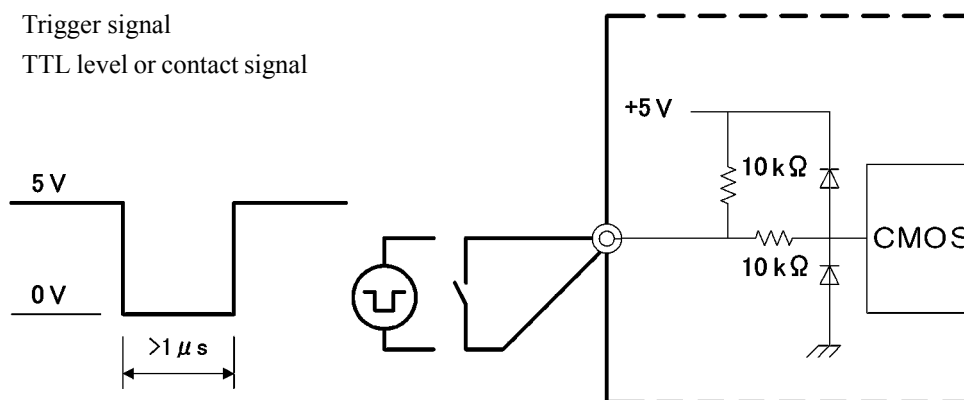


Figure 6-1 Simple Equivalent Circuit of Trigger Input Terminal

6.6 Command Reference

This section describes the command reference of this instrument.

6.6.1 Selecting the Command Language

This instrument can be operated remotely by using the following commands.

The command selection procedure is shown below.

1. Select the **I/F** category from the **MENU** mode and press ∇ to enter the selection layer.
2. Select the LANG parameter by pressing $\leftarrow$ and $\rightarrow$, and press ∇ to enter in the entry layer.
3. Select from ADC, SCPI, and R6452 by pressing $\leftarrow$ and $\rightarrow$, and press **ENTER** to apply the setting.

ADC: Command used in ADC CORPORATION

SCPI: SCPI Command

R6452: Command used in ADC CORPORATION's R6452
For more information on operations, refer to "R6452 Series Digital Multimeter Operation Manual".

Selectable command languages are different depending on the interface.

The relationships between command languages and interfaces are shown below:

	ADC	SCPI	R6452
GPIO	☐	☐	☐
USB	☐	☐	-
RS-232	☐	-	☐

6.6.1 Selecting the Command Language

6.6.1.1 Compatibility

If R6452 is selected as the command language, this instrument executes the R6452 command.

CAUTION:

1. *In the compatibility mode, operations such as measurement timing may be different.*
2. *Commands, which are related to non-existent functions in this instrument such as the memory card control, cannot be used.*

Command enabled only when R6452 is selected	
Trigger	E
Sample mode	Mn
Buzzer	BZn BZ?
Status mask setting	MSn
Device initialization	Z
Device information	IDN?
Status clear.	CS
Status byte output request (Only RS-232)	SB?
Setting channel specification	Dn

Command disabled when R6452 is selected, because ER -010 occurs	
Trigger source selection	TRSn
Status-related	MSR? MSEn MSE? QSR? QSEn QSE? OSR? OSEN OSE? *PSCn *PSC?
Comparator result buzzer	BPn BP?
PASS condition range setting	LOPn LOP? MIPn MIP? HIPn HIP?
Calculation between 2 measurements	HCLn
Output data element specification	ODEn

CAUTION: *When using commands for R6452, ensure the following:
When the 10 A range of the current function is specified, the command can be used but the operation in this instrument is as follow: the 10 A range of the Bch current measurement function is set.*

6.6.2 Output Data Format

1. Measurement data output format (Single display)

With header and output data element

○ ○ ○ ◎ S±DD...DDE±DD, □ □ ... □ △ △
 1 2 3 4 5 6 7

Without header and output data element

±DD...DDE±DD △ △
 3 4 7

2. Measurement data output format (Dual display)

With header and output data element

○ ○ ○ ◎ S±DD...DDE±DD, □ □ ... □, ○ ○ ○ ◎ S±DD...DDE±DD, □ □ ... □ △ △
 1 2 3 4 5 6 5 1 2 3 4 5 6 7

Without header and output data element

±DD...DDE±DD, ±DD...DDE±DD △ △
 3 4 5 3 4 7

3. Measurement data memory output format

With header and output data element

○ ○ ○ ◎ S±DD...DDE±DD, □ □ ... □, ○ ○ ○ ◎ S±DD...DDE±DD, □ □ ... □, ... △ △
 1 2 3 4 5 6 5 1 2 3 4 5 6 7

Without header and output data element

±DD...DDE±DD, ±DD...DDE±DD, ... △ △
 3 4 5 3 4 5 7

- | | | |
|---|-----------|--|
| 1 | ○ ○ ○ ◎ | Main header of 3 characters (○ ○ ○) and sub header (◎) |
| 2 | S | Space |
| 3 | Mantissa | Polarity, decimal point, and 6-digit number |
| 4 | Exponent | E, polarity, and 2-digit number |
| 5 | “,” | Comma |
| 6 | □ □ ... □ | Output data element *1 |
| 7 | △ △ | Block delimiter |

*1: This is not added when R6452 is selected as the command language.

6.6.2 Output Data Format

4. Header

Description		Main header
Measurement function	DC voltage measurement (DCV)	DCV
	AC voltage measurement (ACV)	ACV
	Resistance measurement ($2W\Omega$)	R2W
	DC current measurement (DCI)	DCI
	AC current measurement (ACI)	ACI
	AC voltage (AC+DC coupling) measurement (ACV(AC+DC))	ADV
	AC current (AC+DC coupling) measurement (ACI(AC+DC))	ADI
	Diode measurement (DIODE)	DOD
	Low power $2W\Omega$ (LP- $2W\Omega$)	R2L
	Continuity test (CONT)	RCT
	Frequency measurement (FREQ)	FRQ
	Temperature measurement (TEMP)	TC_
	Bch DC voltage measurement (DCV-Bch)	BDV
	Bch DC current measurement (DCI-Bch)	BDI
	Bch AC current measurement (ACI-Bch)	BAI
	Bch AC current (AC+DC coupling) measurement (ACI(AC+DC)-Bch)	BCI

Description			Sub header	Priority
Error information	Out of range		O	1
	Calculation	Scaling error and OL for the calculation between 2 measurements	O	1
		Err D (dB/dBm calculation error and zero division for the calculation between 2 measurements)	E	2
Calculation result information	Comparator result	HIGH	H	3
		GO	P	3
		LOW	L	3
	Calculation between 2 measurements (Only 2nd display side)		D	3
	MAX data		M	4
	MIN data		I	4
	AVE data		A	4
	dB data		B	5
	dBm data		W	5
	Scaling calculation data		S	6
	NULL calculation data		N	7
Other than those above			—	8

5. Output data element

Calculation between 2 measurements	Measurement for 1st display + Measurement for 2nd display	ADD
	Measurement for 1st display - Measurement for 2nd display	SUB
	Measurement for 1st display * Measurement for 2nd display	MUL
	Measurement for 1st display / Measurement for 2nd display	DIV
	OFF	OFF

6.6.2 Output Data Format

6. Measurement data (when an error occurs)

 $\pm \text{○○○○○○○○} \text{E} \pm \text{○○}$ (5 1/2-digit display)

Overload (OL)	$\pm 9.99999\text{E}+37$
OVER for the scaling calculation result or calculation result between 2 measurements	$\pm 9.99999\text{E}+36$
dB/dBm calculation error or zero division error for the calculation between 2 measurements	$\pm 9.99999\text{E}+35$

DCV-Ach	200 mV	$\pm \text{ddd.dddE-03}$
	2000 mV	$\pm \text{dddd.ddE-03}$
	20 V	$\pm \text{dd.dddE+00}$
	200 V	$\pm \text{ddd.dddE+00}$
	1000 V	$\pm \text{dddd.ddE+00}$
DCV-Bch	200 mV	$\pm \text{ddd.dddE-03}$
	2000 mV	$\pm \text{dddd.ddE-03}$
	20 V	$\pm \text{dd.dddE+00}$
	200 V	$\pm \text{ddd.dddE+00}$
ACV-Ach	200 mV	$\pm \text{ddd.dddE-03}$
	2000 mV	$\pm \text{dddd.ddE-03}$
	20 V	$\pm \text{dd.dddE+00}$
	200 V	$\pm \text{ddd.dddE+00}$
	700 V	$\pm 0\text{ddd.ddE+00}$
ACV (AC+DC)-Ach	200 mV	$\pm \text{ddd.dddE-03}$
	2000 mV	$\pm \text{dddd.ddE-03}$
	20 V	$\pm \text{dd.dddE+00}$
	200 V	$\pm \text{ddd.dddE+00}$
	700 V	$\pm 0\text{ddd.ddE+00}$
2WΩ-Ach	200 Ω	$\pm \text{ddd.dddE+00}$
	2000 Ω	$\pm \text{dddd.ddE+00}$
	20 kΩ	$\pm \text{dd.dddE+03}$
	200 kΩ	$\pm \text{ddd.dddE+03}$
	2000 kΩ	$\pm \text{dddd.ddE+03}$
	20 MΩ	$\pm \text{dd.dddE+06}$
	200 MΩ	$\pm \text{ddd.dddE+06}$

DCI-Ach	2000 nA	$\pm$ dddd.ddE-09
	20 μ A	$\pm$ dd.ddddE-06
	200 μ A	$\pm$ ddd.dddE-06
	2000 μ A	$\pm$ dddd.ddE-06
	20 mA	$\pm$ dd.ddddE-03
	200 mA	$\pm$ ddd.dddE-03
	2000 mA	$\pm$ dddd.ddE-03
DCI-Bch	10 A	$\pm$ dd.ddddE+00
ACI-Ach	200 μ A	$\pm$ ddd.dddE-06
	2000 μ A	$\pm$ dddd.ddE-06
	20 mA	$\pm$ dd.ddddE-03
	200 mA	$\pm$ ddd.dddE-03
	2000 mA	$\pm$ dddd.ddE-03
ACI-Bch	10 A	$\pm$ dd.ddddE+00
ACI (AC+DC)-Ach	200 μ A	$\pm$ ddd.dddE-06
	2000 μ A	$\pm$ dddd.ddE-06
	20 mA	$\pm$ dd.ddddE-03
	200 mA	$\pm$ ddd.dddE-03
	2000 mA	$\pm$ dddd.ddE-03
FREQ-Ach	(<10 Hz)	($\pm$ d.dddddE+00)
	(<100 Hz)	($\pm$ dd.ddddE+00)
	(<1000 Hz)	($\pm$ ddd.dddE+00)
	(<10 kHz)	($\pm$ d.dddddE+03)
	(<100 kHz)	($\pm$ dd.ddddE+03)
	(<1000 kHz)	($\pm$ ddd.dddE+03)
ACI (AC+DC)-Bch	10 A	$\pm$ dd.ddddE+00
LP-2W Ω -Ach	200 Ω	$\pm$ ddd.dddE+00
	2000 Ω	$\pm$ dddd.ddE+00
	20 k Ω	$\pm$ dd.ddddE+03
	200 k Ω	$\pm$ ddd.dddE+03
	2000 k Ω	$\pm$ dddd.ddE+03
	20 M Ω	$\pm$ dd.ddddE+06
DIODE-Ach	1 range	$\pm$ dddd.ddE-03
CONT-Ach	1 range	$\pm$ dddd.ddE+00
TEMP	(K, T)	$\pm$ dddd.ddE+00

(): The Talker format for the frequency measurement function differs depending on the measurement range.

6.6.3 ADC Command Reference

6.6.3 ADC Command Reference

This section describes the ADC command reference of this instrument.

The “Initial value” column shows the status when the instrument is shipped or the *RST command is executed.

The “R6452” column shows a command used when R6452 is selected as the command language.

A “–” shows that there are not any related functions in the R6452.

A blank shows that the same command as the 7352A/E can be used.

CAUTION: *Ensure the following when using the USB interface.*
 When executing the query command, set a wait time of 20 msec after the last command.

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
Measure-ment	Function	F1	DC voltage measurement (DCV-Ach)	○	●	
		F2	AC voltage measurement (ACV-Ach)	○		
		F3	Resistance measurement (2WΩ-Ach)	○		
		F5	DC current measurement (DCI-Ach)	○		
		F6	AC current measurement (ACI-Ach)	○		
		F7	AC voltage (AC+DC coupling) measurement (ACV(AC+DC)-Ach)	○		
		F8	AC current (AC+DC coupling) measurement (ACI(AC+DC)-Ach)	○		
		F12	Bch DC voltage measurement (DCV-Bch)	○	●	
		F13	Diode measurement (DIODE-Ach)	○		
		F20	Low power 2WΩ (LP-2WΩ-Ach)	○		–
		F22	Continuity test (CONT-Ach)	○		
		F35	Bch DC current measurement (DCI-Bch)	○		–
		F36	Bch AC current measurement (ACI-Bch)	○		–
		F37	Bch AC current (AC+DC coupling) measurement (ACI(AC+DC)-Bch)	○		–
		F40	Temperature measurement (TEMP)	○		
		F50	Frequency measurement (FREQ-Ach)	○		

Item	Command	Description	Valid command for the 2nd display setting	Initial value	R6452										
Measurement	Function	F?	Reply: F01 to F03, F05 to F08, F12, F13, F20, F22, F35 to F37, F40, F41, F50	○											
		<table border="1"><tr><td></td><td>Setting command</td><td>Reply</td><td>Description</td></tr><tr><td>Left side display</td><td>[DSP1,]F?</td><td>Fn</td><td>Function for the left side display</td></tr><tr><td>Right side display</td><td>DSP2,F?</td><td>Fn</td><td>Function for the right side display</td></tr></table> A portion enclosed by [] can be omitted.		Setting command	Reply	Description	Left side display	[DSP1,]F?	Fn	Function for the left side display	Right side display	DSP2,F?	Fn	Function for the right side display	
		Setting command	Reply	Description											
	Left side display	[DSP1,]F?	Fn	Function for the left side display											
	Right side display	DSP2,F?	Fn	Function for the right side display											
	Disable function	INHn,m	Sets the disable function. n:1 (DC voltage measurement) to 50 (Frequency measurement) m: 0: Disable state of the function selection 1: Enable state of the function selection		m=0 *1										
		INH?n	Reads the disable state of the function. n:1 (DC voltage measurement) to 50 (Frequency measurement) Reply 0: Disable state of the function selection 1: Enable state of the function selection												
	Trigger	*TRG	Trigger command			*TRG or E									
	Measured data output request (Only RS-232)	MD?													
	Measured data memory	ST0	Store OFF		●	—									
ST1		Store ON			—										
ST?		Reply: ST0 or ST1			—										
IRDn,m		Sets the recall data range. Single measurement mode: n,m:0 to 19999 Dual measurement mode: n,m:0 to 9999 • If the recall data does not exist within the set range, an error occurs. • Parameters cannot be omitted.	○	(0,0)	—										
IRO?		Reads the stored data. When the dual measurement mode is set, the 1st and 2nd displays are read out separately. Reply: Refer to Section 6.3.3 RS-232 Output data format.	○		—										
	IRPO?	Reads the stored data count. Reply: IRPOdddd	○		—										

*1: This is initialized when the instrument is shipped. This cannot be initialized by the *RST command.

6.6.3 ADC Command Reference

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
Measurement	Measured data memory	IRNO?	Reads the stored data range. Reply: IRNO dddddd, dddddd(Single measurement mode) IRNO 0ddddd, 0ddddd(Dual measurement mode) (If the stored data does not exist: IRNO 00000, -0001)	○		—
		ICL	Clears the measurement data memory.			—
Trigger system	Start	INI	Leaves the IDLE state.			—
	Continue	INIC0	CONTINUOUS OFF			—
		INIC1	CONTINUOUS ON		●	—
		INIC?	Reply: INIC0 or INIC1			—
	Trigger abort	ABO	Forcibly moves to the IDLE state.			—
	Trigger source selection	TRS0	IMMEDIATE		●	—
		TRS1	MANUAL			—
		TRS2	EXTERNAL (7352A)			—
		TRS3	BUS			—
		TRS?	Reply: TRS0 to TRS3			—
	Trigger delay	TRDn	n: 0 to 3600 (seconds)	○*2	(0)	—
		TRD?	Reply: TRD±d.ddddE±dd (The position of the decimal point is different depending on the setting.)	○*2		—
	Each trigger delay for the right and left displays	TDE0	OFF		●	—
		TDE1	ON			—
		TDE?	Reply: TDE0 to TDE1			—
	Trigger count	TRNn	n: 1 to 50000 (counts)		(1)	—
		TRN?	Reply: TRNddddd			—
	Sampling count (by one trigger)	SPNn	n: 1 to 16000 (counts)		(1)	—
		SPN?	Reply: SPNddddd			—

*2: This command is valid for the 2nd display setting only when TDE1 is set.

6.6.3 ADC Command Reference

Item	Com-mand	Description										Valid command for the 2nd display setting	Initial value	R6452
Mea-sure-ment condi-tion	Measurement range (Ach)		DCV	ACV	ACV (AC+DC)	2WΩ	DCI	ACI	ACI (AC+DC)	LP-2WΩ	FREQ			
		R0	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	-	○	●	
		R1	-	-	-	-	2000 nA	-	-	-	-	○		
		R2	-	-	-	-	20 μA	-	-	-	-	○		
		R3	200 mV	200 mV	200 mV	200 Ω	200 μA	200 μA	200 μA	200 Ω	(200 mV)	○		
		R4	2000 mV	2000 mV	2000 mV	2000 Ω	2000 μA	2000 μA	2000 μA	2000 Ω	(2000 mV)	○		
		R5	20 V	20 V	20 V	20 kΩ	20 mA	20 mA	20 mA	20 kΩ	(20 V)	○		
		R6	200 V	200 V	200 V	200 kΩ	200 mA	200 mA	200 mA	200 kΩ	(200 V)	○		
		R7	1000 V	700 V	700 V	2000 kΩ	2000 mA	2000 mA	2000 mA	2000 kΩ	(700 V)	○		
		R8	-	-	-	20 MΩ	-	-	-	20 MΩ	-	○		
		R9	-	-	-	200 MΩ	-	-	-	-	-	○		
		R?	Reply: One of R0 and R1 to R9									○		-
	Measurement range (Bch)		DCV-Bch				DCI-Bch	ACI-Bch	ACI (AC+DC)-Bch					
		R0	AUTO				-	-	-			○	●	
		R1	-				-	-	-					
		R2	-				-	-	-					
		R3	200 mV				-	-	-			○		
		R4	2000 mV				-	-	-			○		
		R5	20 V				-	-	-			○		
		R6	200 V				-	-	-			○		
		R7	-				-	-	-					
		R8	-				10 A	10 A	10 A			○		
		R9	-				-	-	-					
		R?	Reply: DCV-Bch: One of R0 and R3 to R6 DCI-Bch/ACI-Bch/ACI(AC+DC)-Bch: R8									○		-
	Range Fix	RX	Switches from the AUTO to MANUAL range.											
	Sampling rate	PR1	FAST											
		PR2	MED											
		PR3	SLOW1										●	
		PR4	SLOW2 (The same as PR3 for the frequency measurement)											-
		PR?	Reply: One of PR1 to PR4											-

6.6.3 ADC Command Reference

Item		Com- mand	Description	Valid command for the 2nd display setting	Initial value	R6452
Mea- sure- ment condi- tion	Number of display digits	RE3	3 1/2-digit display	○		
		RE4	4 1/2-digit display	○		
		RE5	5 1/2-digit display	○	●	
		RE?	Reply: One of RE3 to RE5	○		—
	Auto-zero	AZ0	OFF			—
		AZ1	ON		●	—
		AZ2	ONCE (Auto-zero is set to OFF after this command is executed.)			—
		AZ?	Reply: AZ0 or AZ1			—
	Continuity threshold constant	KOMn	n:1 to 1000 (Ω) setting resolution: 1 Ω		(10)	—
		KOM?	Reply: KOMdddd			—
	Temperature sensor	TCR0	K-type thermocouple		●	—
		TCR1	T-type thermocouple			—
		TCR?	Reply: TCR0 or TCR1			—

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
Calculation	NULL calculation	NL0	OFF	☐	●	
		NL1	ON	☐		
		NL?	Reply: NL0 or NL1	☐		—
		KNLn	NULL constant setting n: -999999.E+6 to +999999.E+6 Setting resolution: 0.00001E-9 Note: The setting is disabled when the NULL calculation is set to OFF.	☐	(0)	
		KNL?	Reply: KNL±d.dddddE±dd *3	☐		—
	Smoothing calculation	SM0	OFF	☐	●	
		SM1	ON	☐		
		SM?	Reply: SM0 or SM1	☐		—
		TIn	Smoothing count n: 2 to 100 (counts)	☐	(10)	
		TI?	Reply: TIddd	☐		—
	Scaling calculation	SC0	OFF	☐	●	
		SC1	ON	☐		
		SC?	Reply: SC0 or SC1	☐		—
		KAn	Constant A (Zero cannot be set). n: -999999.E+6 to +999999.E+6 Setting resolution: 0.00001E-9	☐	(1)	
		KBn	Constant B n: -999999.E+6 to +999999.E+6 Setting resolution: 0.00001E-9	☐	(0)	
		KCn	Constant C n: -999999.E+6 to +999999.E+6 Setting resolution: 0.00001E-9	☐	(1)	
		KAM	Sets Constant A to the measurement value.	☐		
		KBM	Sets Constant B to the measurement value.	☐		
		KCM	Sets Constant C to the measurement value.	☐		
		KA?	Reply: KA±d.dddddE±dd *3	☐		—
		KB?	Reply: KB±d.dddddE±dd *3	☐		—
		KC?	Reply: KC±d.dddddE±dd *3	☐		—

*3: The position of the decimal point in the reply is fixed.

6.6.3 ADC Command Reference

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
Calculation	dB/dBm calculation	DB0	dB calculation OFF	○	●	
		DB1	dB calculation ON For the voltage and current measurement functions, ON can be set.	○		
		DB2	dBm calculation ON For the voltage measurement function, ON can be set.	○		
		DB?	Reply: One of DB0 to DB2	○		—
		KDn	Constant D n:0.00001E-9 to 999999.E+6	○	(1)	
		KDM	Sets the measurement value to constant D.	○		
		KD?	Reply: KD±d.dddddE±dd *3	○		—
	MAX/MIN calculation	MN0	MAX/MIN calculation OFF	○	●	MIN0 MIN1 MIN2
		MN1	MAX/MIN calculation ON	○		
		MN?	Reply: MN0 or MN1	○		—
		MAX?	Reads the MAX value. *3, *4 Reply: ○ ○ ○ M ±d.dddddE±dd	○		—
		MIN?	Reads the MIN value. *3, *4 Reply: ○ ○ ○ I ±d.dddddE±dd	○		—
		AVE?	Reads the AVE value. *3, *4 Reply: ○ ○ ○ A ±d.dddddE±dd	○		—
		AVN?	Reads the measurement count. *3 Reply: AVN±d.dddddE±dd	○		—
	Comparator operation	CO0	OFF		●	
		CO1	ON (If CO1 is set, the single measurement is set.)			
		CO?	Reply: CO0 or CO1			—
		HIIn	Constant HIGH		(0)	
		LOn	Constant LOW n:-999999.E+6 to +999999.E+6 Setting resolution:0.00001E-9		(0)	
		HIM	Sets the measurement value to constant HIGH.			
		LOM	Sets the measurement value to constant LOW.			
		HI?	Reply: HI±d.dddddE±dd *3			—
		LO?	Reply: LO±d.dddddE±dd *3			—

*3: The position of the decimal point in the reply is fixed.

*4: ○ ○ ○: Header of the Talker format for the measurement data

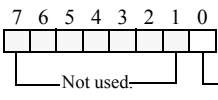
Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
Calculation	PASS condition range setting	LOP0	Calculation result LO is not specified as the PASS condition.		●	—
		LOP1	Calculation result LO is specified as the PASS condition.			—
		LOP?	Reply: LOP0 or LOP1			—
		MIP0	Calculation result GO is not specified as the PASS condition.			—
		MIP1	Calculation result GO is specified as the PASS condition.		●	—
		MIP?	Reply: MIP0 or MIP1			—
		HIP0	Calculation result HI is not specified as the PASS condition.		●	—
		HIP1	Calculation result HI is specified as the PASS condition.			—
		HIP?	Reply: HIP0 or HIP1			—
	Statistical calculation	SIRDn, m	Sets the statistical calculation range and performs the calculation. Single measurement mode: n, m: 0 to 19999 Dual measurement mode: n, m: 0 to 9999 NOTE: 1. If the recall data does not exist within the set range, an error occurs. 2. Parameters cannot be omitted.	○	(0,-1)	—
		SIRD?	Reads the statistical calculation range. Reply: SIRDdddd,dddd (Initial state: SIRD0000,-0001)	○		—
			Reads the statistical calculation result.			
		SCNT?	Reads the number of samples in the measurement memory. Reply: SCNT±d.dddddE±dd *3	○		—
		SMAx?	Reads the maximum value in the measurement memory. Reply: SMAx±d.dddddE±dd *3	○		—
		SMIN?	Reads the minimum value in the measurement memory. Reply: SMIN±d.dddddE±dd *3	○		—
		SAVE?	Reads the average value in the measurement memory. Reply: SAVE±d.dddddE±dd *3	○		—
		SSIG?	Reads the standard deviation value in the measurement memory. Reply: SSIG±d.dddddE±dd *3 However, if the number of samples is 1 or less, SSIG+9.99999E+11 is output.	○		—
		SPTP?	Reads MAX-MIN in the measurement memory. Reply: SPTP±d.dddddE±dd *3	○		—

*3: The position of the decimal point in the reply is fixed.

6.6.3 ADC Command Reference

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
Calculation	Calculation between 2 measurements	MCLn	Calculation between 2 measurements Performs four arithmetic operations between measurement values for the right and left side displays. n: 0: OFF 1: Measurement for the left side display + Measurement for the right side display 2: Measurement for the left side display - Measurement for the right side display 3: Measurement for the left side display * Measurement for the right side display 4: Measurement for the left side display / Measurement for the right side display NOTE: Valid only when the 2nd display is ON.		0	—
		MCL?	Reply: MCLd			—
System	2nd display	DE0	2nd display OFF			—
		DE1	2nd display ON		●	
		DE?	Reply: DE0 or DE1			—
	1st/2nd display selection *5	DSP1	Selects the 1st display.		●	—
		DSP2	Selects the 2nd display. NOTE: Valid in a line where this command is set. The 1st display is selected in a line where this command is omitted.			—
	Output data	SD0	Remotely output the 1st and 2nd displays.		●	
		SD1	Remotely output the 1st display only.			
		SD2	Remotely output the 2nd display only. NOTE: When the 2nd display is OFF, this command is changed to SD0.			
		SD?	Reply: One of SD0 to SD2			—
	Buzzer	BZ0	OFF		●	BZ0 BZ1 BZ2 BZ3 BZ4
		BZ1	ON			
		BZ?	Reply: BZ0 or BZ1			—

*5: This command cannot be used when the calibration mode is set to ON.

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
System	Comparator result buzzer	BP0	OFF		●	—
		BP1	The buzzer sounds when the comparator operation result is FAIL.			—
		BP2	The buzzer sounds when the comparator operation result is PASS.			—
		BP?	Reply: One of BP0 to BP2			—
	Measurement data display	DS0	OFF			
		DS1	ON		●	
		DS?	Reply: DS0 or DS1			—
	Power supply frequency	LF?	Reply: LF0: 50 Hz LF1: 60 Hz			—
	Device initialization	*RST	Parameter initialization			*RST or Z
		C	Device clear			
	Device information	*IDN?	Reply: ADC Corp.,7352x,nnnnnnnnn,mmm nnnnnnnnn:Serial No. mmm:Revision No.			*IDN? or IDN?
	Header	H0	Header OFF			
		H1	Header ON		● *1	
		H?	Reply: H0 or H1			—
	Output data element specification	ODEn			0	—
		ODE?	Reply: ODEddd			—
	Block delimiter *6	DL0	CR/LF+EOI		●	
		DL1	LF			
		DL2	EOI			
		DL?	Reply: One of DL0 to DL2			—
	SRQ	S1	Prohibition of SRQ transmitting			
		S0	Permission for SRQ transmitting		●	
		S?	Reply: S0 or S1			—

*1: This is initialized when the instrument is shipped. This cannot be initialized by the *RST command.

*6: EOI is a GPIB function. EOI is not output in USB. The setting cannot be changed in an RS-232 interface.

6.6.3 ADC Command Reference

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
System	Status	*CLS	Clears each status byte.			*CLS or CS
		*STB?	Reads the status byte register Reply: ddd			
		*SREn	Sets the service request enable register (SRER). n: 0 to 255 (However, bit6 cannot be set.)			—
		*SRE?	Reply: ddd			—
		*ESR?	Reads the standard event status register (SESR). Reply: ddd			—
		*ESEn	Sets the standard event status enable register (SESER). n:0 to 255			—
		*ESE?	Reply: ddd			—
		MSR?	Reads the measurement event register (MER). Reply: ddddd	○		—
		MSEn	Sets the measurement event enable register (MEER). n:0 to 65535	○		—
		MSE?	Reply: ddddd	○		—
		QSR?	Reads the questionable event register (QER). Reply: ddddd	○		—
		QSEn	Sets the questionable event enable register (QEER). n:0 to 65535	○		—
		QSE?	Reply: ddddd	○		—
		OSR?	Reads the operation event register (OER). Reply: ddddd			—
		OSEn	Sets the operation event enable register (OEER). n:0 to 65535			—
		OSE?	Reply: ddddd			—
		*PSCn	n:±32767 If the power is turned on when “n” is set to any value except zero, the following registers are cleared. Even if the power is turned on when “n” is set to zero, the following registers are not cleared. • Service request enable register • Standard event status enable register			—
		*PSC?	Reply: 0 or 1 (When any value except for zero is set.)			—
	Option	*OPT?	Reads the option information. Reply: 0 No option			—

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
System	Operation complete	*OPC	Sets the operation complete bit (bit0) of the Standard Event Status Register after all operations are complete.			—
		*OPC?	Reply: 1 (After all operations are complete.)			—
		*WAI	Waits for the completion of all operations. (GPIB only)			—
	Self-Test	*TST?	Performs the self- test and reads the result. (It takes time to perform the self-test. Read the result after the *WAI command is executed.) Reply 0 : Pass 1 : Fail			—
	Calibration	CAL0	Sets the calibration mode to OFF. (Writes the calibration factor when leaving the calibration mode.)		●	—
		CAL1	Sets the calibration mode to ON.			—
		CAL?	Reply: CAL0 or CAL1			—
		XOUT	Cancels (sets to OFF) the calibration mode. (Does not write the calibration factor.)			—
		PCn	Enters the STD value. (Displayed count value) n:0 to ±999999			—
		XDTn	Enters the STD value. (Displayed value) n: Displayed value on STD			—
		CMNT "str"	Sets the calibration information. Enclose the character strings by using (") or ('). Up to 50 characters (one byte alphanumeric characters) can be stored.			—
		CMNT?	Reply: CMNT"xxxx · · · xxx" Up to 50 characters			—
	Setting parameter	*SAVn	n:0 to 3 Saves the setting parameters in area [n] of the nonvolatile memory.			—
		SINI	Sets the values when shipping to all area from [0] to [3].			—
		*RCLn	n:0 to 3 Loads the setting parameters from area [n] of the non-volatile memory.			—
		RINI	Loads the values when shipping as the setting parameters.			—

Item		Command	Description	Valid command for the 2nd display setting	Initial value	R6452
System	Error record reading	ERR?	Reads the error details. Reply: $\pm$ddd,"xxxxxxxxxx" • Up to 20 error queues can be stored. The error is output according to the FIFO system. • If more than 20 errors occur, the last error is replaced with -350, "Queue overflow". Any errors after this are not saved. • If no error occurs, +000, "No error" is returned.			—

6.6.4 SCPI Command Reference

This section describes the syntax to explain each SCPI command.

The syntax, which is used when the command is sent from the external controller to this instrument, is described. The syntax consists of a command section and a parameter section. The command and parameter sections are delimited by a space.

Commas (,) are used to delimit two or more parameters. The ellipsis (...) displayed between commas shows that the parameters are omitted at that position.

For example, the description "<numeric value 1>,...,<numeric value 4>" shows "<numeric value 1>,<numeric value 2>,<numeric value 3>,<numeric value 4>".

The part, which is written by lowercase alphabetical characters in the syntax, shows that it can be omitted.

For example, ":CALibration:ABORt" can be abbreviated to ":CAL:ABOR".

The symbols used in the syntax are defined as follows:

< >	Shows a parameter required for sending a command
[]	Shows that the command is optional. It can be omitted.
{ }	Shows that only one item need be selected from multiple items.
	Written in curly brackets {..} and used as a delimiter for multiple items.

When the parameter is a numeric type or a character (string) type, it is enclosed in angle brackets (<>).

If the parameter is optional, it is enclosed in braces { }.

Optional items are delimited by (|) and the multiple descriptions in one optional item are delimited by (,).

Example 1: {ON,1|OFF,0}

This example shows that "ON" has the same meaning of "1", "OFF" has the same meaning of "0", and the following four options: ON, 1, OFF, and 0 can be selected.

If the range of the value to be entered is restricted, the range is shown as [Minimum value| Default value| Maximum value] or [Minimum value| Maximum value].

In this manual, parameter types are described in the following formats:

<int_d>	Shows a numeric value that can be entered in each format of NR1, NR2, and NR3 and is rounded to an integer in this instrument. "d" shows the number of digits.
<real_d>	Shows a numeric value that can be entered in each format of NR1, NR2, and NR3 and is rounded to a real number in this instrument. "d" shows the number of decimals.
<str>	Shows a character string or alphanumeric symbols enclosed in quotation (') or double quotation (") marks.
<func>	Shows the same character string as the query reply to the [:SENSe]:FUNctIon command.

Example 2: {MAXimum|MINimum|MEASurement}

If MAXimum is selected, the maximum value is set. If MINimum is selected, the minimum value is set.

If MEASurement is selected, the measurement is performed and the measured value is set.

6.6.4 SCPI Command Reference

NOTE: When using the SCPI commands via USB, note the following point.
 After sending a setting command, send a query command and receive its response.
 Sending more than one setting command in succession may cause unintended behavior or an error to occur.
 As query command for setting commands, error check by ":SYSTem:ERRor?" and completion check by "*OPC?" are generally used.

1. Measurement function and trigger system relevant command

Item		SCPI command	Description	Initial value
Measurement	Measurement function	[:SENSe]:FUNction "<function>" [:(@<ch>)]	"<function>":	
			"VOLTage:DC" DC voltage measurement (DCV)	●
			"VOLTage:AC" AC voltage measurement (ACV)	
			"RESistance" Resistance measurement (2WΩ)	
			"CURRent:DC" DC current measurement (DCI)	
			"CURRent:AC" AC current measurement (ACI)	
			"VOLTage:ACDC" AC voltage (AC+DC coupling) measurement (ACV(AC+DC))	
			"CURRent:ACDC" AC current (AC+DC coupling) measurement (ACI(AC+DC))	
			"DIODE" Diode measurement (DIODE)	
			"RESistance:LPOWer" Low power resistance measurement (LP-2WΩ)	
			"CONTinuity" Continuity test (CONT)	
			"FREQuency" Frequency measurement (FREQ)	
			"VOLTage:BDC" Bch DCV measurement (DCV-Bch)	●
			"CURRent:BDC" Bch DCI measurement (DCI-Bch)	
			"CURRent:BAC" Bch ACI measurement (ACI-Bch)	
			"CURRent:BACDC" Bch ACI(AC+DC) measurement (ACI(AC+DC)-Bch)	
			"TEMPerature" Temperature measurement (TEMP)	
			<ch> 1: 1st display 2: 2nd display	
		[:SENSe]:FUNction? [:(@<ch>)]	Reply: "VOLT:DC", "VOLT:AC", "RES", "CURR:DC", "CURR:AC", "VOLT:ACDC", "CURR:ACDC", "DIOD", "RES:LPOW", "CONT", "FREQ", "VOLT:BDC", "CURR:BAC", "CURR:BACD", "TEMP"	

Item		SCPI command	Description	Initial value
Measurement	Measurement function	:CONFigure:<function> [{<real> MAXimum MINimum} [,(@<ch>)] *1	Switches to the specified measurement function. <real>: Range setting ($0 \leq \text{Setting value} \leq \text{Maximum value}$ for the measurement range) (For the frequency measurement, the maximum value is 750 V.) This setting is not allowed for the continuity, diode, and temperature measurements.	
		:CONFigure? [(@<ch>)]	Reply: "Function $\pm d.dddddE\pm dd$ " " $\pm d.dddddE\pm dd$ " shows a measurement range.	
	Invalid function	:INHibit:FUNCTION:<function> {ON,1 OFF,0}	Sets the invalid function.	
		:INHibit:FUNCTION:<function>?	Reads a status in which the invalid function is set. Reply: 1 or 0 (1: Invalid, 0: Valid)	
	Trigger	*TRG	Trigger command	
	Measurement operation	:MEASure?	Sets the DCV measurement function and reads the measured data. Reply: $\pm dddd.ddE\pm dd$ For more information, refer to 6.6.3 RS232 Output data format	
		:MEASure:<function>? *1	Performs the measurement operation once and reads the measured value. For more information, refer to 6.6.3 RS232 Output data format.	
		:READ?	Initializes the trigger system and reads the measured data. For more information, refer to 6.6.3 RS232 Output data format.	
		:FETCh?	Only reads the measured data. Reply: $\pm dddd.ddE\pm dd$ For more information, refer to 6.6.3 RS232 Output data format.	
	Measurement data memory store ON/OFF	:TRACe:STATe {ON,1 OFF,0}	ON, 1: ON OFF, 0: OFF	●
		:TRACe:STATe?	Reply: 1 or 0	
	Measurement data memory reading	:TRACe:NUMBer {<real> MAXimum MINimum}, {<real> MAXimum MINimum} [,(@<ch>)]	Sets the recall data range. <real>: 0 to 19999	
		:TRACe:DATA? [(@<ch>)]	Reads the stored data. (xxx shows a function.) xxx_ $\pm d.dddddE\pm dd$,...	
		:TRACe:DATA:NUMBer? [(@<ch>)]	Reads the stored data range. Reply: dddd,dddd	0, -1
		:TRACe:DATA:POINts? [(@<ch>)]	Reads the stored data count. Reply: dddd	0
	Measurement memory initialization	:TRACe:CLEar		

*1: <function>: Specify a function of "measurement function", but do not enclose in double quotes.

6.6.4 SCPI Command Reference

Item		SCPI command	Description	Initial value
Trigger system	Start	:INITiate	Exits the IDLE state and moves to the trigger waiting state.	
	Continue	:INITiate:CONTinuous	ON, 1: CONTINUOUS ON	
		{ON,1 OFF,0}	OFF, 0: CONTINUOUS OFF	●
		:INITiate:CONTinuous?	Reply: 1 or 0	
	Trigger abort	:ABORt	Forcibly moves to the IDLE state.	
	Trigger source selection	:TRIGger:SOURce	IMMediate	●
		{IMMediate MANual EXTernal BUS}	MANual	
			EXTernal (7352A)	
			BUS	
		:TRIGger:SOURce?	Reply: IMM, MAN, EXT, or BUS	
	Trigger delay	:TRIGger:DELay	<real>: 0 to 3600 (seconds)	(0)
		{<real> MAXimum MINimum} [,(@<ch>)] *2		
		:TRIGger:DELay?	Reply: +d.ddddE±dd	
		[{MAXimum MINimum}] [,(@<ch>)] *2		
	Each trigger delay for the right and left displays	:TRIGger:DELay:EACH	ON, 1: Each trigger delay ON	
		{ON,1 OFF,0}	OFF, 0: Each trigger delay OFF	●
		:TRIGger:DELay:EACH?	Reply: 1 or 0	
	Trigger count	:TRIGger:COUNt	<real>: 1 to 50000	
		{<real> MAXimum MINimum}		
		:TRIGger:COUNt?	Reply: ddddd	
		[{MAXimum MINimum}]		
	Sampling count (1 trigger)	:SAMPle:COUNt	<int>: 1 to 16000	
		{<int> MAXimum MINimum}		
		:SAMPle:COUNt?	Reply: ddddd	
		[{MAXimum MINimum}]		

*2: [,(@<ch>)] in the trigger delay can only be set when each trigger delay for the right and left displays is set to ON.

2. Measurement condition command

Item	SCPI command	Description	Initial value
Measure- ment condi- tion	Measurement range	[[:SENSe]:<function>:RANGe :AUTO {ON,1 OFF,0}[,(@<ch>)]	Auto-range setting
		ON, 1: Auto-range ON	●
		OFF, 0: Auto-range OFF This setting is not allowed for the continuity, diode, fre- quency, and temperature measurements.	
		[[:SENSe]:<function>:RANGe: AUTO?[(@<ch>)]	Reply: 1 or 0
		[[:SENSe]:<function>:RANGe {<real> MAXimum MINimum} [,(@<ch>)] *1, *3	Fixed range setting <real>: $0 \leq \text{Setting value} \leq \text{Maximum value for the measurement range}$
		[[:SENSe]:<function>:RANGe? {MAXimum MINimum} [,(@<ch>)] *1, *3	Reply: +d.dddddE $\pm$ dd The reply is the maximum value for the specified range.
	For more information on the range configuration and settable maximum value, see the table on the next page.		
	Sampling rate	[[:SENSe]:<function>:SRATe {FAST MED SLOW SSLow} *1	FAST: FAST
			MED: MED
			SLOW: SLOW1
			SSLow: SLOW2
			The setting is applied to all functions.
		[[:SENSe]:<function>:SRATe? *1	Reply: FAST, MED, SLOW, or SSL
	Number of dis- play digits	[[:SENSe]:<function>:DIGits {3 4 5 MAXimum MINimum} [,(@<ch>)] *1	3: 3 1/2-digit display
			4: 4 1/2-digit display
			5: 5 1/2-digit display
			The setting is applied to all functions.
		[[:SENSe]:<function>:DIGits? {MAXimum MINimum} [,(@<ch>)] *1, *3	Reply: 3, 4, or 5
	Auto-zero	[[:SENSe]:ZERO:AUTO {ON,1 OFF,0 ONCE}	ON, 1: ON
			OFF, 0: OFF
			ONCE: (Auto-zero is set to OFF after executing once.)
		[[:SENSe]:ZERO:AUTO?	Reply: 1 or 0
	Continuity threshold constant	[[:SENSe]:CONTInuity:THReshold {<real> MAXimum MINimum}	<real>: 1 to 1000 (Ω) setting resolution: 1 Ω
		[[:SENSe]:CONTInuity:THReshold? {MAXimum MINimum}	Reply: dddd

*1: <function>: Specify a function of “measurement function”, but do not enclose in double quotes.

*3: If omitting [{MAXimum|MINimum}], do not set “,” followed by (@<ch>).

6.6.4 SCPI Command Reference

Item		SCPI command	Description	Initial value
Measurement condition	Temperature sensor	[:SENSe]:TEMPerature:TCouple:TYPE {K T}	K: K-type thermocouple	●
			T: T-type thermocouple	
		[:SENSe]:TEMPerature:TCouple:TYPE?	Reply: K or T	

Range configuration and settable maximum value

Ach									Bch				Measurement range notation
DCV	ACV	ACV (AC+DC)	2WΩ	DCI	ACI	ACI (AC+DC)	LP-2WΩ	Freq	DCV	DCI	ACI	ACI (AC+DC)	
				2000 nA									+1.99999E-06
				20 μA									+1.99999E-05
				200 μA	200 μA	200 μA							+1.99999E-04
				2000 μA	2000 μA	2000 μA							+1.99999E-03
				20 mA	20 mA	20 mA							+1.99999E-02
200 mV	200 mV	200 mV		200 mA	200 mA	200 mA		(200 mV)	200 mV				+1.99999E-01
2000 mV	2000 mV	2000 mV		2000 mA	2000 mA	2000 mA		(2000 mV)	2000 mV				+1.99999E-00
-	-	-						-	-	10 A	10 A	10 A	+1.09999E+01
20 V	20 V	20 V						(20V)	20 V				+1.99999E+01
200 V	200 V	200 V	200 Ω				200 Ω	(200V)	200 V				+1.99999E+02
-	700 V	700 V	-				-	(700V)					+7.49999E+02
1000 V			-				-						+1.09999E+03
			2000 Ω				2000 Ω						+1.99999E+03
			20 kΩ				20 kΩ						+1.99999E+04
			200 kΩ				200 kΩ						+1.99999E+05
			2000 kΩ				2000 kΩ						+1.99999E+06
			20 MΩ				20 MΩ						+1.99999E+07
			200 MΩ										+1.99999E+08

NOTE: The range setting is not allowed for the continuity, diode, and temperature measurements. If the range setting is executed, an error occurs.

3. Measurement condition command

Item		SCPI command	Description	Initial value
Calculation	NULL calculation	[:SENSe]:<function>:REfERENCE:STATe {ON,1 OFF,0}[,(@<ch>)] *1	ON, 1: ON	
			OFF, 0: OFF	●
		[:SENSe]:<function>:REfERENCE:STATe? [(@<ch>)] *1	Reply: 1 or 0	
		[:SENSe]:<function>:REfERENCE {<real> MAXimum MINimum}[,(@<ch>)] *1	NULL constant setting <real>: -999999.E+6 to +999999.E+6 Setting resolution: 0.00001E-9 Note: The setting is disabled when the NULL calculation is set to OFF.	(0)
	Smoothing calculation	[:SENSe]:<function>:REfERENCE? [{MAXimum MINimum}][,(@<ch>)] *3	Reply: $\pm d.dddddE \pm dd$	
		:CALCulate:SMOothing {ON,1 OFF,0}[,(@<ch>)]	ON, 1: ON	
			OFF, 0: OFF	●
		:CALCulate:SMOothing? [(@<ch>)]	Reply: 1 or 0	
		:CALCulate:SMOothing:POINts {<int> MAXimum MINimum}[,(@<ch>)]	Smoothing count <int>: 2 to 100 (counts)	(10)
		:CALCulate:SMOothing:POINts? [(@<ch>)]	Reply: ddd	
	Scaling calculation	:CALCulate:SCALing {ON,1 OFF,0}[,(@<ch>)]	ON, 1: ON	
			OFF, 0: OFF	●
		:CALCulate:SCALing? [(@<ch>)]	Reply: 1 or 0	
		:CALCulate:SCALing:A {<real> MAXimum MINimum MEASurement}[,(@<ch>)]	Constant A (Zero cannot be set) <real>: 0.00001E-9 to +999999.E+6	(1)
		:CALCulate:SCALing:B {<real> MAXimum MINimum MEASurement}[,(@<ch>)]	Constant B <real>: 0.00001E-9 to +999999.E+6	(0)
		:CALCulate:SCALing:C {<real> MAXimum MINimum MEASurement}[,(@<ch>)]	Constant C <real>: 0.00001E-9 to +999999.E+6	(1)
		:CALCulate:SCALing:A? [{MAXimum MINimum}][,(@<ch>)] *3	Reply: $\pm d.dddddE \pm dd$	
		:CALCulate:SCALing:B? [{MAXimum MINimum}][,(@<ch>)] *3		
		:CALCulate:SCALing:C? [{MAXimum MINimum}][,(@<ch>)] *3		

*1: <function>: Specify a function of "measurement function", but do not enclose in double quotes.

*3: If omitting [{MAXimum|MINimum}], do not set ",", followed by (@<ch>).

6.6.4 SCPI Command Reference

Item		SCPI command	Description	Initial value
Calculation	dB/dBm calculation	:CALCulate:DB {DB DBM OFF} [,(@<ch>)]	DB: dB calculation ON For the voltage and current measurement functions, ON can be set.	
			DBM: dBm calculation ON For the voltage measurement function, ON can be set.	
			OFF: dB/dBm calculation OFF	●
		:CALCulate:DB?[(@<ch>)]	Reply: DB, DBM, or OFF	
		:CALCulate:DB:D {<real> MAXimum MINimum MEASurement} [,(@<ch>)]	Constant D <real>: 0.00001E-9 to 999999.E+6	(1)
		:CALCulate:DB:D? [{MAXimum MINimum}] [,(@<ch>)] *3	Reply: $\pm d.dddddE \pm dd$	
	MAX•MIN calculation	:CALCulate:AVERage {ON,1 OFF,0} [,(@<ch>)]	ON, 1: ON OFF, 0: OFF	●
		:CALCulate:AVERage?[(@<ch>)]	Reply: 1 or 0	
		:CALCulate:AVERage:MAXimum? [(@<ch>)]	MAX value reading Reply: $\pm d.dddddE \pm dd$	
		:CALCulate:AVERage:MINimum? [(@<ch>)]	MIN value reading Reply: $\pm d.dddddE \pm dd$	
		:CALCulate:AVERage:AVERage? [(@<ch>)]	AVE value reading Reply: $\pm d.dddddE \pm dd$	
		:CALCulate:AVERage:COUNt?[(@<ch>)]	Measurement count reading Reply: $\pm d.dddddE \pm dd$	
	Comparator calculation	:CALCulate:LIMit {ON,1 OFF,0}	ON, 1: ON OFF, 0: OFF	
		:CALCulate:LIMit?	Reply: 1 or 0	
		:CALCulate:LIMit:UPPer {<real> MAXimum MINimum MEASurement}	HIGH constant <real>: 0.00001E-9 to 999999.E+6 Setting resolution:0.000001E-9	
		:CALCulate:LIMit:LOWer {<real> MAXimum MINimum MEASurement}	LOW constant <real>: 0.00001E-9 to 999999.E+6 Setting resolution:0.000001E-9	
		:CALCulate:LIMit:UPPer? [{MAXimum MINimum}]	Reply: $\pm d.dddddE \pm dd$	
		:CALCulate:LIMit:LOWer? [{MAXimum MINimum}]		

*3: If omitting [{MAXimum|MINimum}], do not set “,” followed by (@<ch>).

6.6.4 SCPI Command Reference

Item		SCPI command	Description	Initial value
Calculation	PASS condition range setting	:CALCulate:LIMit:PASS:LOWer {ON,1 OFF,0}	ON, 1: Calculation result LOW is specified as the PASS condition. OFF, 0: Calculation result LOW is not specified as the PASS condition.	
		:CALCulate:LIMit:PASS:LOWer?	Reply: 1 or 0	
		:CALCulate:LIMit:PASS:MED {ON,1 OFF,0}	ON, 1: Calculation result GO is specified as the PASS condition. OFF, 0: Calculation result GO is not specified as the PASS condition.	
		:CALCulate:LIMit:PASS:MED?	Reply: 1 or 0	
		:CALCulate:LIMit:PASS:UPPer {ON,1 OFF,0}	ON, 1: Calculation result HI is specified as the PASS condition. OFF, 0: Calculation result HI is not specified as the PASS condition.	
		:CALCulate:LIMit:PASS:UPPer?	Reply: 1 or 0	
	Statistical calculation	:CALCulate:STATistics <int>,<int>[,(@<ch>)]	Specifies the calculation range and performs the calculation. <int>: 0 to 19999	0, -1
		:CALCulate:STATistics? [(@<ch>)]	Reply: dddd,dddd	
		:CALCulate:STATistics:COUNt? [(@<ch>)]	Reads the number of samples. Reply: +d.dddddE+dd	
		:CALCulate:STATistics:MAXimum? [(@<ch>)]	Reads the maximum value. Reply: ±d.dddddE+dd	
		:CALCulate:STATistics:MINimum? [(@<ch>)]	Reads the minimum value. Reply: ±d.dddddE+dd	
		:CALCulate:STATistics:AVERage? [(@<ch>)]	Reads the average value. Reply: ±d.dddddE+dd	
		:CALCulate:STATistics:DEVIation? [(@<ch>)]	Reads the standard deviation. Reply: ±d.dddddE+dd	
		:CALCulate:STATistics:MAXMin? [(@<ch>)]	Reads the “maximum value - minimum value.” Reply: ±d.dddddE+dd	
	Calculation between 2 measurements	:CALCulate:DDISplay {OFF ADDition SUBTraction MULTiply DIVision}	Performs the calculation between 2 measurements. OFF: Calculation OFF ADDition: Measurement for 1st display + Measurement for 2nd display SUBTraction: Measurement for 1st display - Measurement for 2nd display MULTiply: Measurement for 1st display * Measurement for 2nd display DIVision: Measurement for 1st display / Measurement for 2nd display NOTE: Valid only when the 2nd display is ON.	
		:CALCulate:DDISplay?	Reply: OFF, ADD, SUBT, MULT, DIV	

*1: <function>: Specify a function of “measurement function”, but do not enclose in double quotes.

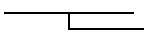
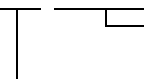
6.6.4 SCPI Command Reference

4. System relevant command

Item		SCPI command	Description	Initial value
System	2nd display	:DISPlay:WINDow2:STATe {ON,1 OFF,0}	ON, 1: ON	●
			OFF, 0: OFF	
			NOTE: Valid only when the 2nd display is ON.	
	Measurement data display	:DISPlay {OFF,0 ON,1}	ON, 1: ON	●
			OFF, 0: OFF	
			Reply: 1 or 0	
	Output data	:FORMat:READing:MCHannel {BOTH 1 2}	BOTH: 1st and 2nd displays	●
			1: Only the 1st display	
			2: Only the 2nd display	
			NOTE: Valid only when the 2nd display is ON.	
	Buzzer	:CALCulate:LIMit:BEEPer {OFF FAIL PASS}	Reply: BOTH, 1, or 2	
			OFF: OFF	●
			FAIL: The buzzer sounds when the comparator calculation result is FAIL.	
			PASS: The buzzer sounds when the comparator calculation result is PASS.	
		:CALCulate:LIMit:BEEPer?	Reply: OFF, FAIL, or PASS	
		:SYSTem:BEEPer:STATe {ON,1 OFF,0}	ON, 1: Buzzer ON	
			OFF, 0: Buzzer OFF	●
	Power supply frequency	:SYSTem:BEEPer:STATe?	Reply: 1 or 0	
	Initialization	:SENSe]:LFRQency?	Reply: 50 or 60	
	Device information	*RST	Parameter initialization	
	Output data element specification	*IDN?	Reply: ADC Corp., 7352X nnnnnnnnn, mmm nnnnnnnnn: Serial number mmm: Revision number	
			DDISplay: Specifies the calculation output between 2 measurements.	
			NONE: No output	●
	Block delimiter	:FORMat:ELEMents {DDISplay NONE}	Reply: DDIS or NONE	
		:FORMat:ELEMents?	CRLF: CRLF+EOI	●
			LF: LF	
			EOI: EOI	
		:SYSTem:GPIB:DELimiter:BLOCK?	Reply: CRLF, LF, or EOI	

Item		SCPI command	Description	Initial value
System	Status	*CLS	Clears each status byte.	
		*STB?	Reads the status byte register Reply: ddd	
		*SRE <int>	Sets the service request enable register. <int>: 0 to 255 (However, bit6 cannot be set.)	
		*SRE?	Reply: ddd	
		*ESR?	Reads the standard event status register. Reply: ddd	
		*ESE <int>	Sets the standard event status enable register. <int>: 0 to 255	
		*ESE?	Reply: ddd	
		:STATus:MEASurement:EVENT? [(@<ch>)]	Reads the measurement event register. Reply: dddddd	
		:STATus:MEASurement:ENABLE <int>[,(@<ch>)]	Sets the measurement event enable register. <int>: 0 to 65535	
		:STATus:MEASurement:ENABLE? [(@<ch>)]	Reply: dddddd	
		:STATus:OPERation:EVENT?	Reads the operation event register. Reply: dddddd	
		:STATus:OPERation:ENABLE <int>	Sets the operation event enable register. Reply: dddddd	
		:STATus:OPERation:ENABLE?	Reply: dddddd	
		:STATus:QUESTionable:EVENT? [(@<ch>)]	Reads the questionable event register. Reply: dddddd	
		:STATus:QUESTionable:ENABLE <int>[,(@<ch>)]	Sets the questionable event enable register. <int>: 0 to 65535	
		:STATus:QUESTionable:ENABLE? [(@<ch>)]	Reply: dddddd	
		*PSC <int>	<int>: -32767 to +32767 If "int" is set to any value except for zero, SRER and SESER are cleared when the power is turned on.	
		*PSC?	Reply: 0 or 1 (When any value except for zero is set.)	
	Option	*OPT?	Reads the option information. Reply: 0 No option	
	Operation complete	*OPC	Sets the operation complete bit (bit0) of the Standard Event Status Register after all operations are complete.	
		*OPC?	Reply: 1 (After all operations are complete.)	
		*WAI	Waits for the completion of all operations. (GPIB only)	
	Self-Test	*TST?	Performs the self- test and reads the result. (It takes time to perform the self-test. Read the result after executing the *WAI command.) Reply 0 : Pass 1 : Fail	

6.6.4 SCPI Command Reference

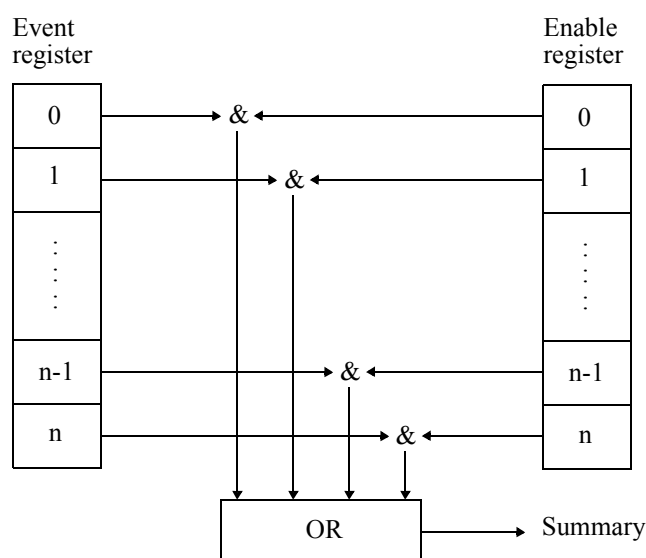
Item		SCPI command	Description	Initial value
System	Calibration	:CALibration:SECure:STATe {ON,1 OFF,0}	ON, 1: Sets the calibration mode to ON. OFF, 0: Sets the calibration mode to OFF. (Writes the calibration factor when leaving the calibration mode.)	●
		:CALibration:SECure:STATe?	Reply: 1 or 0	
		:CALibration:ABORt	Cancels (sets to OFF) the calibration mode. (Does not write the calibration factor.)	
		:CALibration <int>	Enters the STD value. (Displayed count value) <int>: 0 to ±999999	
		:CALibration:VALue <real>	Enters the STD value. (Displayed value) <real>: Displayed value on STD	
		:CALibration:STRing <"str">	Sets the calibration information. <"str">: Enclose the character strings by using (""). Up to 50 characters (one byte alphanumeric characters) can be stored.	
		:CALibration:STRing?	Reply: "xxxxx . . . xxx"  Up to 50 characters	
	Setting parameter	*SAV <int>	<int>: 0 to 3 Saves the setting parameters in area [int] of the non-volatile memory.	
		*RCL <int>	<int>: 0 to 3 Loads the setting parameters from area [int] of the non-volatile memory.	
	Error log reading	:SYSTem:ERRor?	Reads the error details. Reply: ±ddd,"xxxxxxxxxx"  Error character strings (Up to 80 characters) Error code • Up to 20 error queues can be stored. The error is output according to the FIFO system. • If more than 20 errors occur, the last error is replaced with -350, "Queue overflow." Any errors after this are not saved. • If no error occurs, +000, "No error" is returned.	
	SCPI version	:SYSTem:VERSion?	Reads SCPI version. Reply: 1991.0	

6.6.5 Status Register Structure

This instrument has a layered status register structure that is compliant with IEEE standard 488.2-1987 and can send various statuses of this instrument to the controller. This section describes the operation model of status structure and allocation of events.

1. Status register

This instrument adopts the model of the status register defined in IEEE standard 488.2-1987. The status register consists of the event register and enable register.



- **Event register**
The event register latches and retains statuses correspondent to each event. (Sometimes the changes are retained.)
Once this register is set, the setting value is kept until it is read by a query or cleared by *CLS.
Data cannot be written into the event register.
- **Enable register**
The enable register specifies which bit in the event register is set as an effective status to generate a summary. The enable register is ANDed with the event register and the OR of the result is generated as a summary. The summary is written into the status byte register.
Data can be written into the enable register.

This instrument uses the following five types of status registers:

- Status Byte Register (STB)
- Standard Event Status Register (SESR)
- Measurement Event Register (MER)
- Questionable Event Register (QER)
- Operation Event Register (OER)

6.6.5 Status Register Structure

The status register structure in this instrument is shown in Figure 6-2 and Figure 6-3.

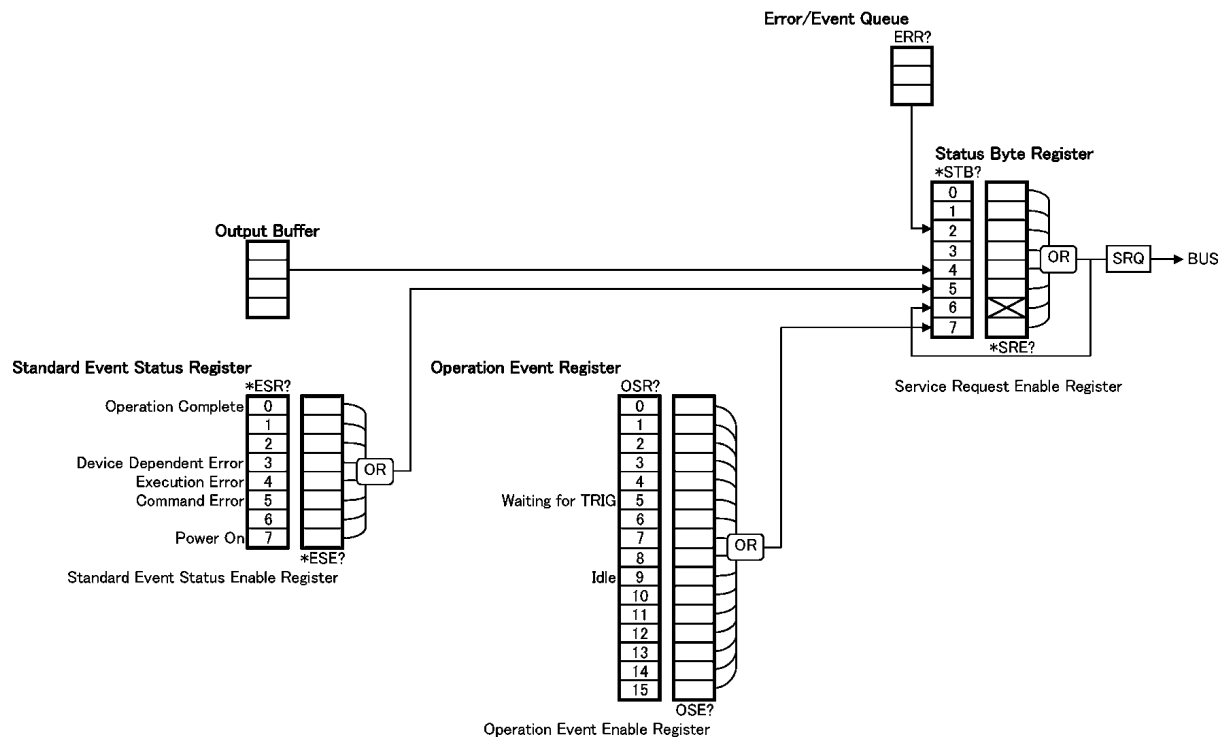


Figure 6-2 Status Register Structure (1/2)

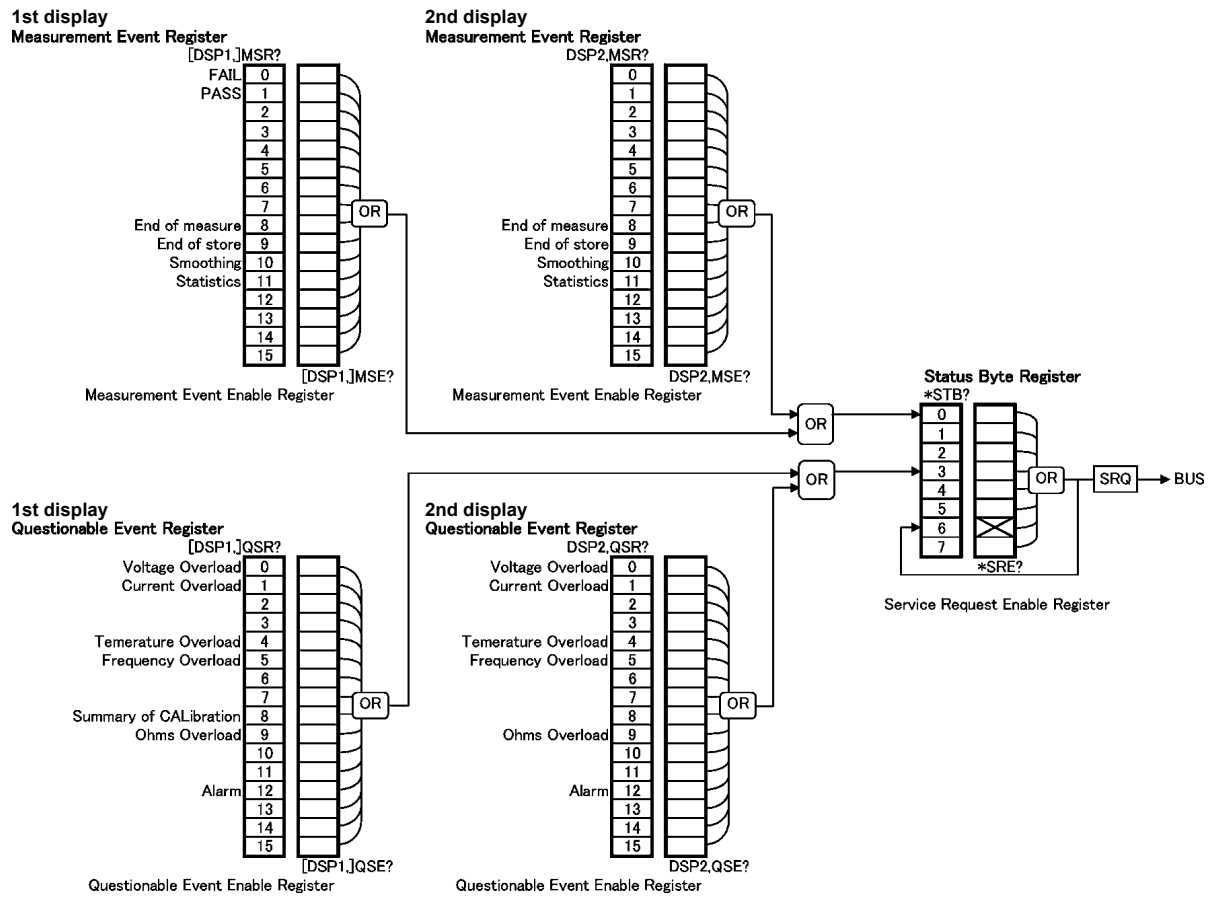


Figure 6-3 Status Register Structure (2/2)

6.6.5 Status Register Structure

2. Event enable register

Each event register has an enable register that determines which bit is to be enabled. The enable register sets the corresponding bit in decimal.

- Service Request Enable Register (SRER) setting: *SRE
- Standard Event Status Enable Register (SESER) setting: *ESE
- Measurement Event Enable Register (MEER): MSE
- Questionable Event Enable Register (QEER): QSE
- Operation Event Enable Register (OEER): OSE

3. Status byte register

The status byte register summarizes the information from the status register. A summary of this status byte register is sent to the controller as a service request. Therefore, the status byte register operates slightly differently than the status register structure. This section describes the status byte register.

The status byte register structure is shown in Figure 6-4.

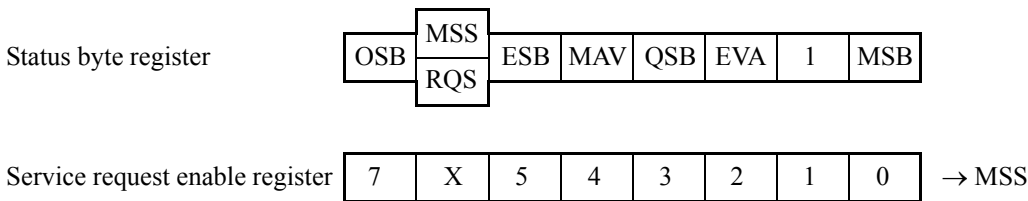


Figure 6-4 Status Byte Register Structure

This status byte register follows the status register except for the following three points:

- The summary of the status byte register is written to bit6 of the status byte register.
- Bit6 of the enable register is always enabled and cannot be changed.
- Bit6 (MSS) of the status byte register writes the RQS of the service request.

This register responds to the serial polling from the controller. When the register responds to the serial polling, bit0 to bit5, bit7, and the RQS of the status byte register are read, and then the RQS is reset to 0. No other bits are cleared until each factor is set to 0.

The status byte register, RQS, and MSS can be cleared by executing “*CLS”. Consequently, the SRQ line is set to FALSE.

A description of each bit in the status byte register is shown in Table 6-4.

Table 6-4 Status Byte Register (STB)

bit	Name	Description
0	MSB Measurement Summary Bit	ON: When any event of Measurement Event Register occurs and Measurement Event Register is set to 1, this bit is set to 1 if the corresponding bit of the Measurement Event Enable Register is set to 1. OFF: This bit is set to 0 when Measurement Event Register is cleared by being read.
1	Not used	Always 0
2	EAV Error Available	ON: This bit is set to 1 when the error information is stored in Error Queue. OFF: This bit is set to 0 when the error information is read from Error Queue and no data has existed in Error Queue.
3	QSB Questionable Summary Bit	ON: When any event of Questionable Event Register occurs and Questionable Event Register is set to 1, this bit is set to 1 if the corresponding bit of the Questionable Event Enable Register is set to 1. OFF: This bit is set to 0 when Questionable Event Register is cleared by being read.
4	MAV Message Available	ON: This bit is set to 1 when the output data is entered in the output buffer. OFF: This bit is set to 0 when the data is read from the output buffer.
5	ESB Standard Event Status	ON: When any event of SESR occurs and SESR is set to 1, this bit is set to 1 if the corresponding bit of the SESER is set to 1. OFF: This bit is set to 0 when SESR is cleared by being read (*ESR?).
6	MSS Master Summary	ON: When any event of STB occurs, this bit is set to 1 if the corresponding bit of the SRER is set to 1.
	RQS Request Service	ON: If MSS is set to 1, SRQ is generated and RQS is set to 1. OFF: When STB is read in the serial polling, RQS is set to 0.
7	OSB Operation Summary Bit	ON: When any event of Operation Event Register occurs and Operation Event Register is set to 1, this bit is set to 1 if the corresponding bit of the Operation Event Enable Register is set to 1. OFF: This bit is set to 0 when Operation Event Register is cleared by being read.

Conditions which clear the status byte register

- When the power is turned on.
- When the *CLS command is executed. However, the MAV bit is not cleared if data exists in the output buffer.
- The status byte register is not cleared by executing *STB? command.

Conditions which clear the service request enable register

- When the power is turned on. (When the PSC flag is set to 1)
- When the *SRE0 command is executed.

6.6.5 Status Register Structure

4. Standard event status register

Allocation in the standard event register is shown in Table 6-5.

Table 6-5 Standard Event Status Register (ESR)

bit	Name	Description
0	OPC Operation Complete	ON: This bit is set to 1 when all operations are complete after receiving the *OPC command.
1	Not used	Always 0
2	Not used	Always 0
3	DDE Device Dependent Error	ON: This bit is set to 1 when the error, which depends on the device, occurs.
4	EXE Execution Error	ON: This bit is set to 1 when the received command cannot be executed. This bit is set to 1 when the command parameter contains an error.
5	CME Command Error	ON: This bit is set to 1 when the received command spelling is incorrect.
6	Not used	Always 0
7	PON Power On	ON: This bit is set to 1 when the power is turned on.

Conditions which clear the standard event status register

- When the power is turned on.
- When the *CLS command is executed.
- When the *ESR? command is executed.

Conditions which clear the standard event status enable register

- When the power is turned on. (When the PSC flag is set to 1)
- When the *ESE0 command is executed.

5. Measurement event register

Allocation in the measurement event register is shown in Table 6-6.

Table 6-6 Measurement Event Register (MER)

bit	Name	Description
0	FL FAIL	ON: This bit is set to 1 when the comparator calculation result satisfies the FAIL conditions.
1	PS PASS	ON: This bit is set to 1 when the comparator calculation result satisfies the PASS conditions.
2	Not used	Always 0
3	Not used	Always 0
4	Not used	Always 0
5	Not used	Always 0
6	Not used	Always 0
7	Not used	Always 0
8	EOM End of measure	ON: This bit is set to 1 when the measurement is complete.
9	EOS End of store	ON: This bit is set to 1 when measurement data cannot be stored in the measurement memory any longer.
10	SM Smoothing complete	ON: This bit is set to 1 when the smoothing count reaches the specified number.
11	STAT Statistics	ON: This bit is set to 1 when the statistical processing is complete.
12	Not used	Always 0
13	Not used	Always 0
14	Not used	Always 0
15	Not used	Always 0

Conditions which clear the measurement event register

- When the power is turned on.
- When the *CLS command is executed.
- When the :STATus:MEASurement[:EVENT]? command is executed.
- When the MSR? command is executed.

Conditions which clear the measurement event enable register

- When the power is turned on.
- When the :STATus:MEASurement:ENABle 0 command is executed.
- When the MSE0 command is executed.

6.6.5 Status Register Structure

6. Questionable event register

Allocation in the questionable event register is shown in Table 6-7.

Table 6-7 Questionable Event Register (QER)

bit	Name	Description
0	Voltage Overload	ON: This bit is set to 1 when OL occurs in the voltage or diode measurement.
1	Current Overload	ON: This bit is set to 1 when OL occurs in the current measurement.
2	Not used	Always 0
3	Not used	Always 0
4	Temperature Overload	ON: This bit is set to 1 when OL occurs in the temperature measurement.
5	Frequency Overload	ON: This bit is set to 1 when OL occurs in the frequency measurement.
6	Not used	Always 0
7	Not used	Always 0
8	Summary of Calibration	ON: This bit is set 1 when the default calibration values or the calibration values acquired in previous power ON is used due to the calibration data SUM failure in the power ON check.
9	Ohms Overload	ON: This bit is set to 1 when OL occurs in the resistance measurement.
10	Not used	Always 0
11	Not used	Always 0
12	Alarm	ON: This bit is set to 1 when an alarm occurs in the measurement.
13	Not used	Always 0
14	Not used	Always 0
15	Not used	Always 0

Conditions which clear the questionable event register

- When the power is turned on.
- When the *CLS command is executed.
- When the :STATus:QUESTionable[:EVENTt]? command is executed.
- When the QSR? command is executed.

Conditions which clear the measurement event enable register

- When the power is turned on.
- When the :STATus:QUESTionable:ENABle 0 command is executed.
- When the QSE0 command is executed.

7. Operation event register

Allocation in the operation event register is shown in Table 6-8.

Table 6-8 Operation Event Register (OER)

bit	Name	Description
0	Not used	Always 0
1	Not used	Always 0
2	Not used	Always 0
3	Not used	Always 0
4	Not used	Always 0
5	Waiting for TRIG	ON: This bit is set to 1 when this instrument is in the Trigger waiting state.
6	Not used	Always 0
7	Not used	Always 0
8	Not used	Always 0
9	Idle	ON: This bit is sets to 1 when this instrument is in the Idle state.
10	Not used	Always 0
11	Not used	Always 0
12	Not used	Always 0
13	Not used	Always 0
14	Not used	Always 0
15	Not used	Always 0

Conditions which clear the operation event register

- When the power is turned on.
- When the *CLS command is executed.
- When the :STATus:OPERation[:EVENT]? command is executed.
- When the OSR? command is executed.

Conditions which clear the operation event enable register

- When the power is turned on.
- When the :STATus:OPERation:ENABle 0 command is executed.
- When the OSE0 command is executed.

6.7 Sample Programs

6.7 Sample Programs

6.7.1 Sample Programs (GPIB commands used in ADC CORPORATION)

This section describes example programs in which a computer controls this instrument through a GPIB interface.

[Operation check environment]

Computer:	DELL OPTIPLEX GX170L(Pentium®4 CPU 2.80GHz)
GPIB hardware:	GPIB-USB-HS manufactured by NATIONAL INSTRUMENTS
Module:	Niglobal.bas,Vbib-32.bas (Included software in PCI-GPIB)
Language:	Microsoft Excel Visual Basic for Application

NOTE: The GPIB address of the 7352A is set to 1.

Example 1 Sets the DCV-Ach 2 V range for the left side display and the ACV-Ach 2 V range for the right side display, and then performs the measurement in the free run state. Detects the measurement end by polling the status byte, reads the measured data from the 7352A, and then displays the data in cells.

Dim DMM_ADR As Integer	'Declares the GPIB address variable of the 7352A.
Dim dmm As Integer	'Declares the variable of the device descriptor.
Dim dt As String * 100	'Declares the variable of the buffer used for receiving the GPIB data.
DMM_ADR = 1	'GPIB address of the 7352A
Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)	'Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1)	'Sets the transmitting and receiving addresses individually.
Call ibwrt(dmm, "*RST" & vbLf)	'Initializes the 7352A.
Call ibwrt(dmm, "H1" & vbLf)	'Sets the header of the output data to ON.
Call ibwrt(dmm, "DSP1,F1" & vbLf)	'Sets the measurement function for the left side display to DCV.
Call ibwrt(dmm, "DSP1,R4" & vbLf)	'Sets the measurement range for the left side display to 2 V.
Call ibwrt(dmm, "DSP2,F2" & vbLf)	'Sets the measurement function for the right side display to ACV.
Call ibwrt(dmm, "DSP2,R4" & vbLf)	'Sets the measurement range for the right side display to 2 V.
Call ibwrt(dmm, "PR2" & vbLf)	'Sets the sampling rate to MED.
Call ibwrt(dmm, "*CLS" & vbLf)	'Clears the status byte.
Do	'Reads the status byte of the measurement end.
Call ibwrt(dmm, "*STB?" & vbLf)	'Requests the content of the status byte.
Call ibrd(dmm, dt)	'Writes the content of the status byte into the variable dt.
dt = dt And 16	'Performs the AND operation in bit4 (MAV).
Loop While (dt <> 16)	
Call ibwrt(dmm, "SD1" & vbLf)	'Sets the output data to the left side display.
Call ibrd(dmm, dt)	'Reads the measured data on the left side display.
Cells(1, 1) = "" & Left(dt, 18)	'Substitutes the measured value on the left side display for a cell.

6.7.1 Sample Programs (GPIB commands used in ADC CORPORATION)

<code>Call ibwrt(dmm, "SD2" & vbLf)</code>	'Sets the output data to the right side display.
<code>Call ibrd(dmm, dt)</code>	'Reads the measured data on the right side display.
<code>Cells(2, 1) = "" & Left(dt, 18)</code>	'Substitutes the measured value on the right side display for a cell.
<code>Call ibonl(dmm, 0)</code>	'Terminates.

6.7.1 Sample Programs (GPIB commands used in ADC CORPORATION)

Example 2 Sets the DCV-Ach 2 V range for the left side display and the ACV-Ach 2 V range for the right side display, specifies BUS as the trigger source, sets the sample count to 10, sets the measurement data memory to ON, and then triggers to start the measurement. Detects the measurement end by the SRQ interruption, reads the measurement memory data from the 7352A, and displays the data in cells.

```

Dim DMM_ADR As Integer           'Declares the GPIB address variable of the 7352A.
Dim dmm As Integer              'Declares the variable of the device descriptor.
Dim dt As String * 200          'Declares the variable of the buffer used for receiving the GPIB data.
Dim s As Integer                'The variable in which the serial polling result is stored

DMM_ADR = 1                      'GPIB address of the 7352A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm) 'Initializes the GPIB I/F.

Call ibconfig(dmm, IbcUnAddr, 1) 'Sets the transmitting and receiving addresses individually.
Call ibconfig(0, IbcAUTOPOLL, 0) 'Disables the automatic serial polling.
Call ibwrt(dmm, "**RST" & vbLf)    'Initializes the 7352A.

Call ibwrt(dmm, "S0" & vbLf)      'Permits SRQ transmitting.
Call ibwrt(dmm, "DSP1,F1" & vbLf) 'Sets the measurement function for the left side display to DCV.
Call ibwrt(dmm, "DSP1,R4" & vbLf) 'Sets the measurement range for the left side display to 2 V.
Call ibwrt(dmm, "DSP2,F2" & vbLf) 'Sets the measurement function for the right side display to ACV.
Call ibwrt(dmm, "DSP2,R4" & vbLf) 'Sets the measurement range for the right side display to 2 V.
Call ibwrt(dmm, "PR2" & vbLf)     'Sets the sampling rate to MED.

Call ibwrt(dmm, "SPN10" & vbLf)   'Sets the sampling count to 10.
Call ibwrt(dmm, "TRS3" & vbLf)    'Specifies "BUS" as the trigger source.

Call ibwrt(dmm, "ST1" & vbLf)     'Sets the measurement data memory to ON.
Call ibwrt(dmm, "ICL" & vbLf)     'Measurement data memory initialization

Call ibwrt(dmm, "*CLS" & vbLf)    'Clears the status byte.
Call ibwrt(dmm, "DSP2,MSE256" & vbLf) 'Sets the measurement end flag of MSR for the right side display.

Call ibwrt(dmm, "**SRE1" & vbLf)  'Sets the measurement end flag of SRE.
Call ibwrt(dmm, "**TRG" & vbLf)   'Triggers.

Call ibwait(dmm, RQS Or TIMO)     'Waits for the SRQ transmission.

If (ibsta And TIMO) Then           'Judges whether a time-out occurs.
    Call MsgBox("SRQ Time Out", vbOKOnly, "Error") 'Sends a message if a time-out occurs.

    Call ibwrt(dmm, "DS1" & vbLf) 'Sets the display to ON.
    Call ibonl(dmm, 0)            'Terminates.
    Exit Sub
Else
    Call ibrsp(dmm, s)            'Reads the status.
End If

Call ibwrt(dmm, "IRD0,9" & vbLf) 'Sets the data reading range for the left side display.
Call ibwrt(dmm, "IRO?" & vbLf)   'Data reading for the left side display
Call ibrd(dmm, dt)                'Reads the measured data on the left side display.
Cells(1, 1) = "" & dt           'Substitutes the measured value on the left side display for a cell.

Call ibwrt(dmm, "DSP2,IRD0,9" & vbLf) 'Sets the data reading range for the right side display.

```

6.7.1 Sample Programs (GPIB commands used in ADC CORPORATION)

<code>Call ibwrt(dmm, "DSP2,IRO?" & vbLf)</code>	'Data reading for the right side display
<code>Call ibrd(dmm, dt)</code>	'Reads the measured data on the right side display.
<code>Cells(2, 1) = "" & dt</code>	'Substitutes the measured value on the right side display for a cell.
<code>Call ibonl(dmm, 0)</code>	'Terminates.

6.7.2 Sample Programs (SCPI commands for GPIB)

6.7.2 Sample Programs (SCPI commands for GPIB)

This section describes example programs in which a computer controls this instrument through a GPIB interface.

[Operation check environment]

Computer:	DELL OPTIPLEX 170L(Pentium®4 CPU 2.80GHz)
GPIB hardware:	GPIB-USB-HS manufactured by NATIONAL INSTRUMENTS
Module:	Niglobal.bas,Vbib-32.bas (Included software in PCI-GPIB)
Language:	Microsoft Excel Visual Basic for Application

NOTE: The GPIB address of the 7352A is set to 1.

Example 1 Sets the DCV-Ach 2 V range for the left side display and the ACV-Ach 2 V range for the right side display, sets the trigger source to BUS, and then triggers to start the measurement. Detects the measurement end by polling the status byte, reads the measured data from the 7352A, and then displays the data in cells.

```
Dim DMM_ADR As Integer           'Declares the GPIB address variable of the 7352A.
Dim dmm As Integer              'Declares the variable of the device descriptor.
Dim dt As String * 100          'Declares the variable of the buffer used for receiving the GPIB data.

DMM_ADR = 1                     'GPIB address of the 7352A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)
                                'Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1) 'Sets the transmitting and receiving addresses individually.
Call ibwrt(dmm, "*RST" & vbLf)   'Initializes the 7352A.

Call ibwrt(dmm, ":SENSE:FUNCTION 'VOLTAGE:DC', (@1)" & vbLf)
                                'Sets the measurement function for the left side display to DCV.
Call ibwrt(dmm, ":SENSE:VOLTAGE:DC:RANGE 1.99999, (@1)" & vbLf)
                                'Sets the measurement range for the left side display to 2 V.
Call ibwrt(dmm, ":SENSE:VOLTAGE:DC:SRATE MED" & vbLf)
                                'Sets the sampling rate to MED.
Call ibwrt(dmm, ":SENSE:FUNCTION 'VOLTAGE:AC', (@2)" & vbLf)
                                'Sets the measurement function for the right side display to ACV.
Call ibwrt(dmm, ":SENSE:VOLTAGE:AC:RANGE 1.99999, (@2)" & vbLf)
                                'Sets the measurement range for the right side display to 2 V.
Call ibwrt(dmm, ":SENSE:VOLTAGE:AC:SRATE MED" & vbLf)
                                'Sets the sampling rate to MED.
Call ibwrt(dmm, ":TRIGGER:SOURCE BUS" & vbLf)
                                'Sets the trigger source to "BUS".
Call ibwrt(dmm, ":INITIATE:CONTINUOUS ON" & vbLf)
                                'Sets CONTINUOUS ON.

Call ibwrt(dmm, "*CLS" & vbLf)  'Clears the status byte.
Call ibwrt(dmm, "*TRG" & vbLf)  'Triggers.

Do
    Call ibwrt(dmm, "*STB?" & vbLf) 'Requests the content of the status byte.
    Call ibrd(dmm, dt)               'Writes the content of the status byte into the variable dt.
    dt = dt And 16                   'Performs the AND operation in bit4 (MAV).
Loop While (dt <> 16)
```

6.7.2 Sample Programs (SCPI commands for GPIB)

```
Call ibwrt(dmm, ":FORMAT:READING:MCHANNEL 1" & vbLf)
                                                    'Sets the output data to the left side display.
Call ibwrt(dmm, ":FETCH?" & vbLf)
                                                    'Requests the measured data on the left side display.
Call ibrd(dmm, dt)
                                                    'Reads the measured data on the left side display.
Cells(1, 1) = "" & Left(dt, 18)
                                                    'Substitutes the measured value on the left side display for a cell.

Call ibwrt(dmm, ":FORMAT:READING:MCHANNEL 1" & vbLf)
                                                    'Sets the output data to the right side display.
Call ibwrt(dmm, ":FETCH?" & vbLf)
                                                    'Requests the measured data on the right side display.
Call ibrd(dmm, dt)
                                                    'Reads the measured data on the right side display.
Cells(2, 1) = "" & Left(dt, 18)
                                                    'Substitutes the measured value on the right side display for a cell.

Call ibonl(dmm, 0)
                                                    'Terminates.
```

6.7.2 Sample Programs (SCPI commands for GPIB)

Example 2 Sets the DCV-Ach 2 V range for the left side display and the ACV-Ach 2 V range for the right side display, specifies BUS as the trigger source, sets the sample count to 10, sets the measurement data memory to ON, and then triggers to start the measurement. Detects the measurement end by the SRQ interruption, reads the measurement memory data from the 7352A, and then displays the data in cells.

```
Dim DMM_ADR As Integer      'Declares the GPIB address variable of the 7352A.
Dim dmm As Integer         'Declares the variable of the device descriptor.
Dim dt As String * 200     'Declares the variable of the buffer used for receiving the GPIB data.
Dim s As Integer           'The variable in which the serial polling result is stored

DMM_ADR = 1                'GPIB address of the 7352A

Call ibdev(0, DMM_ADR, 0, T10s, 1, 0, dmm)
                           'Initializes the GPIB I/F.
Call ibconfig(dmm, IbcUnAddr, 1)
                           'Sets the transmitting and receiving addresses individually.
Call ibconfig(0, IbcAUTOPOLL, 0)
                           'Disables the automatic serial polling.
Call ibwrt(dmm, "*RST" & vbLf)
                           'Initializes the 7352A.

Call ibwrt(dmm, ":SENSE:FUNCTION 'VOLTAGE:DC', (@1)" & vbLf)
                           'Sets the measurement function for the left side display to DCV.
Call ibwrt(dmm, ":SENSE:VOLTAGE:DC:RANGE 1.99999, (@1)" & vbLf)
                           'Sets the measurement range for the left side display to 2 V.
Call ibwrt(dmm, ":SENSE:VOLTAGE:DC:SRATE MED" & vbLf)
                           'Sets the sampling rate to MED.
Call ibwrt(dmm, ":SENSE:FUNCTION 'VOLTAGE:AC', (@2)" & vbLf)
                           'Sets the measurement function for the right side display to ACV.
Call ibwrt(dmm, ":SENSE:VOLTAGE:AC:RANGE 1.99999, (@2)" & vbLf)
                           'Sets the measurement range for the right side display to 2 V.
Call ibwrt(dmm, ":SENSE:VOLTAGE:AC:SRATE MED" & vbLf)
                           'Sets the sampling rate to MED.

Call ibwrt(dmm, ":SAMPLE:COUNT 10" & vbLf)
                           'Sets the sampling count to 10.
Call ibwrt(dmm, ":TRIGGER:SOURCE BUS" & vbLf)
                           'Sets the trigger source to "BUS".
Call ibwrt(dmm, ":INITIATE:CONTINUOUS ON" & vbLf)
                           'Sets CONTINUOUS ON.

Call ibwrt(dmm, ":TRACE:STATE ON" & vbLf)
                           'Enables the measurement memory.
Call ibwrt(dmm, ":TRACE:CLEAR" & vbLf)
                           'Clears the measurement memory.

Call ibwrt(dmm, "*CLS" & vbLf)
                           'Clears the status byte.
Call ibwrt(dmm, ":STATUS:MEASUREMENT:ENABLE 256, (@2)" & vbLf)
                           'Sets the measurement end flag of MSR for the right side display.
Call ibwrt(dmm, "*SRE 1" & vbLf)
                           'Sets the measurement end flag of SRE.
Call ibwrt(dmm, "*TRG" & vbLf)
                           'Triggers.

Call ibwait(dmm, RQS Or TIMO)
                           'Waits for the SRQ transmission.

If (ibsta And TIMO) Then
    Call MsgBox("SRQ Time Out", vbOKOnly, "Error")
    'Sends a message if a time-out occurs.
    Call ibonl(dmm, 0)
    'Terminates.
    Exit Sub
Else
    Call ibrsp(dmm, s)
    'Reads the status byte.
End If
```

6.7.2 Sample Programs (SCPI commands for GPIB)

```
Call ibwrt(dmm, ":TRACE:NUMBER 0,9, (@1)" & vbLf)
'Sets the recall data range for the left side display. (0 to 9)
Call ibwrt(dmm, ":TRACE:DATA? (@1)" & vbLf)
'Requests the measured data on the left side display.
Call ibrd(dmm, dt)
'Substitutes the measured value on the left side display for a variable.
Cells(1, 1) = "" & dt
'Substitutes the measured value on the left side display for a cell.

Call ibwrt(dmm, ":TRACE:NUMBER 0,9, (@2)" & vbLf)
'Sets the recall data range for the right side display. (0 to 9)
Call ibwrt(dmm, ":TRACE:DATA? (@2)" & vbLf)
'Requests the measured data on the right side display.
Call ibrd(dmm, dt)
'Substitutes the measured value on the right side display for a variable.
Cells(2, 1) = "" & dt
'Substitutes the measured value on the right side display for a cell.

Call ibonl(dmm, 0)
'Terminates.
```

6.7.3 Sample Programs (RS-232)

6.7.3 Sample Programs (RS-232)

This section describes example programs in which a computer controls this instrument through an RS-232 interface.

Computer: DELL OPTIPLEX 170L(Pentium®4 CPU 2.80GHz)
 Control used: Microsoft Communications Control (RS-232 control)
 Language: Microsoft Excel Visual Basic for Application

Example 1 Sets the 2WΩ resistance measurement 20 kΩ range for the left side display, detects the measurement end by the measurement event register, reads the measured data from the 7352A, and then displays the data in cells.

```
Private Sub Com_RS232_Sample1_Click
    'Sample 1
    Dim dt As String * 20      'Declares the variable of the buffer used for receiving the RS-232
                                'data.
    Dim stb As String          'Declares the variable of the buffer used for receiving the status byte.
    Dim sts As Integer         'Declares the variable of the status byte after numerical conversion.

    With UserForm1
        With .MSComm1
            .CommPort = 1      'Uses COM1.
            .Settings = "9600,N,8,1" '9600 bps, Non parity, Data length: 8 bits, and Number of stop bits: 1
            .InputLen = 0      'Reads all of the buffers when the Input property is used.
            .PortOpen = True   'Opens the port.
            .InBufferCount = 0 'Clears the receiving buffer.
            .DTREnable = True  'Enables DTR.

            dt = Space(20)     'Initializes the buffer for receiving data

            .Output = "*RST" & vbCrLf 'Initializes the 7352A.
            Call rx_prompt       'Receives a prompt.

            .Output = "H0,F3,R5,PR2,TRS3,DE0" & vbCrLf
            'Sets the header of the output data to OFF.
            'Sets the measurement function for the left side display to 2WΩ.
            'Sets the measurement range for the left side display to 20 kΩ.
            'Sets the sampling rate to MED.
            'Sets the trigger source to "BUS".
            'Sets the right side display to OFF.
            Call rx_prompt      'Receives a prompt.

            .Output = "*TRG" & vbCrLf 'Triggers.
            Call rx_prompt       'Receives a prompt.

            Do
                .Output = "MSR?" & vbCrLf 'Sends the command for reading the measurement event register.
                Call rx_data(stb, 5)      'Reads the measurement event register (5 characters) and stores them
                'in stb.
                sts = Val(stb) And 256    'Converts the 5 characters, which are stored in 'stb, to a numerical
                'value and stores the value in sts.
            Loop While (sts <> 256)        'Repeats until EOM is set.
            .Output = "MD?" & vbCrLf     'Sends the command for reading the measured data.
            Call rx_data(dt, 14)         'Reads the measured data (14 characters) and stores them in dt$.
            Cells(1, 1) = "" & Left(dt, 12) 'Displays the measured data in cells.
        End With
    End With
End Sub
```

```

        .PortOpen = False                'Closes the port.
    End With
End Sub
-----
Private Sub rx_prompt()                  'Subroutine for receiving a prompt
    Dim prompt As String

    Do
        If UserForm1.MSComm1.InBufferCount >= 5 Then
            'If there is received data, reads it.
            prompt = UserForm1.MSComm1.Input
            'Reads data from the receiving buffer.
            If InStr(prompt, ">") Then
                'Checks whether ">" is received.
                Exit Do
                'Exits Do While Loop.
            End If
        End If
    Loop
End Sub
-----
Private Sub rx_data(dt As String, dt_len As Integer)
    'Subroutine for receiving data and a prompt

    Do
        If UserForm1.MSComm1.InBufferCount >= (5 + dt_len) Then
            'If there is received data, reads it.
            dt$ = UserForm1.MSComm1.Input
            'Reads data from the receiving buffer.
            If InStr(dt, ">") Then
                'Checks whether ">" is received.
                dt = Mid(dt, 2, dt_len)
                'Separates a prompt from received data.
                Exit Do
                'Exits Do While Loop.
            End If
        End If
    Loop
End Sub

```

6.7.4 Sample Programs (USB)

6.7.4 Sample Programs (USB)

This section describes example programs in which a computer controls this instrument through a USB interface.

[Operation check environment]

Computer:	DELL OPTIPLEX 170L(Pentium®4 CPU 2.80GHz)
Module:	asub.bas (Software included in the USB driver for the ADC CORPORATION's instrument)
Language:	Microsoft Excel Visual Basic for Application

NOTE: *The USB ID of the 7352A/E is set to 1.*

Example 1 Sets the DCV-Ach 20 V range for the left side display and performs the measurement in the free run state. Reads the measured data from the 7352A/E and displays the data in cells.

```

Dim DMM As Long           'USB handle
Private Const OK As Integer = 0 'Definition of OK

Private Sub Com_usb_sample1_Click()
    'Sample 1
    Dim myID As Long        'Declares the USB ID variable of the 7352A/E.
    Dim rstr As String      'Declares the variable of the buffer used for receiving the USB data.
    Dim ret As Long         'Declares the variable of the driver return value.

    myID = 1                'Sets USB ID to 1.

    ret = ausb_start(10)    'USB initialization, time-out: 10 seconds
    If ret <> OK Then        'If the USB initialization is performed incorrectly,
        MsgBox "USB initialization error", vbExclamation
        GoTo err_exit
    End If

    Call mSecSleep(100)     'Waits for the USB initialization (100 ms).

    ret = ausb_open(DMM, myID) 'Opens the DMM whose MyID is set to 1 and acquires the USB handle.
    If ret <> OK Then        'If the device is opened incorrectly,
        MsgBox "device OPEN error", vbExclamation
        GoTo err_exit
    End If

    Call ausbwrt(DMM, "*RST") 'Initializes the instrument.
    Call ausbwrt(DMM, "H1")   'Sets the header of the output data to ON.
    Call ausbwrt(DMM, "DE0")  'Sets the right side display to OFF.
    Call ausbwrt(DMM, "DSP1,F1") 'Sets the measurement function for the left side display to DCV.
    Call ausbwrt(DMM, "DSP1,R5") 'Sets the measurement range for the left side display to 20 V.
    Call ausbwrt(DMM, "PR4")  'Sets the sampling rate to SLOW2.

    Call ausbrd(DMM, rstr)    'Reads the measured data.
    Cells(1, 1) = "" & rstr  'Substitutes the measured value for a cell.

err_exit:

    ret = ausb_close(DMM)    'Closes the device.

```



```

If ret <> OK Then                                'If the device is closed incorrectly,
    MsgBox "device CLOSE error", vbExclamation
End If

ret = ausb_end()                                'USB terminates.
If ret <> OK Then                                'If the USB terminates incorrectly,
    MsgBox "USB end error", vbExclamation
End If
End Sub
-----
Private Sub ausbwrt(id As Long, command As String)
    'Subroutine for transmitting the command
    Dim ret As Long                             'Declares the variable of the driver return value.

    ret = ausb_write(id, command)               'Transmits the command.
    If ret <> OK Then
        MsgBox "transmission error", vbExclamation
        GoTo err_exit
    End If

Exit Sub
err_exit:

    ret = ausb_close(DMM)
    If ret <> OK Then
        MsgBox "device CLOSE error", vbExclamation
    End If

    ret = ausb_end()
    If ret <> OK Then
        MsgBox "USB end error", vbExclamation
    End If
End Sub
-----
Private Sub ausbrd(id As Long, dt As String)
    'Subroutine for receiving data
    Dim ret As Long                             'Declares the variable of the driver return value.
    Dim siz As Long

    ret = ausb_read(id, dt, 50, siz)             'Receives data.
    If ret = OK Then
        dt = Left$(dt, siz - 1)
    Else
        MsgBox "receiving error", vbExclamation
        GoTo err_exit
    End If

Exit Sub
err_exit:

    ret = ausb_close(DMM)
    If ret <> OK Then
        MsgBox "device CLOSE error", vbExclamation
    End If

    ret = ausb_end()
    If ret <> OK Then
        MsgBox "USB end error", vbExclamation
    End If
End Sub

```


7. SPECIFICATIONS

7.1 Specifications

Unless otherwise specified, the measurement accuracy of this instrument is guaranteed for one year under the following conditions: Temperature; $23 \pm 5^{\circ}\text{C}$, relative humidity; 85% or less (75% or less in a resistance measurement of larger than $20\text{ M}\Omega$ and low power resistance measurement of larger than $2\text{ M}\Omega$)

Temperature coefficient:

For the 4 1/2-digit display, the digit term is reduced to 1/10 of its specification.

7.1.1 DC Voltage Measurement (DCV)

7.1.1.1 Ach DC Voltage Measurement (DCV-Ach)

Input terminal	Range	Maximum display		Resolution		Input impedance	Measurement accuracy *1 $\pm(\% \text{ of reading} + \text{digits})$			Temperature coefficient $\pm(\text{ppm of reading} + \text{digits})/^{\circ}\text{C}$	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
V Ω Hz-COM A	200 mV	199.99	199.999	10 μV	1 μV	More than 1 G Ω	0.012+4	0.012+6	0.012+6	15+0.85	15+2
	2000 mV	1999.9	1999.99	100 μV	10 μV	More than 1 G Ω	0.011+2	0.011+2	0.011+2	15+0.2	15+1.5
	20 V	19.999	19.9999	1 mV	100 μV	10 M $\Omega \pm 1\%$	0.015+2	0.015+4	0.015+4	20+0.25	20+1.5
	200 V	199.99	199.999	10 mV	1 mV	10 M $\Omega \pm 1\%$	0.015+2	0.015+4	0.015+4	20+0.25	20+1.5
	1000 V	1099.9	1099.99	100 mV	10 mV	10 M $\Omega \pm 1\%$	0.015+2	0.015+4	0.015+4	20+0.25	20+1.5

*1: For Auto-zero ON For Auto-zero OFF, 1 digit is added to the digit term.

Maximum allowable applied voltage

Between V Ω Hz and COM A terminals	1000 V _{peak}
Between COM A terminal and the chassis	500 V
Between COM A and COM B terminals	200 V

Noise rejection ratio

		FAST	MED/SLOW1, 2
Effective common mode noise rejection ratio (Unbalanced impedance of 1 k Ω)	DC	Approx. 130 dB	Approx. 130 dB
	50/60 Hz $\pm 0.08\%$	Approx. 60 dB	Approx. 120 dB
Normal mode noise rejection ratio	50/60 Hz $\pm 0.08\%$	0 dB	Approx. 60 dB

7.1.1 DC Voltage Measurement (DCV)

7.1.1.2 Bch DC Voltage Measurement (DCV-Bch)

Input terminal	Range	Maximum display		Resolution		Input impedance	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
V-COM B	200 mV	199.99	199.999	10 μ V	1 μ V	More than 1 G Ω	0.012+4	0.012+7	0.012+6	15+0.85	15+2
	2000 mV	1999.9	1999.99	100 μ V	10 μ V	More than 1 G Ω	0.011+2	0.011+5	0.011+2	15+0.2	15+1.5
	20 V	19.999	19.9999	1 mV	100 μ V	10 M Ω $\pm$ 1%	0.015+2	0.015+5	0.015+5	20+0.25	20+1.5
	200 V	199.99	199.999	10 mV	1 mV	10 M Ω $\pm$ 1%	0.015+2	0.015+5	0.015+5	20+0.25	20+1.5

*1: For Auto-zero ON For Auto-zero OFF, 1 digit is added to the digit term.

Maximum allowable applied voltage

Between V and COM B terminals	200 V _{peak}
Between COM B terminal and the chassis	500 V
Between COM A and COM B terminals	200 V

Noise rejection ratio

		FAST	MED/SLOW1, 2
Effective common mode noise rejection ratio (Unbalanced impedance of 1 k Ω)	DC	Approx. 130 dB	Approx. 130 dB
	50/60 Hz $\pm$ 0.08%	Approx. 60 dB	Approx. 120 dB
Normal mode noise rejection ratio	50/60 Hz $\pm$ 0.08%	0 dB	Approx. 60 dB

7.1.2 AC Voltage Measurement

7.1.2.1 Ach AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)

Measurement method	True RMS measurement, RMS display
Input range	5% or more of a full scale
Crest factor	3:1 at a full scale (This is restricted to the maximum allowable applied voltage.)
Temperature coefficient	(1/10 of measurement accuracy that includes the additional error)/°C in each range and frequency range
Response time	Approx. 1 s (Time until the measurement value reaches within 0.1% of the final value in the same range)

ACV-Ach

Input terminal	Range	Maximum display		Resolution		Input impedance
		FAST/MED	SLOW1, 2	FAST/MED	SLOW1, 2	
VΩHz-COM A	200 mV	199.99	199.999	10 μV	1 μV	1 MΩ ±2%, 140 pF or less
	2000 mV	1999.9	1999.99	100 μV	10 μV	
	20 V	19.999	19.9999	1 mV	100 μV	
	200 V	199.99	199.999	10 mV	1 mV	
	700 V	749.9	749.99	100 mV	10 mV	

Range	Measurement accuracy *1 ±(% of reading + digits)				
	20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz
200 mV	0.38+140	0.11+120	0.1+100	0.25+150	0.7+240
2000 mV	0.38+140	0.11+120	0.1+100	0.2+150	0.6+240
20 V	0.38+140	0.11+120	0.1+100	0.2+150	0.6+240
200 V	0.38+140	0.11+120	0.1+100	0.2+150	0.6+240
700 V	0.38+100	0.11+100	0.1+100	—	—

*1: For a sine-wave input When RATE is set to FAST or MED, the digit term is reduced to 1/10 of its specification.

7.1.2 AC Voltage Measurement

ACV(AC+DC)-Ach

Input terminal	Range	Maximum display		Resolution		Input impedance
		FAST/MED	SLOW1, 2	FAST/MED	SLOW1, 2	
VΩHz-COM A	200 mV	199.9	199.99	100 μV	10 μV	1 MΩ ±12%, 140 pF or less
	2000 mV	1999	1999.9	1000 μV	100 μV	
	20 V	19.99	19.999	10 mV	1000 μV	
	200 V	199.9	199.99	100 mV	10 mV	
	700 V	749	749.9	1000 mV	100 mV	

Range	Measurement accuracy *1 ±(% of reading + digits)				
	20 Hz - 45 Hz	45 Hz - 100 Hz	100 Hz - 20 kHz	20 kHz - 50 kHz	50 kHz - 100 kHz
200 mV	0.38+14	0.11+14	0.1+14	0.25+15	0.7+24
2000 mV	0.38+14	0.11+14	0.1+14	0.2+15	0.6+24
20 V	0.38+14	0.11+14	0.1+14	0.2+15	0.6+24
200 V	0.38+14	0.11+14	0.1+14	0.2+15	0.6+24
700 V	0.38+10	0.11+10	0.1+10	—	—

*1: For sine-wave input When RATE is set to FAST or MED, the digit term is reduced to 1/10 of its specification.

Additional error depending on the crest factor (For a non sine-wave input voltage)

1 - 2	0 + 0.05	±(% of reading + % of range)
2 - 3	0 + 0.05	

Additional error to the ACV(AC+DC)-Ach measurement

	FAST	MED	±(% of reading)
20 - 45 Hz	1.24%	0.72%	
45 - 100 Hz	0.06%	—	

Maximum allowable applied voltage

Between VΩHz and COM A terminals	700 Vrms, 1000 Vpeak, 2.2×10^7 V · Hz
Between COM A terminal and the chassis	500 V
Between COM A and COM B terminals	200 V

7.1.3 Ach Resistance Measurement (2WΩ-Ach and LP-2WΩ-Ach)

7.1.3 Ach Resistance Measurement (2WΩ-Ach and LP-2WΩ-Ach)

2WΩ resistance measurement

Input terminal	Range	Maximum display		Resolution		Measurement current	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
VΩHz-COM A	200 Ω	199.99	199.999	10 mΩ	1 mΩ	1 mA	0.02+4	0.02+9	0.02+8	20+1	20+2
	2000 Ω	1999.9	1999.99	100 mΩ	10 mΩ	1 mA	0.02+2	0.02+5	0.014+3	15+0.25	15+1.5
	20 kΩ	19.999	19.9999	1 Ω	100 mΩ	100 μA	0.02+2	0.02+5	0.014+3	15+0.25	15+1.5
	200 kΩ	199.99	199.999	10 Ω	1 Ω	10 μA	0.02+2	0.02+5	0.02+5	20+0.25	20+1.5
	2000 kΩ	1999.9	1999.99	100 Ω	10 Ω	1 μA	0.03+2	0.03+10	0.03+10	35+2	35+5
	20 MΩ	19.999	19.9999	1 kΩ	100 Ω	100 nA	0.2+2	0.2+10	0.2+10	155+2	155+5
	200 MΩ	199.99	199.999	10 kΩ	1 kΩ	10 nA	1.5+2	1.5+10	1.5+10	1500+2	1500+5

Low power resistance measurement (LP-2WΩ)

Input terminal	Range	Maximum display		Resolution		Measurement current	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
VΩHz-COM A	200 Ω	199.99	199.999	10 mΩ	1 mΩ	1 mA	0.02+4	0.02+9	0.02+8	20+1	20+5
	2000 Ω	1999.9	1999.99	100 mΩ	10 mΩ	100 μA	0.03+4	0.03+9	0.03+8	20+1	20+3
	20 kΩ	19.999	19.9999	1 Ω	100 mΩ	10 μA	0.03+4	0.03+9	0.03+8	20+1	20+3
	200 kΩ	199.99	199.999	10 Ω	1 Ω	1 μA	0.03+4	0.03+9	0.03+8	30+1	30+3
	2000 kΩ	1999.9	1999.99	100 Ω	10 Ω	100 nA	0.2+4	0.2+12	0.2+12	150+2	150+5
	20 MΩ	19.999	19.9999	1 kΩ	100 Ω	10 nA	1.5+5	1.5+50	1.5+50	1500+2	1500+5

*1: For Auto-zero ON For Auto-zero OFF, 2 digits are added to the digit term. In addition, the resistance of the measurement cable and an offset error of 0.3 Ω are added.

Response time: 200 MΩ = 2 s
(Time until the measurement value reaches within 0.1% of the final value)
20 MΩ = 0.5 s
(Time until the measurement value reaches within 0.1% of the final value)

Voltage between input terminals without connections:
7.5 V or less

Maximum allowable applied voltage

Between VΩHz and COM A terminals	1000 V _{peak}
Between COM A terminal and the chassis	500 V
Between COM A and COM B terminals	200 V

7.1.4 DC Current Measurement (DCI)

7.1.4 DC Current Measurement (DCI)

7.1.4.1 Ach DC Current Measurement (DCI-Ach)

Input terminal	Range	Maximum display		Resolution		Resistance between terminals (A fuse resistance is included.)	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
mA-COM A	2000 nA	1999.9	1999.99	100 pA	10 pA	11.5 kΩ or less	0.15+35	0.15+40	0.15+40	150+4	150+4
	20 μA	19.999	19.9999	1 nA	100 pA		0.1+7	0.1+7	0.1+5	100+1	100+1.6
	200 μA	199.99	199.999	10 nA	1 nA	102 Ω or less	0.04+20	0.04+40	0.04+40	40+4	40+5
	2000 μA	1999.9	1999.99	100 nA	10 nA		0.04+2	0.04+7	0.04+5	40+1	40+1.6
	20 mA	19.999	19.9999	1 μA	100 nA	2 Ω or less	0.04+20	0.04+40	0.04+40	40+4	40+5
	200 mA	199.99	199.999	10 μA	1 μA		0.05+2	0.05+7	0.05+6	50+0.6	50+3
	2000 mA	1999.9	1999.99	100 μA	10 μA		0.15+2	0.15+7	0.15+6	150+0.6	150+3

*1: For Auto-zero ON For Auto-zero OFF, 2 digits are added to the digit term.

	Maximum allowable applied current	Input protection	Fuse replacement
Between mA and COM A terminals	2000 mA	2 A/250 V fast-blow fuse which is compliant with IEC60127 sheet1	On the rear panel

7.1.4.2 Bch DC Current Measurement (DCI-Bch)

Input terminal	Range	Maximum display		Resolution		Resistance between terminals (A fuse resistance is included.)	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
A-COM B	10 A	11.999	11.9999	1 mA	100 µA	0.03 Ω or less	0.2+2	0.2+7	0.2+5	200+0.6	200+3

*1: For Auto-zero ON For Auto-zero OFF, 2 digits are added to the digit term.

	Maximum allowable applied current	Input protection	Fuse replacement
Between A and COM B terminals	10 A	Protected by the 15 A/250 V fast-blow fuse with 10000 A breaking capacity.	Contact ADC CORPORATION to repair.

7.1.5 AC Current Measurement (ACI, ACI(AC+DC))

7.1.5 AC Current Measurement (ACI, ACI(AC+DC))

7.1.5.1 Ach AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)

Measurement method	True RMS measurement, RMS display
Input range	5% or more of a full scale
Crest factor	3:1 at a full scale
Temperature coefficient	(1/10 of measurement accuracy that includes the additional error)/°C in each range and frequency range
Response time	Approx. 1 s (Time until the measurement value reaches within 0.1% of the final value in the same range)

ACI-Ach

Input terminal	Range	Maximum display		Resolution		Resistance between terminals (A fuse resistance is included.)
		FAST/MED	SLOW1, 2	FAST/MED	SLOW1, 2	
mA-COM A	200 μ A	199.99	199.999	10 nA	1 nA	102 Ω or less
	2000 μ A	1999.9	1999.99	100 nA	10 nA	
	20 mA	19.999	19.9999	1 μ A	100 nA	2 Ω or less
	200 mA	199.99	199.999	10 μ A	1 μ A	
	2000 mA	1999.9	1999.99	100 μ A	10 μ A	0.6 Ω or less

Range	Measurement accuracy *1 $\pm$ (% of reading + digits)		
	20 Hz - 45 Hz	45 Hz - 1 kHz	1 kHz - 5 kHz
200 μ A	0.4+200	0.3+200	0.3+100
2000 μ A	0.5+200	0.35+100	0.35+200
20 mA	0.4+200	0.3+200	0.3+100
200 mA	0.4+200	0.3+200	0.3+100
2000 mA	0.5+200	0.35+100	0.7+200

*1: For sine-wave input When RATE is set to FAST or MED, the digit term is reduced to 1/10 of its specification.

ACI(AC+DC)-Ach

Input terminal	Range	Maximum display		Resolution		Resistance between terminals (A fuse resistance is included.)
		FAST/MED	SLOW1, 2	FAST/MED	SLOW1, 2	
mA-COM A	200 μ A	199.9	199.99	100 nA	10 nA	102 Ω or less
	2000 μ A	1999	1999.9	1 μ A	100 nA	
	20 mA	19.99	19.999	10 μ A	1 μ A	2 Ω or less
	200 mA	199.9	199.99	100 μ A	10 μ A	
	2000 mA	1999	1999.9	1 mA	100 μ A	0.6 Ω or less

7.1.5 AC Current Measurement (ACI, ACI(AC+DC))

Range	Measurement accuracy *1 $\pm$ (% of reading + digits)		
	20 Hz - 45 Hz	45 Hz - 1 kHz	1 kHz - 5 kHz
200 μ A	0.4+20	0.3+20	0.3+10
2000 μ A	0.5+20	0.35+10	0.35+20
20 mA	0.4+20	0.3+20	0.3+10
200 mA	0.4+20	0.3+20	0.3+10
2000 mA	0.5+20	0.35+10	0.7+20

*1: For sine-wave input When RATE is set to FAST or MED, the digit term is reduced to 1/10 of its specification.

Additional error depending on the crest factor (For a non sine-wave input current)
 $\pm$ (% of reading + % of range)

Range	Crest factor	
	1 - 2	2 - 3
200 μ A	0+0.05	0+0.15
2000 μ A	0+0.05	0+0.15
20 mA	0+0.05	0+0.15
200 mA	0+0.05	0.1+0.15
2000 mA	0+0.05	0.1+0.15

Additional error to the ACI(AC+DC)-ACh measurement

	FAST	MED
20 - 45 Hz	1.24%	0.72%
45 Hz - 1 kHz	0.06%	—

$\pm$ (% of reading)

	Maximum allowable applied current	Input protection	Fuse replacement
Between mA and COM A terminals	2 A	2 A/250 V fast-blow fuse which is compliant with IEC60127 sheet1	On the rear panel

7.1.5 AC Current Measurement (ACI, ACI(AC+DC))

7.1.5.2 Bch AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)

Measurement method	True RMS measurement, RMS display
Input range	5% or more of a full scale
Crest factor	3:1 at a full scale
Temperature coefficient	(1/10 of measurement accuracy that includes the additional error)/°C in each range and frequency range
Response time	Approx. 1 s (Time until the measurement value reaches within 0.1% of the final value in the same range)

ACI-Bch

Input terminal	Range	Maximum display		Resolution		Resistance between terminals (A fuse resistance is included.)
		FAST/MED	SLOW1, 2	FAST/MED	SLOW1, 2	
A-COM B	10 A	11.999	11.9999	1 mA	100 μ A	0.03 Ω or less

Range	Measurement accuracy *1 $\pm$ (% of reading + digits)		
	20 Hz - 45 Hz	45 Hz - 1 kHz	1 kHz - 5 kHz
10 A	0.5+200	0.5+200	0.7+200

*1: For sine-wave input When RATE is set to FAST or MED, the digit term is reduced to 1/10 of its specification.

ACI(AC+DC)-Bch

Input terminal	Range	Maximum display		Resolution		Resistance between terminals (A fuse resistance is included.)
		FAST/MED	SLOW1, 2	FAST/MED	SLOW1, 2	
A-COM B	10 A	11.99	11.999	10 mA	1000 μ A	0.03 Ω or less

Range	Measurement accuracy *1 $\pm$ (% of reading + digits)		
	20 Hz - 45 Hz	45 Hz - 1 kHz	1 kHz - 5 kHz
10 A	0.5+20	0.5+20	0.7+20

*1: For sine-wave input When RATE is set to FAST or MED, the digit term is reduced to 1/10 of its specification.

Additional error depending on the crest factor (For a sine-wave input current)
 $\pm$ (% of reading + % of range)

Range	Crest factor	
	1 - 2	2 - 3
10 A	0+0.05	0.07+0.15

7.1.5 AC Current Measurement (ACI, ACI(AC+DC))

Additional error to the ACI(AC+DC)-Bch measurement

	FAST	MED
20 - 45 Hz	1.24%	0.72%
45 Hz - 1 kHz	0.06%	—

 $\pm$ (% of reading)

	Maximum allowable applied current	Input protection	Fuse replacement
Between A and COM B terminals	10 A	Protected by the 15 A/250 V fast-blow fuse with 10000 A breaking capacity.	Contact ADC CORPORATION to repair.

7.1.6 Ach Frequency Measurement (FREQ-Ach)

7.1.6 Ach Frequency Measurement (FREQ-Ach)

Measurement method Reciprocal

Input terminal	Measurement frequency range	Measurement accuracy
VΩHz-COM A	10 Hz - 300 kHz	0.02% of reading

A frequency over the above range is displayed but not guaranteed.

Input signal condition For sine-wave

Input signal voltage range 100 mVrms to 700 Vrms and 10% of each voltage range or more
(However, the input signal is restricted to the maximum allowable applied voltage.)

Gate time

Sampling rate	Gate time	Measurement frequency range	Maximum measurement period	Maximum display
SLOW	1000 ms	1 Hz to 300 kHz	2.2 s	999999
MED	100 ms	10 Hz to 300 kHz	220 ms	99999
FAST	10 ms	100 Hz to 300 kHz	22 ms	9999

Maximum allowable applied voltage

Between VΩHz and COM A terminals	700 Vrms, 1000 Vpeak, 2.2×10^7 V · Hz
Between COM A terminal and the chassis	500 V
Between COM A and COM B terminals	200 V

7.1.7 Temperature Measurement (TEMP)

		Thermocouple	
		K(CA)	T(CC)
Input terminal	Range	-50°C to 1370°C	-50°C to 400°C
TEMP HI - LO	Maximum display	13700	4000
	Resolution	0.1°C	0.1°C
	Measurement accuracy	±0.15%±3°C	±0.15%±3°C

Maximum allowable applied voltage

Between TEMP (HI/LO) and COM A terminals	200 Vpeak
Between TEMP (HI/LO) and COM B terminals	200 Vpeak
Between TEMP (HI) and TEMP (LO) terminals	36 Vpeak
Between TEMP (HI/LO) terminal and the chassis	500 V

Compliant thermocouple standard: JIS C1602

Cold junction compensation: Internal

7.1.8 Ach Diode Measurement-Ach

Input terminal	Range	Maximum display		Resolution		Measurement current	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
VΩHz-COM A	2000 mV	1999.9	1999.99	100 μV	10 μV	1 mA	0.014+2	0.014+5	0.014+3	15+0.25	15+1.5

*1: For Auto-zero ON For Auto-zero OFF, 2 digits are added to the digit term. In addition, an offset error, which is obtained by multiplying (the resistance of the measurement cable + 0.3 Ω) by 1 mA, is added.

Other specifications are the same as the resistance measurement function.

7.1.9 Ach Continuity Measurement-Ach

7.1.9 Ach Continuity Measurement-Ach

Input terminal	Range	Maximum display		Resolution		Measurement current	Measurement accuracy *1 ±(% of reading + digits)			Temperature coefficient ±(ppm of reading + digits)/°C	
		FAST	MED/ SLOW1, 2	FAST	MED/ SLOW1, 2		FAST	MED	SLOW1, 2	Auto-zero ON	Auto-zero OFF
VΩHz-COM A	2000 Ω	1999.9	1999.99	100 mΩ	10 mΩ	1 mA	0.014+2	0.014+5	0.014+3	15+0.25	15+1.5

*1: For Auto-zero ON For Auto-zero OFF, 2 digits are added to the digit term.

Continuity judgment value: 1 Ω to 1000 Ω

Other specifications are the same as the resistance measurement function.

7.1.10 Calculation Functions

- NULL calculation
Display value (NULL) = Measured value - NULL constant
- Smoothing calculation
Display value (SM) = Moving average over a specified number of measurements
- Comparator calculation
Display (HIGH) $\leftarrow$ HIGH setting value < Measured value
Display (LOW) $\leftarrow$ Measurement value < LOW setting value
Display (GO) $\leftarrow$ LOW setting value $\leq$ Measured value $\leq$ HIGH setting value
- Scaling calculation
Display (SCL) = (Measurement value - B)/A $\times$ C
A, B, and C are constants. (Setting values)
- MAX and MIN calculation
Display value (MAX) = Maximum measured value after the calculation starts
Display value (MIN) = Minimum measured value after the calculation starts
Display value (AVE) = Averaged value after the calculation starts (Remote output only)
- dB and dBm calculation
dB display = $20 \log (\text{Measured value}/D)$
dBm display value = $10 \log ((\text{Measured value})^2/D)/10^{-3}$
D is a constant. (Set by the user)
- Statistical calculation

Number of samples	Display value (SAMPLE) = Number of measured values in the specified range of the measurement memory
Maximum value	Display value (MAX) = Maximum measured value in the specified range of the measurement memory
Minimum value	Display value (MIN) = Minimum measured value in the specified range of the measurement memory
Average value	Display value (AVE) = Averaged value in the specified range of the measurement memory
Standard deviation	Display value (SIGMA) = Standard deviation in the specified range of the measurement memory
Dispersion	Display value (P-P) = ((Maximum measured value) - (Minimum measured value)) in the specified range of the measurement memory
- Calculation between 2 measurements
Display value (M1+M2) = Left side display: M1 + Right side display: M2
Display value (M1-M2) = Left side display: M1 - Right side display: M2
Display value (M1*M2) = Left side display: M1 * Right side display: M2
Display value (M1/M2) = Left side display: M1 / Right side display: M2

7.2 Interface Specifications

7.2 Interface Specifications

Item	Specifications
Remote control	
Remote command	Compliance with the command format for ADC, SCPI and R6452A.
Interface	
USB	
Standard	Full speed Compliance with USB2.0
Connector	Type B
GPIB (7352A)	
Standard	Compliance with IEEE488.2-1987
Connector	24-pin Anphenol
Interface functions	SH1, AH1, T5, L4, SR1, RL1, PP0, DC1, DT1, C0, E2
Output format	ASCII
Addressing	31 kinds of talker/listener addresses can be specified from the front panel.
EIA232 (7352A)	
Standard	Compliance with EIA232 (RS-232)
Connector	D-Sub9 pin
Baud rate	9600, 4800, 2400, 1200, 600, 300
Parity	Even number (EVEN), odd number (ODD), or none
Number of data bits	7 bits or 8 bits
Number of stop bits	1 bit or 2 bits
Echo	ON, OFF
External trigger signal (7352A)	
Connector	BNC
Signal level	TTL Detecting the falling edge
Pulse width	1 μ s or more

7.3 General Specifications

Item	Specifications																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Operation environment	Ambient temperature: 0°C to +50°C However, 0 to +45°C in cases where 1 A or more and 5 A or more are simultaneously applied to Ach and Bch respectively for the current measurement. Relative humidity: 85% or less no condensation																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Storage environment	Ambient temperature: -25°C to +70°C																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Warm-up time	60 minutes or more																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Display	Dual 6-digit and 17-segment fluorescent display																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Range switching	Automatic and manual																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Input method	Floating																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Maximum allowable applied voltage	<table><tr><td rowspan="8">Chassis</td><td colspan="3">Ach INPUT terminal</td><td colspan="5">Bch INPUT terminal</td><td>TEMP LO</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>TEMP HI</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>36</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td>200</td></tr><tr><td colspan="3"></td><td colspan="5"></td><td></td></tr></table>	Chassis	Ach INPUT terminal			Bch INPUT terminal					TEMP LO									TEMP HI									36									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									200									
Chassis	Ach INPUT terminal			Bch INPUT terminal					TEMP LO																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
									TEMP HI																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
									36																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
									200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
									200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
									200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
									200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
									200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
								200																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									

7.4 Accessories

Table 7-1 Accessories

Name	Model	Remarks
Input cable	CC010001	Standard accessory
	A01001	Shielded cable 33 VAC and 70 VDC or less
Alligator clip adapter	CC015001	33 VAC and 70 VDC or less
RS-232 cable	A01265	
Terminal adapter	TR1111	
JIS rack-mount set	A02263	
	A02264	Twins
EIA rack-mount set	A02463	
	A02464	Twins
Panel-mount set	A02039	
	A02040	Twins
Sheath type thermocouple	TR1101-100	T(CC) type
	TR1101-130	K(CA) type

8. MAINTENANCE

This chapter describes the following information which relates to the maintenance of this instrument.

- 8.1 Replacing Fuses
- 8.2 Cleaning
- 8.3 Calibration
- 8.4 Replacing Parts with Limited Life
- 8.5 Product Disposal and Recycle
- 8.6 Storage
- 8.7 Transportation
- 8.8 Notes Regarding Repair, Replacement, and Periodic Calibration
- 8.9 System Recovery Procedure
- 8.10 Self-Test
- 8.11 Error Message List

8.1 Replacing Fuses

This instrument includes a power fuse and protection fuse. Replace them according to the following procedure.

CAUTION:

1. *To prevent fire and electrical shock, replace with a fuse listed in Table 3-1. Never use an incompatible fuse and short-circuit the fuse holder.*
 2. *Whether a fuse has blown cannot be checked by observation. Measure the resistance of the fuse and judge whether the fuse has blown. (If the resistance is 15 Ω or less, the fuse has not blown.)*
 3. *Do not cut any of the internal ground wiring in this instrument or disconnect this instrument from ground. This instrument may become unsafe to operate.*
-

8.1.1 Replacing a Power Fuse

The power fuse is placed in the AC power supply connector on the rear panel.

For more information on how to check and replace the power fuse, refer to Section 3.4.2, “Changing the Power Supply Voltage.”

8.1.2 Replacing a Protection Fuse

8.1.2 Replacing a Protection Fuse

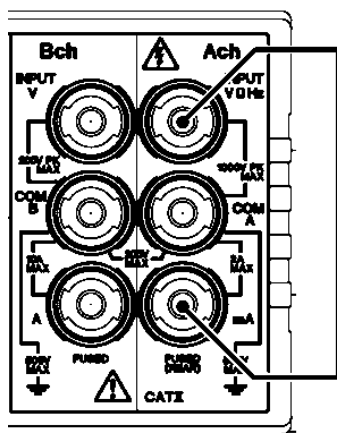
To prevent fire and injuries when an over-current is input to the mA current input terminal, a fast-blow fuse is included on the rear panel of this instrument.

8.1.2.1 How to Check Protection Fuses

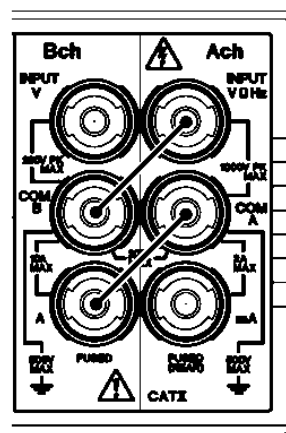
Measure the resistance of the protection fuse and check that the fuse is not open.

This instrument can measure the resistance of the protection fuse by connecting the terminals as shown in Figure 8-1.

- a. For the mA terminal
 - Connect the mA and Ω terminals.
 - Set the resistance measurement function and measure the resistance of the protection fuse.
- b. For the A terminal
 - Connect the A and COM A terminals, and connect the COM B and V Ω Hz terminals.
 - Set the resistance measurement function and measure the resistance of the protection fuse.



a. For the mA terminal (Ach)



b. For the 10 A terminal (Bch)

Figure 8-1 How to Check Protection Fuses

8.1.2.2 How to Replace Protection Fuses

1. Rotate the fuse cap in the counter-clockwise direction by using a flathead screwdriver and remove the fuse.
2. Replace with a fuse, which is compliant with the standard and listed in Table 3-1, and secure the fuse cap.

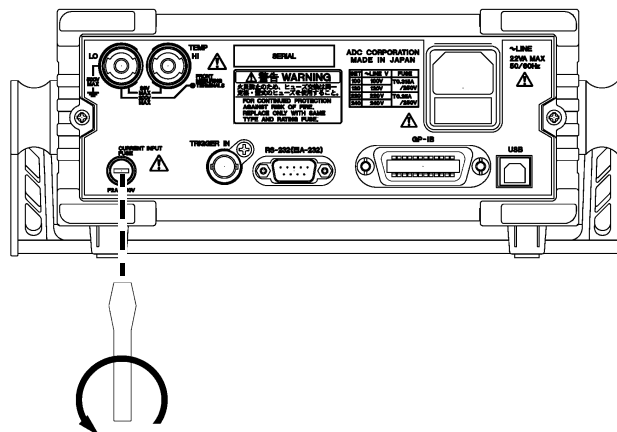


Figure 8-2 Input Terminal Protection Fuse

A protection fuse for the A current input terminal is installed inside this instrument.

Measure the resistance between the A and COM B terminals. If the protection fuse is open, contact ADC CORPORATION sales representative for repair of this instrument.

8.2 Cleaning

8.2 Cleaning

This section describes the cleaning procedure of this instrument and warning information.

WARNING: *To prevent accidents caused by electric shocks, remove all cables from this instrument.
Never attempt to remove the cover to clean the inside of this instrument.*

8.2.1 Cleaning the Outside

Use a soft or damp cloth to clean this instrument.

NOTE:

1. *Do not allow water inside this instrument or lint on this instrument.*
 2. *Do not use an organic solvent such as benzene or acetone for cleaning because it degrades the plastic.*
-

8.2.2 Cleaning Others

Take appropriate precautions to protect this instrument from dust.

WARNING: *Periodically remove any dust which builds up on the AC power connector and plug. The tracking phenomenon, which can be caused by wet dust, may cause a fire.*

8.3 Calibration

To maintain the measurement accuracy of this instrument, perform calibration at least once a year; warranty period.

8.3.1 Preparing for Calibration

1. Power supply
Use an AC power supply in a voltage range listed in Table 3-4 and frequency of 50 Hz or 60 Hz.
2. Environment
Perform calibration in following conditions.
Temperature: $+23 \pm 3^{\circ}\text{C}$
Relative Humidity: 70% or less
An area free from dust, vibration, wind, and noise
3. Warm-up time
Perform calibration after allowing a warm up time of 60 minutes or more.
Sufficient warm-up time is also required for the instrument used for calibration.

8.3.2 Calibration Standards

8.3.2 Calibration Standards

Table 8-1 Calibration Standards

Calibration standard	Operation range	Accuracy
DC voltage	50 mV	± 20 ppm
	190 mV - 1000 V	± 15 ppm
AC voltage	5 mVrms 1 kHz sine-wave	± 1400 ppm
	10 mVrms - 100 mVrms 1 kHz sine-wave	± 1000 ppm
	190 mVrms - 700 Vrms 1 kHz sine-wave	± 200 ppm
Resistance	190 Ω	± 50 ppm
	1.9 k Ω - 190 k Ω	± 20 ppm
	1.9 M Ω	± 50 ppm
	19 M Ω - 100 M Ω	± 1000 ppm
DC current	1.9 μ A	± 530 ppm
	19 μ A	± 200 ppm
	190 μ A - 19 mA	± 100 ppm
	190 mA - 1.9 A	± 150 ppm
	10 A	± 1000 ppm
AC current	10 μ A 1 kHz sine-wave	± 2500 ppm
	19 μ A 1 kHz sine-wave	± 1500 ppm
	50 μ A 1 kHz sine-wave	± 1000 ppm
	100 μ A - 190 mA 1 kHz sine-wave	± 500 ppm
	0.5 A - 10 A 1 kHz sine-wave	± 1000 ppm

8.3.3 Calibration Point

1. Perform calibration for each range of each measurement function. For more information on relationships between functions and ranges, refer to Table 5-2.
2. First, calibrate the zero point in each range calibration.
3. In the DC voltage measurement and DC current measurement, calibrate the zero point and full-scale point.
4. In the resistance measurement, calibrate the zero point and full-scale point.
5. In the low-power resistance measurement, calibrate the zero point and full-scale point.
6. In the AC voltage measurement and AC current measurement, calibrate the zero point, calibration point1, calibration point2, and calibration point3.
7. In the ACV(AC+DC) and ACI(AC+DC) measurements, no calibration is required because the instrument is calibrated when the calibration is performed in the DC voltage and AC voltage measurements.
8. In the continuity and diode measurements, no calibration is required because the instrument is calibrated when the calibration is performed in the DC voltage and resistance measurements.
9. In the frequency measurement, no calibration is required.
10. Calibration for the temperature measurement is performed by using the DC voltage. The zero and full scale points are calibrated.

Recommended input range: +5% to -10% for each calibration point

Table 8-2 Calibration Point (1 of 2)

Measurement function	Range	Zero	Calibration point1	Calibration point2	Calibration point3
DCV-Ach	200 mV	Short	190 mV		
	2000 mV	Short	1900 mV		
	20 V	Short	19 V		
	200 V	Short	190 V		
	1000 V	Short	1000 V		
ACV-Ach	200 mV	Short	5 mV 1 kHz	19 mV 1 kHz	190 mV 1 kHz
	2000 mV	Short	50 mV 1 kHz	190 mV 1 kHz	1900 mV 1 kHz
	20 V	Short	500 mV 1 kHz	1900 mV 1 kHz	19 V 1 kHz
	200 V	Short	5 V 1 kHz	19 V 1 kHz	190 V 1 kHz
	700 V	Short	7 V 1 kHz	70 V 1 kHz	700 V 1 kHz

8.3.3 Calibration Point

Table 8-2 Calibration Point (2 of 2)

Measurement function	Range	Zero	Calibration point1	Calibration point2	Calibration point3
2WΩ-Ach	200 Ω	Short	190 Ω		
	2000 Ω	Short	1900 Ω		
	20 kΩ	Short	19 kΩ		
	200 kΩ	Short	190 kΩ		
	2000 kΩ	Short	1900 kΩ		
	20 MΩ	Short	19 MΩ		
	200 MΩ	Short	100 MΩ		
LP-2WΩ-Ach	200 Ω	Short	190 Ω		
	2000 Ω	Short	1900 Ω		
	20 kΩ	Short	19 kΩ		
	200 kΩ	Short	190 kΩ		
	2000 kΩ	Short	1900 kΩ		
	20 MΩ	Short	19 MΩ		
DCI-Ach	2000 nA	Open	1900 nA		
	20 μA	Open	19 μA		
	200 μA	Open	190 μA		
	2000 μA	Open	1900 μA		
	20 mA	Open	19 mA		
	200 mA	Open	190 mA		
	2000 mA	Open	1900 mA		
ACI-Ach	200 μA	Open	10 μA 1 kHz	19 μA 1 kHz	190 μA 1 kHz
	2000 μA	Open	50 μA 1 kHz	190 μA 1 kHz	1900 μA 1 kHz
	20 mA	Open	500 μA 1 kHz	1900 μA 1 kHz	19 mA 1 kHz
	200 mA	Open	5 mA 1 kHz	19 mA 1 kHz	190 mA 1 kHz
	2000 mA	Open	50 mA 1 kHz	190 mA 1 kHz	1900 mA 1 kHz
DCV-Bch	200 mV	Short	190 mV		
	2000 mV	Short	1900 mV		
	20 V	Short	19 V		
	200 V	Short	190 V		
DCI-Bch	10 A	Open	10 A		
ACI-Bch	10 A	-	100 mA 1 kHz	500 mA 1 kHz	10 A 1 kHz
TEMP-Bch		Short	50 mV		

8.3.4 Calibration Procedure

Perform calibration in following procedure.

1. Set the calibration mode.
2. Short-circuit the Ach VΩHz and COM A terminals, the Bch V and COM B terminals, the TEMP HI and LO terminals on the rear panel, and then open the mA and A terminals.
3. Calibrate the zero point in a function and range to be calibrated.
4. Connect the calibration standard to the input terminal.
5. Calibrate the full-scale point in a function and range to be calibrated.
6. Cancel the calibration mode.

Setting the calibration mode

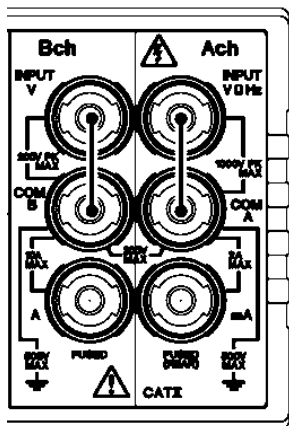
1. Select ON from 'CAL' in **SYSTEM** in **MENU**.
2. Press **ENTER** to apply the setting.
3. Press **EXIT** to exit from MENU.

The CAL indicator is turned on and indicates that the calibration mode is set.

4. Set a function to be calibrated.

Calibrating the zero point

5. Short-circuit the Ach VΩHz and COM A terminals, the Bch V and COM B terminals, the TEMP HI and LO terminals on the rear panel, and then open the mA and A terminals.



6. Set a range to be calibrated.

8.3.4 Calibration Procedure

7. Pressing the **HOLD** key displays the measured value as a setting value.
8. Adjust the setting value to the calibration value by using the $\triangle$, ∇ , $\triangleleft$, and $\triangleright$ keys.

NOTE: Press the **EXIT** key if calibration must be cancelled due to a mistake such as entering an incorrect number.

9. Press the **ENTER** key to execute the calibration.
10. Repeat steps 6 to 9 when calibrating other ranges.

Calibrating the full-scale point

11. Connect the calibration standard to the input terminal.
12. Set a range to be calibrated.
For multiple calibration points, calibrate them in the order of calibration point1, 2, and 3.
13. Input a voltage equivalent to the calibration point.
14. Pressing the **HOLD** key displays the measured value as a setting value.
15. Adjust the setting value to the calibration value by using the $\triangle$, ∇ , $\triangleleft$, and $\triangleright$ keys.

NOTE: Press the **EXIT** key if the calibration must be cancelled due to a mistake such as entering an incorrect number.

16. Press the **ENTER** key to execute the calibration.
17. Repeat steps 12 to 16 when calibrating other ranges.
18. Repeat steps 4 to 17 when calibrating other functions.

Canceling the calibration mode

19. Select OFF from 'CAL' in **SYSTEM** in **MENU**.
20. Press **ENTER** to apply the setting.
21. Press **EXIT** to exit from MENU.
The CAL indicator is turned off and indicates that the calibration mode is canceled.

NOTE:

1. The **AUTO**, **RATE**, **TRIG**, **NULL**, and **SM** keys are disabled in the calibration mode.
Only **I/F** and **SYSTEM** can be selected in **MENU**.
 2. Before turning off the power, cancel the calibration mode.
Calibration data is stored in internal nonvolatile memory and written in it at once when the calibration mode is canceled.
-

8.4 Replacing Parts with Limited Life

8.4 Replacing Parts with Limited Life

Parts with limited life used in this instrument are listed in the following table.

Contact an ADC CORPORATION sales representative to replace any parts, referring to the limited lives listed in the following table.

Note that the life span may be shorter than that described depending on the operating environment, frequency of use, and storage environment.

MEMO: *The table shows the life spans of parts and the number of times parts can be used before requiring replacement, but they are not guaranteed.*

Table 8-3 Parts with Limited Life

Name	Expected lifespan
Relays for switching functions and ranges	1,000,000 operations
Panel keys	500,000 operations
Fluorescent display tube	20,000 hours
USB connector	1,500 operations

8.5 Product Disposal and Recycle

Disposal of this product should comply with the regulations and laws that are established by your country and municipality.

When treating this product, separately collect components according to this chapter to prevent the spread of substances, which may be harmful to humans, and to protect the global environment.

Components, which must be separately collected, are shown in the following table.

The treatment of this product should comply with the relevant laws of your country and waste-disposal regulations of your company.

Substance/Component	Used/ Not used	Location	Unit	Component
Polychlorinated biphenyls (PCB) containing capacitors	Not used	-	-	-
Mercury containing components	Not used	-	-	-
Batteries	Not used	-	-	-
Printed circuit boards	Used	Main frame	MAIN	Printed circuit board
Toner cartridges	Not used	-	-	-
Plastic containing brominated flame retardants	Used	Main frame	BPL-034166X03 BPL-034166X04 BPL-034167X03	Connector, diode, zener diode, photocoupler, FET, analog IC, logic IC, FLASH, and transistor
Asbestos waste and components which contain asbestos	Not used	-	-	-
Cathode ray tubes	Not used	-	-	-
Chlorofluorocarbons (CFC), Hydrochlorofluorocarbons (HCFC), Hydrofluorocarbons (HFC) or Hydrocarbons (HC)	Not used	-	-	-
Gas discharge lamps	Not used	-	-	-
Liquid crystal displays of a surface greater than 100 square centimeters	Not used	-	-	-
External electric cables	Used	Main frame		Power cable Input probe
Components containing refractory ceramic fibers	Not used	-	-	-

8.5 Product Disposal and Recycle

Substance/Component	Used/ Not used	Location	Unit	Component
Components containing radioactive substances	Not used	-	-	-
Electrolyte capacitors containing substances of concern (height > 25 mm, diameter > 25 mm or proportionately similar volume)	Not used	-	-	-
Arsenic and Arsenic compounds	Used	Main frame	Electronic components	Photocoupler and logic IC
Nickel and Nickel compounds	Used	Main frame		Electronic components and mechanical parts
Lead and Lead compounds	Used	Main frame	BPL-034166X03 BPL-034166X04 BPL-034167X03	Lead solder used for assembling electronic components on printed circuit boards
Polyvinyl chloride (PVC)	Used	Main frame		PVC resin components
Antimony and Antimony compounds	Used	Main frame		Electronic components

8.6 Storage

If not using this instrument for a long time, put it in a carton case and store it in an area away from dust and direct sunlight.

Storage temperature: -25°C to $+70^{\circ}\text{C}$

8.7 Transportation

When transporting this instrument, use the packing materials used for the shipping of this instrument or equivalent materials (thickness of carton case is 5 mm or more).

Packing procedure

1. Place the cushioning materials inside the carton case and cover all sides of this instrument with cushioning materials.
2. Place accessories in the carton case and then place the cushioning materials.
3. Close the carton case and bind it with packing strings.

8.8 Notes Regarding Repair, Replacement, and Periodic Calibration

8.8.1 Before Asking for Repair

Before asking for repair, refer to and check Table 8-4.

Table 8-4 Before Asking for Repair

Problem	Check item	Solution
The power supply cannot be turned on.	Check the power cable.	Change the power cable.
	Check the power fuse.	Replace the power fuse.
Key malfunction.	Check whether the RMT indicator is on (in remote operation).	Press the LOCAL key to return to the local state.
	Panel lock functions.	Release the panel lock.
Voltage and resistance cannot be measured.	Check that input cables are correctly connected between the VΩHz and COM A terminals, and between the V and COM B terminals.	Securely connect input cables between the VΩHz and COM A terminals, and between the V and COM B terminals.
	Check whether the input cable is in good condition and not broken.	Change the input cable.
Current cannot be measured.	Input terminals differ depending on the measurement range. Check that the input cable is correctly connected to the current input terminals. 2000 nA and 2000 mA ranges: mA and COM A terminals 10 A range: A and COM B terminals	Securely connect the input cable to the correct terminals.
	Check whether the protection fuse is serviceable and not open. (For more information on how to check, refer to 8.1, “Replacing Fuses”.)	Replace the fuse. For the A terminal, contact ADC CORPORATION or a sales representative to repair this instrument.
	Check whether the input cable is in good condition and not broken.	Change the input cable.
For the high-resistance measurement, the measurement value is not stable.	Induction noise is superimposed on the input cable.	Shield the DUT. Use a shielded input cable.
	Check that the power supply frequency setting is the same as the local commercial power supply frequency.	Perform the measurement with SLOW1 or SLOW2.
Sampling does not operate.	Check whether the HOLD indicator is on (in HOLD operation).	Press the TRIG or HOLD key and check whether sampling operates. (During the remote operation, panel keys do not function. Therefore, after releasing the remote operation, press any panel key.)
	A long trigger delay time is set.	Check the trigger delay time.

8.8.2 Contacting ADC CORPORATION for Repair or Calibration

When sending this instrument to ADC CORPORATION sales representative, attach a tag that indicates the following information.

- Your company name and address
- Name of the person in charge
- Serial number (on the rear panel)
- What work to request. (Repair or periodic calibration)

8.8.3 Address and Phone Number

Contact ADC CORPORATION sales representative.

8.9 System Recovery Procedure

Initialization

The following describes how to initialize the setting parameters.

1. Select **INIT** from the **MENU** mode and press ∇ to enter the selection layer.
2. The 'PARAM' parameter is displayed. Press ∇ to enter the entry layer.
3. Press **ENTER** while 'DFLT' is displayed.

The parameters can be initialized by selecting and executing 'DFLT' when loading the parameter setting.

For more information on how to initialize the parameters, refer to Section 5.10, "Saving and Loading the Measurement Condition Settings."

Initial value when shipping

The following describes how to set all parameters except for the calibration value and calibration memo to the values when shipping.

1. Turn on the power while pressing the **NULL** and **UP** keys.

Quitting MENU

Press **EXIT** to quit the operation in MENU.

MENU is closed even if any layer in MENU is operated.

NOTE: *Changing parameters, which are not appliedd by pressing **ENTER**, is canceled.*

Initial value

The list of conditions, under which the setting parameters are initialized, and initial values are shown in Table 8-5.

Table 8-5 Initial Values of Setting Parameters (1 of 3)

Parameter	Left side display	Right side display	When shipping	Selecting DFLT from INIT in MENU	Selecting P.OFF from 'P.ON.LD'	'LOAD' in PARAM	*RST	*CLS	DCL
Right side display OFF/ON		●	ON	ON	Saved value	Selecting USER0-3	ON		
Function	●	●	DCV	DCV	Saved value	Selecting USER0-3	DCV		
Range	●	●	Highest	Highest	Saved value	Selecting USER0-3	Highest		
Auto-range	●	●	ON	ON	Saved value	Selecting USER0-3	ON		
Number of display digits.	●	●	Maximum	Maximum	Saved value	Selecting USER0-3	Maximum		
Auto-zero		●	ON	ON	Saved value	Selecting USER0-3	ON		
Temperature sensor		●	K	K	Saved value	Selecting USER0-3	K		
NULL calculation	●	●	OFF	OFF	Saved value	Selecting USER0-3	OFF		
NULL constant	●	●	0	0	Saved value	Selecting USER0-3	0		
COMP calculation		●	OFF	OFF	Saved value	Selecting USER0-3	OFF		
HIGH value		●	0	0	Saved value	Selecting USER0-3	0		
LOW value		●	0	0	Saved value	Selecting USER0-3	0		
MAX/MIN calculation	●	●	OFF	OFF	Saved value	Selecting USER0-3	OFF		
Selecting the MAX/MIN display	●	●	NORMAL	NORMAL	Saved value	Selecting USER0-3	NORMAL		
Smoothing calculation	●	●	OFF	OFF	Saved value	Selecting USER0-3	OFF		
Smoothing count	●	●	10	10	Saved value	Selecting USER0-3	10		
Scaling calculation	●	●	OFF	OFF	Saved value	Selecting USER0-3	OFF		
Constant A	●	●	1	1	Saved value	Selecting USER0-3	1		
Constant B	●	●	0	0	Saved value	Selecting USER0-3	0		
Constant C	●	●	1	1	Saved value	Selecting USER0-3	1		
dB calculation	●	●	OFF	OFF	Saved value	Selecting USER0-3	OFF		
Constant D	●	●	1	1	Saved value	Selecting USER0-3	1		
Calculation between 2 measurements		●	OFF	OFF	Saved value	Selecting USER0-3	OFF		
Sampling HOLD		●	OFF	OFF	Saved value	Selecting USER0-3	OFF		
Sampling rate		●	SLOW1	SLOW1	Saved value	Selecting USER0-3	SLOW1		
Trigger delay		●	0	0	Saved value	Selecting USER0-3	0		

Blank: No Action

Saved value: Last setting value when the power is turned off.

8.9 System Recovery Procedure

Table 8-5 Initial Values of Setting Parameters (2 of 3)

Parameter	Left side display	Right side display	When shipping	Selecting DFLT from INIT in MENU	Selecting P.OFF from 'P.ON.LD'	'LOAD' in PARAM	*RST	*CLS	DCL
Each trigger delay for the right and left displays	●		OFF	OFF	Saved value	Selecting USER0-3	0		
Sampling count	●		1	1	Saved value	Selecting USER0-3	1		
Memory store	●		OFF	OFF	Saved value	Selecting USER0-3	OFF		
Continuity threshold constant	●		10 Ω	10 Ω	Saved value	Selecting USER0-3	10 Ω		
Trigger count	●		1	1	Saved value	Selecting USER0-3	1		
Trigger source	●		IMMediate	IMMediate	Saved value	Selecting USER0-3	IMMediate		
PASS/FAIL buzzer	●		OFF	OFF	Saved value	Selecting USER0-3	OFF		
Buzzer	●		OFF	OFF	Saved value	Selecting USER0-3	OFF		
Interface	●		USB	USB	Saved value				
GPIB address	●		1	1	Saved value				
USB ID	●		1	1	Saved value				
Baud rate	●		9600	9600	Saved value				
Data length	●		8	8	Saved value				
Parity	●		None	None	Saved value				
Stop bits	●		1	1	Saved value				
Echo	●		OFF	OFF	Saved value				
Talk only	●		OFF	OFF	Saved value				
Header output	●		ON	ON	Saved value				
Output data element	●		OFF	OFF	Saved value	Selecting USER0-3	OFF		
Output data selection	●		Right and left sides	Right and left sides	Saved value	Selecting USER0-3	Right and left sides		
Panel lock	●		OFF		Saved value				
Disabling function	●		ENA	ENA	Saved value				
PSC flag	●		1	1	Saved value				
Programming language	●		ADC	ADC	Saved value				
Continue setting	●		OFF	OFF	Saved value		*1		
Password	●		0000		Saved value				
Measurement memory	●		Empty	Empty	Empty	Empty	Empty		Empty

Blank: No Action

Saved value: Last setting value when the power is turned off.

*1: By selecting the initial value for continue setting

Table 8-5 Initial Values of Setting Parameters (3 of 3)

Parameter	Left side display	Right side display	When shipping	Selecting DFLT from INIT in MENU	Selecting P.OFF from 'P.ON.LD'	'LOAD' in PARAM	*RST	*CLS	DCL
GPIB input output buffer		●	Empty	Empty	Empty	Empty			Empty
Block delimiter		●	CR/LF+EOI	CR/LF+EOI	CR/LF+EOI	CR/LF+EOI	CR/LF+EOI		
Memory readout range	●	●	0,0	0,0	0,0	0,0			0,0
SRQ		●	OFF	OFF	OFF	OFF			
Display ON/OFF		●	ON	ON	ON	ON	ON		
Calibration mode		●	OFF	OFF	OFF		OFF		
Status register QEER	●	●	0	0	0				
Status register OEER		●	0	0	0				
Status register SESER		●	0	0	*2				
Status register MEER	●	●	0	0	0				
Status register SRER		●	0	0	*2				
Status register QER	●	●	0	0	0			0	
Status register STB		●	0	0	0			0	
Status register OER		●	0	0	0			0	
Status register SESR		●	0	0	0			0	
Status register MER	●	●	0	0	0			0	
Error queue		●	Empty	Empty	Empty			Empty	

Blank: No Action

Saved value: Last setting value when the power is turned off.

*2: By setting the *PSCn command

8.10 Self-Test

8.10 Self-Test

8.10.1 Self-Test

The self-test can be performed from the panel operations or by executing the remote command.

The self-test is performed by selecting 'TEST' from **SYSTEM** category in **MENU** or executing the *TST? remote command.

The test items are shown in Table 8-6.

Table 8-6 Self-Test Items

Item	Can be performed from the Panel or by Remote	Can be performed when the power is turned on
ROM check sum	-	○
RAM read/write check	-	○
Panel display check	-	○
Alarm status check	-	○
Calibration data check	○	○
Parameter check	○	○
Analog to logic communication check	○	○
AD operation check in analog section	○	○
Range operation check in analog section	○	○
Reference voltage check in analog section	○	○
Reference current check in analog section	○	○

8.10.2 Display and Key Tests

Turn on all segments of the display and check whether they light.

Pressing panel keys display their names.

Test according to the following procedure.

- Operation
1. Press **MENU** and select **SYSTEM** by using ◀ and ▶.
 2. Press ◀.
 3. Select 'P/KEY' by using ▶.
 4. Press ◀.
 5. Press **ENTER**.
The display changes after a few seconds.
 6. Press a key to be checked.
 7. To quit the key test, press **SHIFT**.
The display changes from "SHIFT" to "DONE".
 8. Press **EXIT** to exit from MENU.

Display section

MENU	SYSTEM
SYSTEM	BEEP
SYSTEM	P/KEY
P/KEY	ENTER
All segments of the display are turned on.	
KEY	Push
KEY	Key name to be pressed
KEY	SHIFT
KEY	DONE
Measurement state	

8.11 Error Message List

8.11 Error Message List

This section describes the error number and how to deal with error displayed on this instrument.

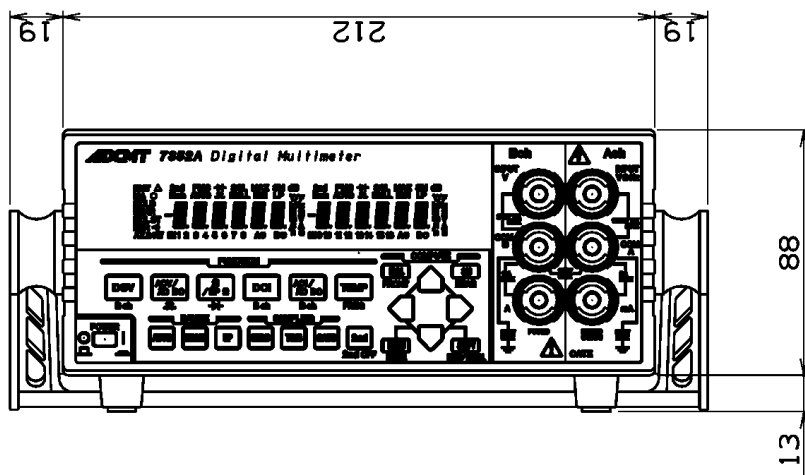
No.	Error display	Error description []: Response to the ERR? command	Solution
1	Err 1	RAM read/write error	Repair
2	Err 2	Display RAM test error or Communication error	Repair
3	Err 3	Calibration data check SUM error	Repair
4	Err 4	ROM check SUM error	Repair
5	Err 6	Parameter check SUM error	Repair
6	Err 16	FLASH memory writing error	Repair
7	Err 101	The function cannot be set. [Inhibited function]	Solution1
8	Err140	RS-232 communication error [RS-232 Communication Error]	Solution10
9	Err 150	USB communication error [Illegal packet received]	Solution2
10	Err 200	Ach Ratio test error of IR1 to IR2 in AD opeeration	Repair
11	Err 201	Ach Ratio test error of IR2 to IR3 in AD operation	Repair
12	Err 205	Ach ACx1 zero measurement test error in analog operation	Repair
13	Err 206	Ach Lx1 zero measurement test error in analog operation	Repair
14	Err 207	Ach Lx10 zero measurement test error in analog operation	Repair
15	Err 208	Ach Lx100 zero measurement test error in analog operation	Repair
16	Err 210	Ach Communication error in Analog	Repair
17	Err 211	Ach RST line error	Repair
18	Err 215	Ach 1/100 ATT measurement test error in analog operation	Repair
19	Err 216	Ach 1/500 ATT measurement test error in analog operation	Repair
20	Err 217	Ach Gx100 measurement test error in analog operation	Repair
21	Err 221	Ach Gx1 measurement test error in analog operation	Repair
22	Err 251	Ach DCFILT reply test error	Repair
23	Err 300	Bch Ratio test error of IR1 to IR2 in AD operation	Repair
24	Err 301	Bch Ratio test error of IR2 to IR3 in AD operation	Repair
25	Err 305	Bch ACx1 zero measurement test error in analog operation	Repair
26	Err 306	Bch Lx1 zero measurement test error in analog operation	Repair
27	Err 307	Bch Lx10 zero measurement test error in analog operation	Repair
28	Err 308	Bch Lx100 zero measurement test error in analog operation	Repair
29	Err 310	Bch Communication error in analog section	Repair

No.	Error display	Error description []: Response to the ERR? command	Solution
30	Err 311	Bch RST line error	Repair
31	Err 315	Bch ATT 1/100 measurement test error in analog operation	Repair
32	Err 351	Bch DCFILT reply test error	Repair
33	Err 500	Calibration data error [CAL data]	Solution3
34	Err 501	ZERO calibration is not performed. [before ZERO]	Solution4
35	Err-102	Command syntax error [Cmd Syntax]	Solution5
36	Err-113	The command cannot be used. [Undefined header]	Solution5
37	Err-121	Unrecognized character as the numerical value [Invalid character in number]	Solution5
38	Err-141	Invalid character string parameter is entered. [Invalid character data]	Solution5
39	Err-151	Error as the character string data (Quotation mark error) [Invalid string data]	Solution5
40	Err-200	Execution error (Command which cannot be executed now) [Execution error]	Solution6
41	Err-213	The INITiate command is ignored because the measurement is performing. [Init ignored]	Solution6
42	Err-214	The READ? or MEAS? command is received when any trigger source except for IMM is set. [Trigger deadlock]	Solution6
43	Err-222	Entered value is outside the setting range or lacking for necessary parameters [Data out of range]	Solution6
44	Err-313	Calibration data is lost. [Calibration memory lost]	Solution7
45	Err-314	Parameters saved by “*SAV” command are lost. [Save/recall memory lost]	Solution8
46	Err-315	Saved parameters are lost. [Configuration memory lost]	Solution6
47	Err-330	Self-test error [Self-test failed]	Repair
48	Err-350	Error queue overflows. [Queue overflow]	Solution9
49	Err d	Calculation error (Refer to Section 5.9.4 dB and dBm Calculations and Section 5.9.8 Calculation between 2 Measurements.)	Solution11

8.11 Error Message List

Solution1	Set the correct function.
Solution2	Check the USB cable connection.
Solution3	Calibrate according to the correct procedure.
Solution4	Perform the ZERO calibration.
Solution5	Send the correct command.
Solution6	Check the settings.
Solution7	Re-calibrate.
Solution8	Re-set.
Solution9	Read out the error queue.
Solution10	Check the RS-232 cable connection
Solution11	Refer to Section 5.9.4 dB and dBm Calculations and Section 5.9.8 Calculation between 2 Measurements.

NOTE: *If the error cannot be fixed by the described solution, contact an ADC CORPORATION sales representative this instrument.*

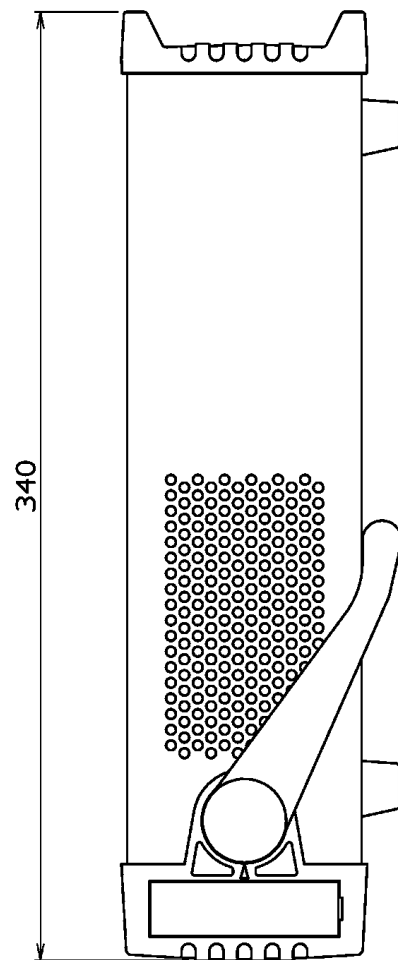
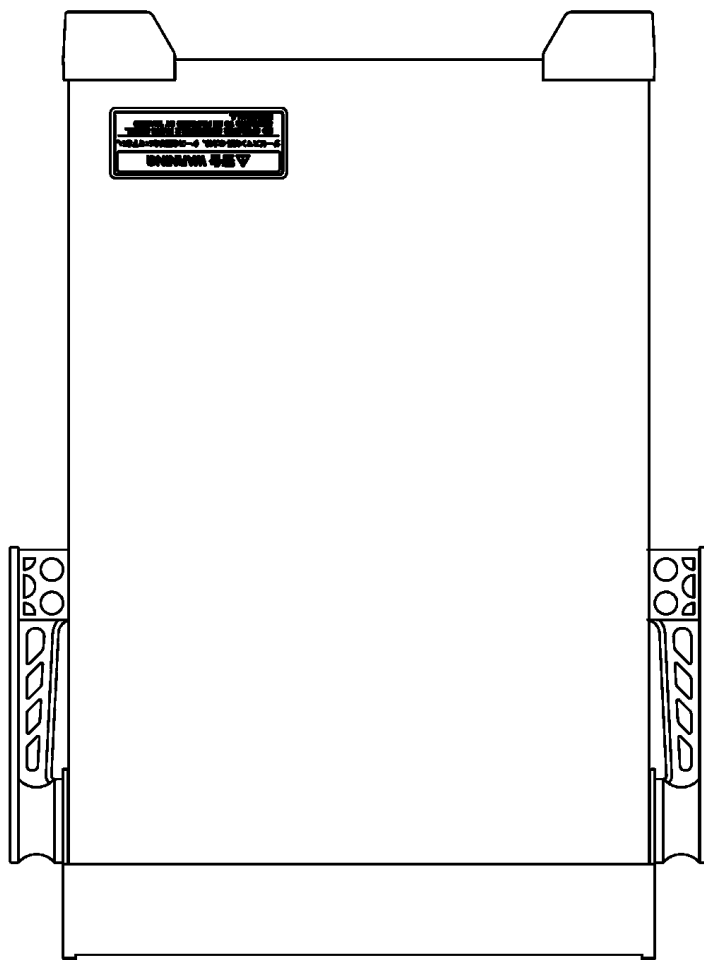


Unit : mm

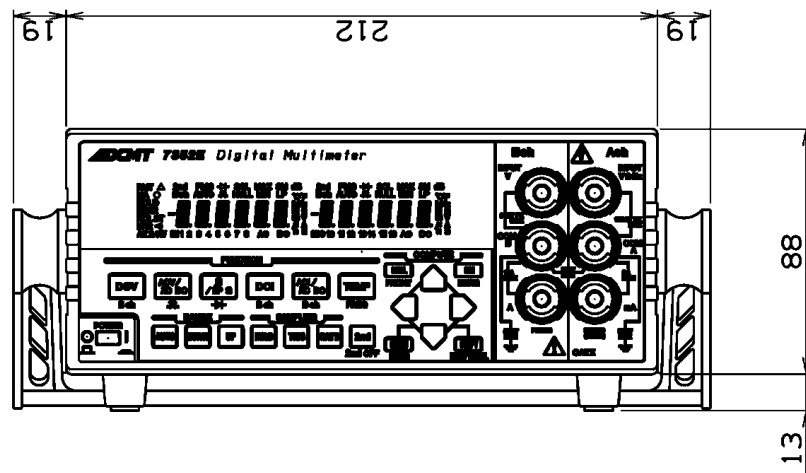
NOTE

This drawing shows external dimensions of this instrument.

The difference in products and options used can cause a change in the appearance of the instrument.



7352A DIMENSIONAL OUTLINE DRAWING

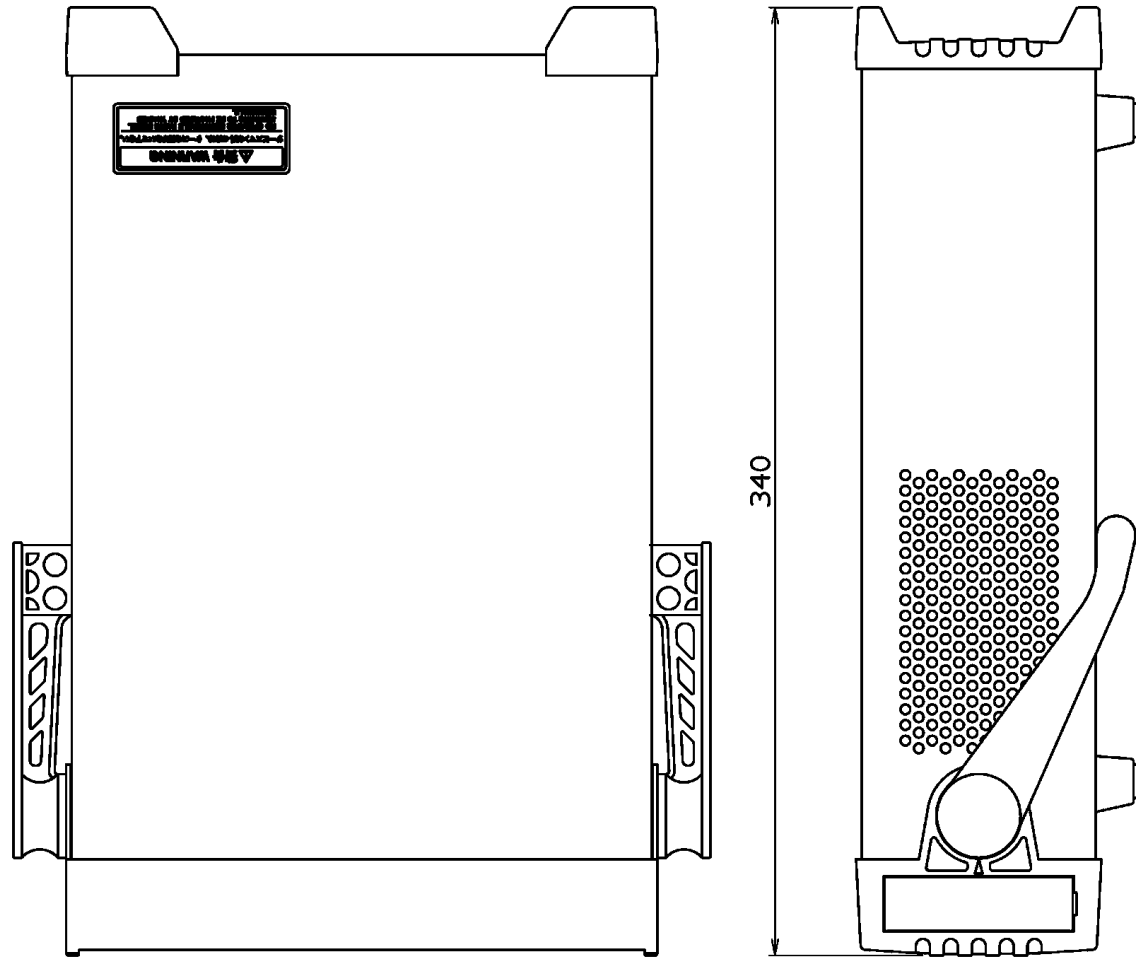


Unit : mm

NOTE

This drawing shows external dimensions of this instrument.

The difference in products and options used can cause a change in the appearance of the instrument.



7352E DIMENSIONAL OUTLINE DRAWING

ALPHABETICAL INDEX

[A]

AC Current Measurement (ACI, ACI(AC+DC))	7-8
AC Voltage Measurement	7-3
Accessories	3-5, 7-18
Ach AC Current Measurement (ACI-Ach and ACI(AC+DC)-Ach)	4-13, 7-8
Ach AC Voltage Measurement (ACV-Ach and ACV(AC+DC)-Ach)	4-9, 7-3
Ach Continuity Measurement	4-14
Ach Continuity Measurement-Ach	7-14
Ach DC Current Measurement (DCI-Ach)	4-12, 7-6
Ach DC Voltage Measurement (DCV-Ach)	4-8, 7-1
Ach Diode Measurement	4-15
Ach Diode Measurement-Ach	7-13
Ach Frequency Measurement (FREQ-Ach)	4-16, 7-12
Ach Measurement Input Terminals	4-4
Ach Resistance Measurement (2W Ω -Ach and LP-2W Ω -Ach)	4-10, 7-5
ADC Command Reference	6-18
Address and Phone Number	8-17
Auto-Loading when Turning on the Power	5-39
Auto-range	5-10
Auto-Zero Operation	5-11

[B]

Basic Operation	4-8
Bch AC Current Measurement (ACI-Bch and ACI(AC+DC)-Bch)	4-20, 7-10
Bch DC Current Measurement (DCI-Bch)	4-19, 7-7
Bch DC Voltage Measurement (DCV-Bch)	4-18, 7-2
Bch Measurement Input Terminals	4-4
Before Asking for Repair	8-16
Buzzer Setting	5-41

[C]

Calculation between 2 Measurements	5-38
Calculation Functions	5-23, 7-15
Calculation Selection Keys	4-3
Calibration	8-5
Calibration Point	8-7
Calibration Procedure	8-9
Calibration Standards	8-6
CAT II (Marking)	4-5

Caution when Connecting Peripherals	3-7
Changing the Measurement Speed and Display Digit	4-22
Changing the Power Supply Voltage	3-9
Checking the Power Supply Voltage	3-8
Cleaning	8-4
Cleaning Others	8-4
Cleaning the Outside	8-4
Command Reference	6-11
Comparator Calculation	5-32
Compatibility	6-12
Connecting the Power Cable	3-11
Connecting to a Personal Computer	6-9
Contacting ADC CORPORATION for Repair or Calibration	8-17
Contents of this Manual	1-1
Continuity Measurement	5-8

[D]

dB and dBm Calculations	5-30
DC Current Measurement (DCI)	7-6
DC Voltage Measurement (DCV)	7-1
Disabling Functions	5-41
Display and Key Tests	8-23
Display Digit Setting	5-12
Display Mode Setting	5-12
Display Section	4-2
Dual Sequential Measurement	5-15
Dual Synchronous Measurement	5-16

[E]

Electromagnetic Interference	2-2
Error Message List	8-24
Error Queue	5-45
External Trigger Terminal (TRIGGER IN) (7352A)	6-10

[F]

Front Panel Description	4-1
Front Panel Disabling Function	5-43

[G]

General Specifications	7-17
GPIO (7352A)	6-2
GPIO Setting	6-4

[H]

How to Check Protection Fuses	8-2
How to Navigate the Menus	5-1

Alphabetical Index

How to Replace Protection Fuses	8-3
How to Use Dual Display	4-26
How to Use Dual Display by Remote Operation	4-27
HOW TO USE THE INTERFACE	6-1

[I]

If a Fault Occurs	2-1
Input Cable	3-5
Inspection on Delivery	3-1
Installation Environment	3-3
Interface Selection	6-1
Interface Specifications	7-16
INTRODUCTION	1-1

[L]

Loading the Setting Parameters	5-39
--------------------------------------	------

[M]

MAINTENANCE	8-1
MAX/MIN Calculation	5-35
Measurement Function and Range Structure	5-9
Measurement Function Keys	4-2
Measurement Functions	4-8, 5-8
Measurement Range and Auto-Range	5-9
Measurement Range Selection Keys	4-3
Memory Clear	5-40
Memory Recall	5-40
Memory Store	5-40
Menu	5-1
Menu List	5-4
MENU OPERATION AND FUNCTION DESCRIPTION	5-1
MENU/ENTER Key	4-3

[N]

Note when Turning on the Power	2-2
Notes Regarding Repair, Replacement, and Periodic Calibration	8-16
NULL Calculation	5-25

[O]

Operating Environment	3-3
Output Data Format	6-13
Overview	6-2, 6-5, 6-9

[P]

Panel Description	4-1
-------------------------	-----

Power Fuse	2-1
Power Requirements	3-8
Power Switch	4-2
PRECAUTIONS	2-1
Precautions in Use of GPIB	6-3
Preparing for Calibration	8-5
Product Disposal and Recycle	8-13
Product Overview	1-2
Protecting Against Electrostatic Discharge	3-4

[Q]

QUICK START	4-1
-------------------	-----

[R]

Rear Panel Description	4-6
Removing the Case	2-1
Replacing a Power Fuse	8-1
Replacing a Protection Fuse	8-2
Replacing Fuses	8-1
Replacing Parts with Limited Life	8-12
Right or Left Display Section Selection Key	4-4
RS-232 Output Data Format	6-8
RS-232 Setting	6-6
RS-232[EIA-232] (7352A)	6-5

[S]

Sample Programs	6-52
Sample Programs (GPIB commands used in ADC CORPORATION)	6-52
Sample Programs (RS-232)	6-60
Sample Programs (SCPI commands for GPIB)	6-56
Sample Programs (USB)	6-62
Sampling Count Setting	5-17
Sampling Operation	5-13
Sampling Selection Keys	4-3
Saving and Loading the Measurement Condition Settings	5-39
Saving the Setting Parameters	5-39
Scaling Calculation	5-28
SCPI Command Reference	6-31
Selecting the Command Language	6-11
Self-Test	8-22
Setting the Measurement Range	4-22
Setting Up the USB Port	6-9
SETUP	3-1
SHIFT/EXIT/LOCAL Key	4-4
Single Measurement	5-14
Smoothing Calculation	5-26
SPECIFICATIONS	7-1

Specifications	7-1
Statistical Calculation	5-37
Status Register Structure	6-43
Storage	8-15
Storing and Recalling Measurement Data	5-40
System Recovery Procedure	8-18
System Settings	5-41

[T]

Temperature Measurement	4-21, 5-8
Temperature Measurement (TEMP)	7-13
Transportation	8-15
Trigger Count Setting	5-22
Trigger Delay (Td)	5-21
Trigger Function	5-18
Trigger Source Selection	5-20
Trigger System Operation	5-19

[U]

USB	6-9
USB Specifications	6-9
USBid Setting	6-9

Other ManualsLib Projects

 www.manualslib.com

 www.manualslib.de

 www.manualslib.es

 www.manualslib.fr

 www.manualslib.nl

 www.manualslib.mx

 www.manualslib.tech 30+ Languages